MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, FLIP-FLOPS AND LATCHES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Dual D-type edge-triggered flip-flop
02	Dual J-K master slave flip-flop
03	Quad 3-state R/S latch
51	Dual D-type edge-triggered flip-flop
52	Dual J-K master slave flip-flop
53	Quad 3-state R/S latch

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
X	F-1 (14-lead, 1/4" x 1/4"), flat package, except A dimension = 0.1" (2.54 mm maximum)
Y	F-2 (14-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm maximum)
Z	F-5 (16-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm maximum)

NOTES:
1. As an exception to 3.5.6.2.3 of MIL-M-38510, for case outlines X, Y, and Z only, the leads of bottom brazed ceramic packages (i.e., configuration 2 of case outlines F-1, F-2, or F-5) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e., finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.
2. For bottom or side braded packages, case outlines X, Y, and Z only, the S_1 dimension may go to .000 inch (.00 mm) mimimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: National Aeronautics and Space Administration, George C. Marshall Space Flight Center, ATTN: EG02, Marshall Space Flight Center, AL 35812, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range (V_{DD} - V_{SS})	
Device types 01, 02, and 03	-0.5 V to +15.5 V
Device types 51, 52, and 53	-0.5 V to +18.0 V
Input current (each input)	±10 mA
Input voltage range	(V_{SS} - 0.5 V) ≤ V_I ≤ (V_{DD} + 0.5 V)
Storage temperature range	-65°C to +175°C
Maximum power dissipation (P_D)	200 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC}):	(See MIL-M-38510, appendix C)
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions:

Device types 01, 02, and 03:	
Supply voltage (V_{DD} - V_{SS})	4.5 V dc to 12.5 V dc
Input low (V_{IL}) voltage range	0-0.85 V dc @ V_{DD} = 5 V; 0-2.0 V dc @ V_{DD} = 10 V 0-2.1 V dc @ V_{DD} = 12.5 V
Input high (V_{IH}) voltage range	3.95-5.0 V dc @ V_{DD} = 5 V; 8-10 V dc @ V_{DD} = 10 V 10.0-12.5 V dc @ V_{DD} = 12.5 V
Device types 51, 52, and 53:	
Supply voltage (V_{DD} - V_{SS}	4.5 V dc to 15 V dc
Input low (V_{IL}) voltage range	0-1.5 V dc @ V_{DD} = 5 V dc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}, 0-2.0 V dc @ 10 V dc, 0-4.0 V dc @ V_{DD} = 15 V dc
Input high (V_{IH}) voltage range	3.5-5.0 V dc @ V_{DD} = 5 V dc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}, 8.0-10.0 V dc @ V_{DD} = 10 V dc, 11.0-15.0 V dc @ V_{DD} = 15 V dc
Load capacitance	50 pF maximum
Ambient operating temperature range	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent, which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.2.3 Truth tables. The truth tables shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III. V_{ICL}, V_{IL}, V_{OH}, $F_{CL(MAX)}$, $T_{TLHCL(MAX)}$, t_p, t_{SHL}, t_{HHL}, t_{HLH}, subgroups 7 and 8 testing requires only a summary of attributes data.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 38 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute sequentially lines 1 through 7 of table II herein.

b. Burn-in test (method 1015 of MIL-STD-883).

1. Static tests (test condition A) using circuit shown on figure 4, or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

2. Dynamic test (test condition D) using circuit shown on figure 5, or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter measurements.

4.2.1 Percent defective allowable (PDA).

a. The class S devices PDA shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining PDA.

c. The PDA for class B devices shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics after burn-in exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in lot initially submitted to burn-in shall be used to determine the percent defective for the lot and lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.3.1 Qualification extension. When authorized by the qualifying activity, if a manufacturer qualifies to a 51, 52, or 53 device type which is manufactured identically to a 01, 02, or 03 device type on this specification, then the 01, 02, or 03 device type may be part I qualified by conducting only group A electrical tests and any electricals specified as additional group C subgroups and submitting data in accordance with MIL-M-38510, appendix D (i.e. groups A, B, C, D, and E tests are not required).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, -55°C < T_C < +125°C, Unless otherwise specified		Device type	Limits		Unit
					Min	Max	
Positive clamping input to V_{DD}	$V_{IC(pos)}$	T_A = 25°C, V_{DD} = GND, V_{SS} = Open, Output = Open, I_I = 1 mA		All		1.5	V dc
Negative clamping input to V_{SS}	$V_{IC(neg)}$	T_A = 25°C, V_{DD} = Open, V_{SS} = GND, Output = Open, I_I = -1 mA		All		-6	V dc
Quiescent supply current	I_{SS}	Any combination of inputs	V_{DD} = 15 V	01,02,03		-2.5	µA
			V_{DD} = 18 V	51,52,53		-2.5	
High level output voltage (Set-reset input)	V_{OH1}	V_{DD} = 5 V dc, I_{OH} = -0.175 mA (See table III)		01, 02, 03	4.5		V dc
	V_{OH2}	V_{DD} = 5 V dc, I_{OH} = 0 (See table III)		01, 02, 03	4.95		V dc
	V_{OH3}	V_{DD} = 12.5 V dc, I_{OH} = 0 (See table III)		01, 02, 03	11.25		V dc
High level output voltage (Data input)	V_{OH4}	V_{DD} = 5 V dc, I_{OH} = 0, Any one input = V_{IL} (See table III)		01, 02	4.95		V dc
	V_{OH5}	V_{DD} = 5 V dc, I_{OH} = 0 (See table III)		51, 52, 53	14.95		V dc
	V_{OH6}	V_{DD} = 15 V dc, I_{OH} = 0 (See table III)		51, 52	14.95		V dc
Low level output voltage (Set-reset input)	V_{OL1}	V_{DD} = 5 V dc, I_{OL} = 0.33 mA (See table III)		01, 02, 03		0.50	V dc
	V_{OL2}	V_{DD} = 5 V dc, I_{OL} = 0 (See table III)		01, 02, 03		0.05	V dc
	V_{OL3}	V_{DD} = 12.5 V dc, I_{OL} = 0 (See table III)		01, 02, 03		1.25	V dc

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, -55°C ≤ T_C ≤ +125°C, unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Low level output voltage (Data input)	V_{OL4}	V_{DD} = 5 V dc, I_{OL} = 0 (See table III)	01, 02		0.05	V dc
	V_{OL5}	V_{DD} = 15 V dc, I_{OL} = 0 (See table III)	51, 52, 53		0.05	V dc
	V_{OL6}	V_{DD} = 15 V dc, I_{OL} = 0 (See table III)	51, 52		0.05	V dc
Input high voltage	V_{IH1}	V_{DD} = 5 V dc, V_O = 4.5 V, $\lvert I_O \rvert \leq 1$ µA	51, 52, 53	3.5		V dc
	V_{IH2}	V_{DD} = 10 V dc, V_O = 9.0 V, $\lvert I_O \rvert \leq 1$ µA	51, 52, 53	7.0		V dc
	V_{IH3}	V_{DD} = 15 V dc, V_O = 13.5 V, $\lvert I_O \rvert \leq 1$ µA	51, 52, 53	11.0		V dc
Input low voltage	V_{IL1}	V_{DD} = 5 V dc, V_O = 4.5 V, $\lvert I_O \rvert \leq 1$ µA	51, 52, 53		1.5	V dc
	V_{IL2}	V_{DD} = 10 V dc, V_O = 9.0 V, $\lvert I_O \rvert \leq 1$ µA	51, 52, 53		3.0	V dc
	V_{IL3}	V_{DD} = 15 V dc, V_O = 13.5 V, $\lvert I_O \rvert \leq 1$ µA	51, 52, 53		4.0	V dc
Output low (sink) current	I_{OL1}	V_{DD} = 5 V dc, V_{OL} = 0.4 V dc	51, 52, 53	0.36		mA
	I_{OL2}	V_{DD} = 15 V dc, V_{OL} = 1.5 V dc	51, 52, 53	2.4		mA
Output high (source) current	I_{OH1}	V_{DD} = 5 V dc, V_{OH} = 4.6 V dc	51, 52, 53	-0.36		mA
	I_{OH2}	V_{DD} = 15 V dc, V_{OH} = 13.5 V dc	51, 52, 53	-2.4		mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, -55°C ≤ T_C ≤ +125°C, Unless otherwise specified		Device type	Limits Min	Limits Max	Unit
Input leakage current 2/	I_{IH}	Measure inputs sequentially	V_{DD} = 15 V dc	01, 02, 03		45	nA
			V_{DD} = 18 V dc	51, 52, 53			
	I_{IL}	Measure inputs sequentially	V_{DD} = 15 V dc	01, 02, 03		-45	nA
			V_{DD} = 18 V dc	51, 52, 53			
Input test voltage	V_{ZAP}	Ci = 100 pF, R_2 = 1.5 kΩ (See 4.5.3)		All	400		V
Input capacitance	Ci	V_{DD} = 0 V dc, f = 1 MHz, T_A = 25°C		All		12	pF
Propagation delay high to low level	t_{PHL}	V_{DD} = 5.0 V dc, C_L = 50 pF		01	35	750	ns
				02	35	865	
				51	25	750	
				52	29	865	
Propagation delay low to high level	t_{PLH}	V_{DD} = 5.0 V dc, C_L = 50 pF		01	35	825	ns
				02	35	940	
				51	28	825	
				52	31	940	
Propagation delay high to low level (Set and reset)	t_{PHL} (R) or (S)	V_{DD} = 5.0 V dc, C_L = 50 pF		01	35	825	ns
				02	35	900	
				03	14	370	
				51	28	825	
				52	30	900	
				53	14	370	
Propagation delay low to high level (Set and reset)	t_{PLH} (R) or (S)	V_{DD} = 5.0 V dc, C_L = 50 pF		01	35	630	ns
				02	35	600	
				03	9	245	
				51	21	630	
				52	20	600	
				53	9	245	
Transition time high to low levels	t_{THL}	V_{DD} = 5.0 V dc, C_L = 50 pF		01	10	450	ns
				02	10	490	
				03	9	245	
				51	15	450	
				52	16	490	
				53	9	245	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, -55°C ≤ T_C ≤ +125°C, Unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Transition time low to high levels	t_{TLH}	V_{DD} = 5.0 V dc, C_L = 50 pF	01	10	525	ns
			02	10	490	
			03	13	360	
			51	18	525	
			52	16	490	
			53	13	360	
Maximum clock frequency	$f_{CL(max)}$	V_{DD} = 5.0 V dc, C_L = 50 pF	01		1	MHz
			02		700	kHz
			51		1	MHz
			52		700	kHz
Maximum clock transition times	t_{TLHCL}	V_{DD} = 5.0 V dc, C_L = 50 pF	01		10	µs
			02			
			51			
			52			
Minimum clock pulse width	t_p	V_{DD} = 5.0 V dc, C_L = 50 pF	01	300		ns
			02			
			51			
			52			
Set-up times	t_{SHL}, t_{SLH}	V_{DD} = 5.0 V dc, C_L = 50 pF	01		225	ns
			02			
			51			
			52			
Hold times	t_{HLH}, t_{HHL}	V_{DD} = 5.0 V dc, C_L = 50 pF	01		225	ns
			02			
			51			
			52			
Output enable time	t_{PZH}, t_{PZL}	V_{DD} = 5.0 V dc, C_L = 50 pF	03		340	ns
			53		240	
Output disable time	t_{PHZ}, t_{PLZ}	V_{DD} = 5.0 V dc, C_L = 50 pF	03		340	ns
			53		240	

1/ Complete terminal conditions shall be as specified in table III.

2/ Input current at one input node.

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TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test methods	Class S device 3/				Class B device 3/			
		Ref. par.	Table 2/ III subgroups	Table 1/ IV delta limits	Test circuit figure	Ref. par.	Table 2/ III subgroups	Table 1/ IV delta limits	Test circuit figure
1	Interim electrical parameters method 5004		1				1		
2	Static burn-in I method 1015	4.2b. 4.5.2			4				
3	Same as line 1		1	Δ					
4	Static burn-in II method 1015	4.2b. 4.5.2			4	4.2b. 4.5.2	4/		4
5	Same as line 1		1*	Δ		4.2d.	1*	Δ	
6	Dynamic burn-in method 1015	4.2b. 4.5.2			5				
7	Same as line 1	4.2d.	1*	Δ					
8	Final electrical parameters method 5004		1*,2,3,7, 9				1*,2,3,7, 9		
9	Group A end point electrical parameters method 5005	4.4.1	1,2,3,4, 7,8,9,10, 11			4.4.1	1,2,3,4, 7,9		
10	Group B end point electrical parameters method 5005	4.4.2	1,2,3,7, 8,9,10,11	Δ					
11	Group C end point electrical parameters method 5005					4.4.3	1,2,3	Δ	
12	Additional group C tests method 5005					4.4.3c.	10,11		
13	Group D end point electrical parameters method 5005	4.4.4	1,2,3			4.4.4	1,2,3		

1/ (Δ) indicates delta limits shall be required on table III subgroup 1, where specified, and delta values shall be computed with reference to the previous interim electrical parameters.

2/ (*) indicates PDA applies to subgroup 1 (see 4.2.1).

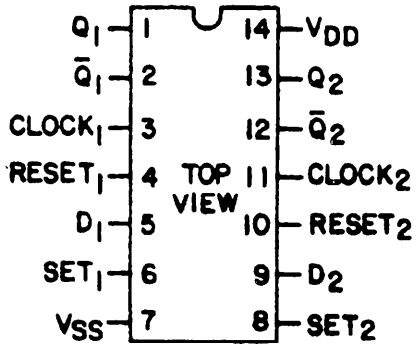
3/ Blank spaces indicate tests are not applicable.

4/ The device manufacturer may, at his option, either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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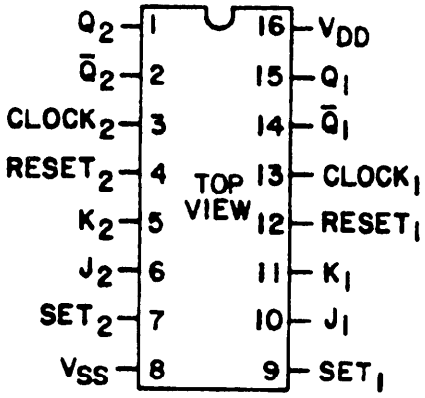
Device types 01 and 51

CASES A, C, D, X AND Y



Device types 02 and 52

CASES E, F AND Z



Device types 03 and 53

CASES E, F AND Z

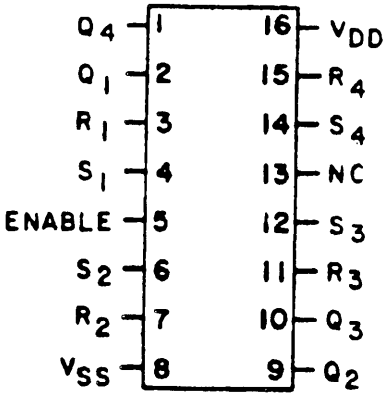
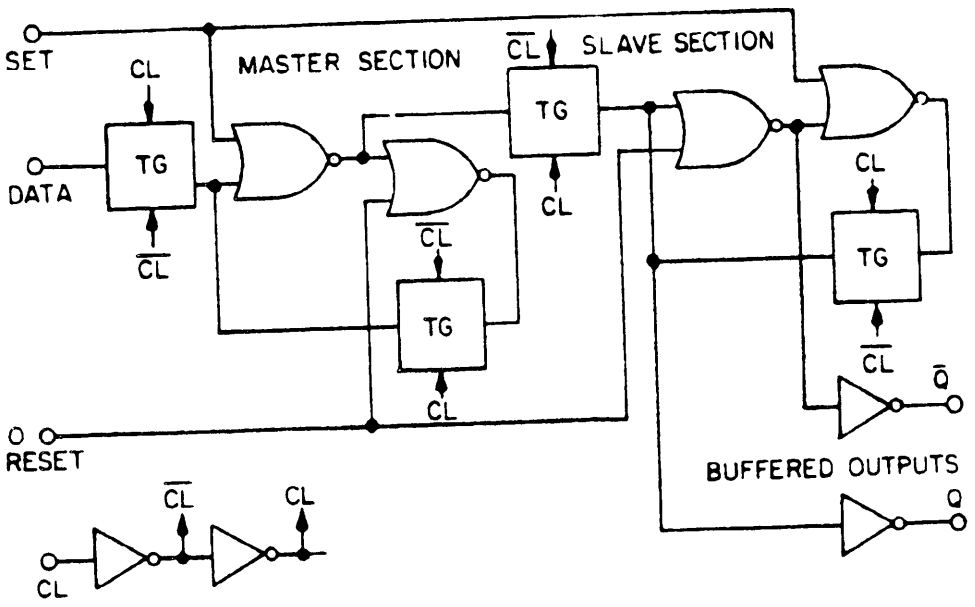


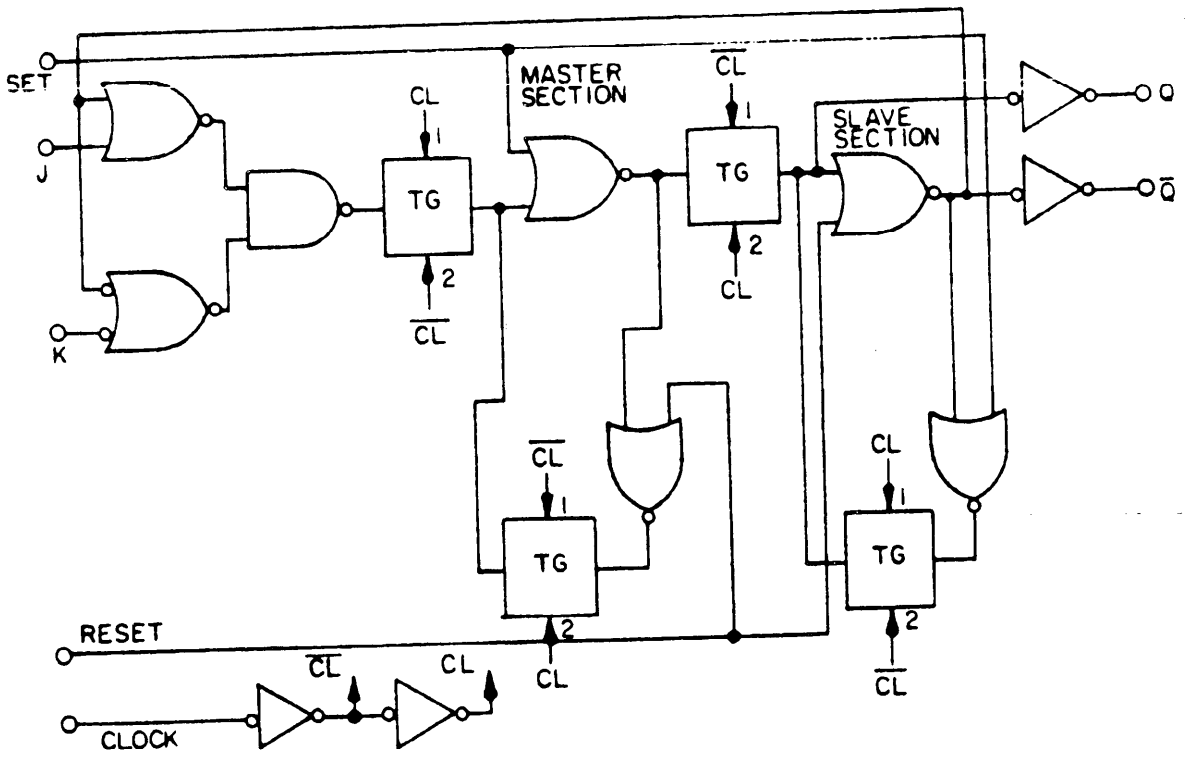
FIGURE 1. Terminal connections.

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Device types 01 and 51



Device types 02 and 52



NOTE: One of two identical flip flops shown.

FIGURE 2. Logic diagram.

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Device types 03 and 53

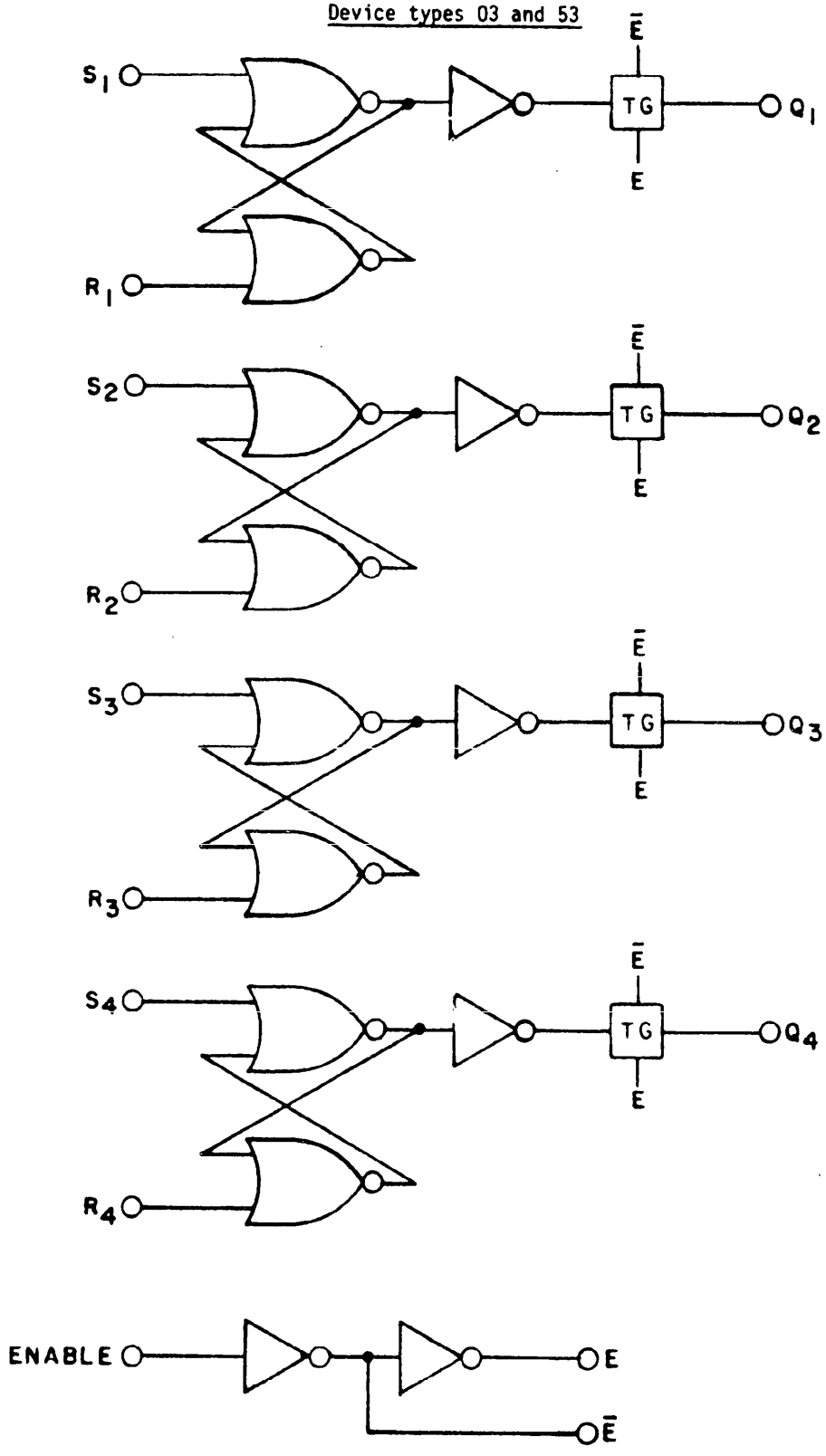


FIGURE 2. Logic diagram - Continued.

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Device types 01 and 51

INPUTS				OUTPUTS		
CLΔ	D	R	S	Q	$\bar{Q}$	
↑	L	L	L	L	H	
↑	H	L	L	H	L	
↓	X	L	L	Q	$\bar{Q}$	NO CHANGE
X	X	H	L	L	H	
X	X	L	H	H	L	
X	X	H	H	H	H	

Positive logic

Δ = Level change
X = Don't care case

Device types 02 and 52

● t_n Inputs						‡ t_{n+1} Outputs		
CLΔ	J	K	S	R	Q	Q	$\bar{Q}$	
↑	H	X	L	L	L	H	L	
↑	X	L	L	L	H	H	L	
↑	L	X	L	L	L	L	H	
↑	X	H	L	L	H	L	H	
↓	X	X	L	L	X	Q	$\bar{Q}$	NO CHANGE
X	X	X	H	L	X	H	L	
X	X	X	L	H	X	L	H	
X	X	X	H	H	X	H	H	

Positive logic

Δ = Level change
X = Don't care
● = t_n refers to the time interval before the positive clock pulse transition
‡ = t_{n+1} refers to the time interval after the positive clock pulse transition

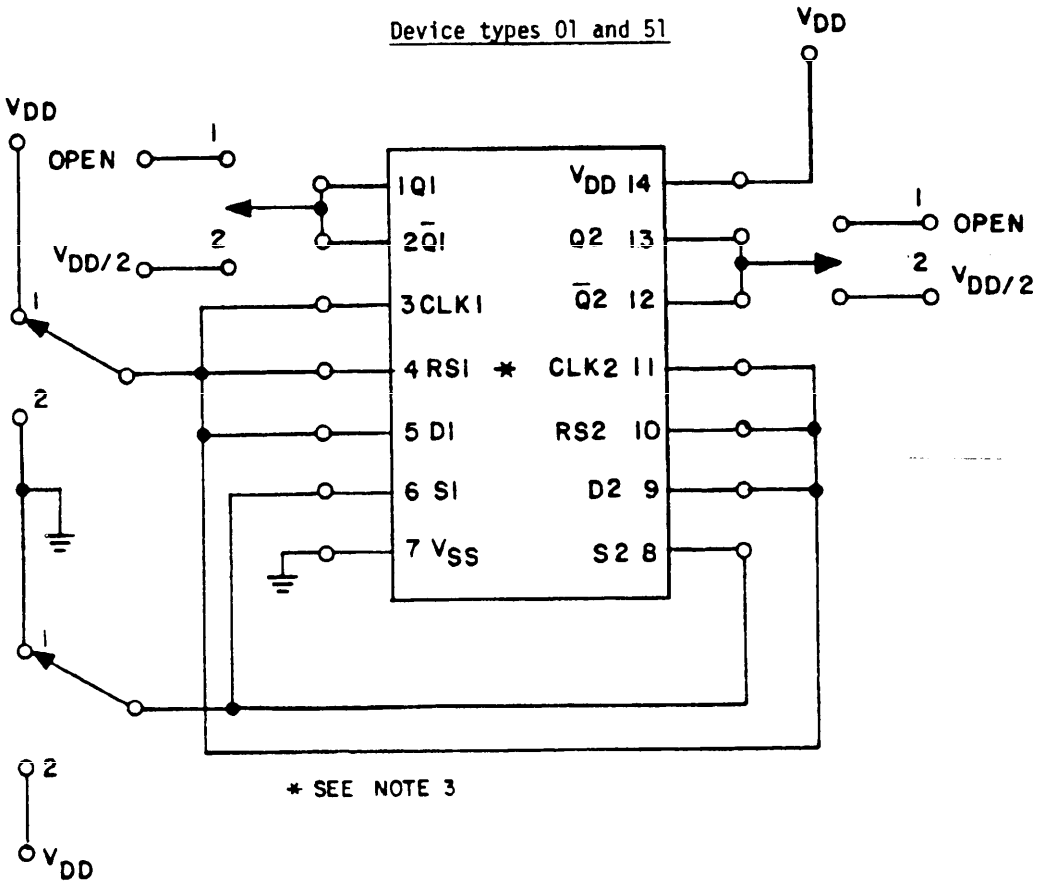
Device types 03 and 53

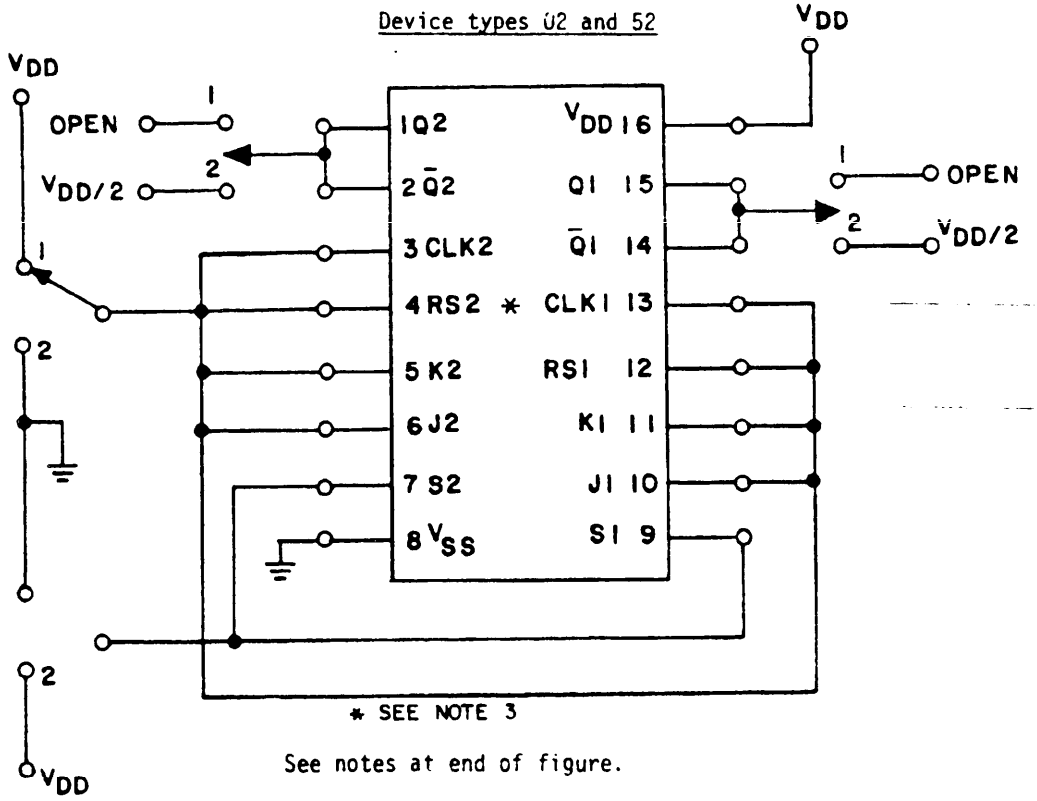
INPUTS			OUTPUT
S	R	E	Q
X	X	L	OC* High impedance
L	L	H	NC+
H	L	H	H
L	H	H	L
H	H	H	Δ High

* Open circuit high impedance
\+ No change
Δ Dominated by S = 1 input (high)
X Don't care

FIGURE 3. Truth tables.

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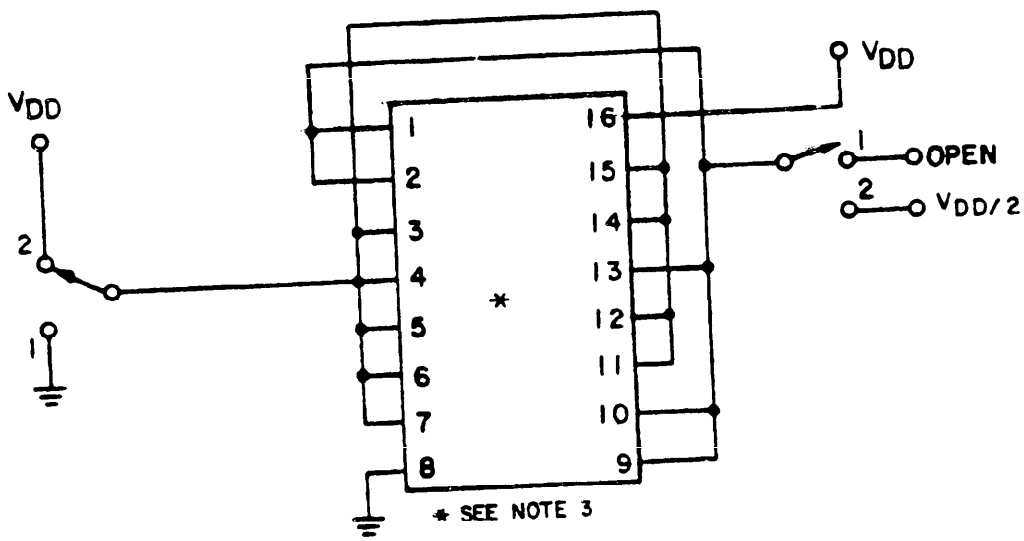


See notes at end of figure.

FIGURE 4. Static burn-in test circuits.

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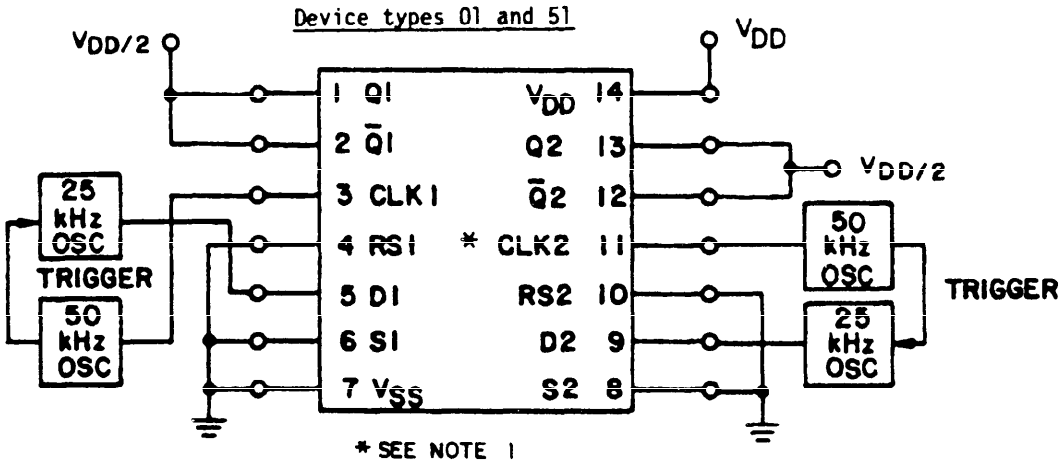
Device types 03 and 53

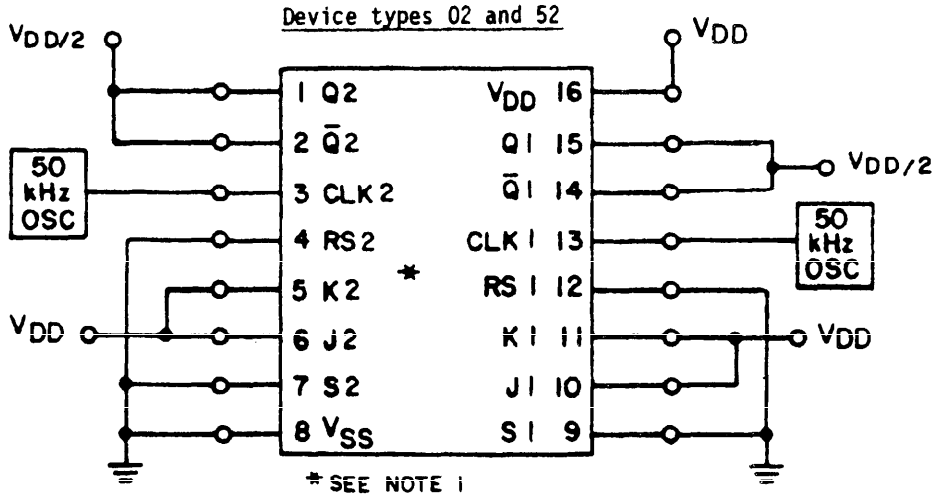


NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.
5. V_{DD} = 12.5 V minimum, 15 V maximum for device types 01, 02, 03.
   V_{DD} = 15 V minimum, 18 V maximum for device types 51, 52, 53.
   V_{DD} = $V_{DD}/2$ ±1.0 V for all devices.
   V_{DD} = 0 V.

FIGURE 4. Static burn-in test circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:03Z
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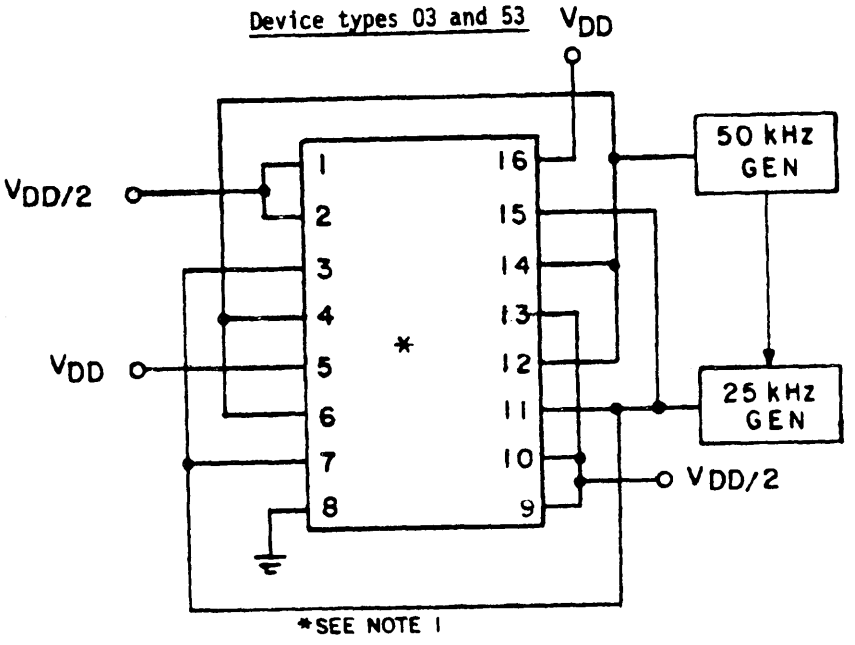




See notes at end of figure.

FIGURE 5. Dynamic burn-in and steady state test circuit.

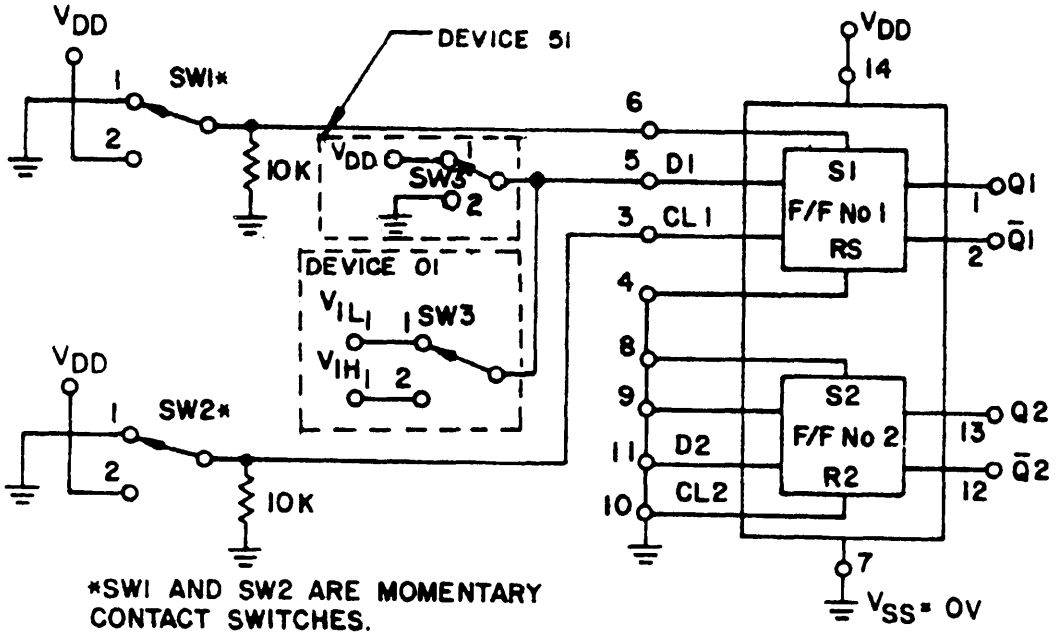
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ or 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz ≤ PRR ≤ 1 MHz.
   c. t_{TLH} and $t_{THL} \leq 1$ µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}
      Maximum = V_{DD} +0.5 V, -10% V_{DD}
3. V_{DD} = 12.5 V minimum, 15.0 V maximum for device types 01, 02, and 03.
   V_{DD} = 15 V minimum, 18 V maximum for device types 51, 52, and 53.
   $V_{DD/2}$ = $V_{DD/2}$ ±1.0 V.
   V_{DD} = 0 V.

FIGURE 5. Dynamic burn-in and operating life test circuit.

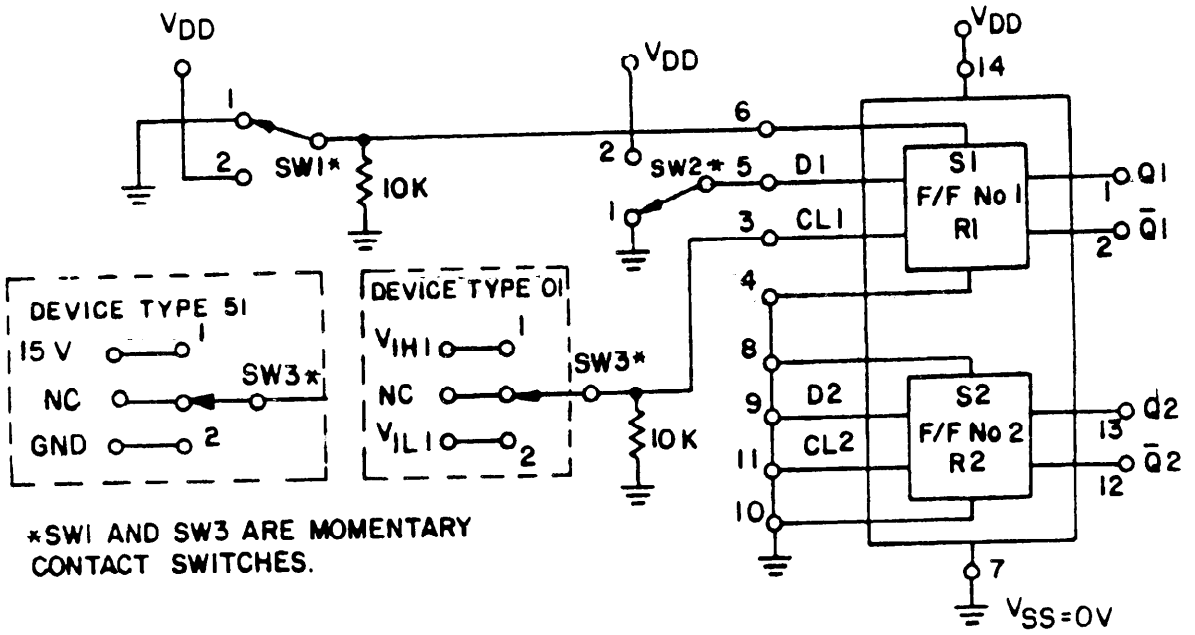
Source: https://assist.dla.mil -- Downloaded: 202610-21T13:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. To implement test numbers 63, 64, 65, and 66 (device type 01), and 47, 48, 49, and 50 (device type 51) place SW3 in the V_{IL1} position. Set the flip-flop by momentarily placing SW1 in position 2. Following the return of SW1 to position 1, momentarily place SW2 in position 2. Measure the output levels at Q and $\overline{Q}$ to insure compliance with table III limits.
2. To implement test numbers 67, 68, 69, and 70 (device type 01), and 51, 52, 53, and 54 (device type 51) set the flip-flop as described in note 1. Place SW3 in the V_{IH1} position. Momentarily place SW2 in position 2. Following the return of SW2 to position 1, measure the output level at Q and $\overline{Q}$ to insure compliance with table III limits.
3. Identical measurements are obtained from either F/F No. 1 or F/F No. 2.

FIGURE 6. Data input high and low test circuit for device types 01 and 51.

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Check the source to verify that this is the current version before use.



NOTES:

1. Test numbers 71 thru 74 (device type 01) and 55 thru 58 (device type 51) implemented by the following step by step sequence:

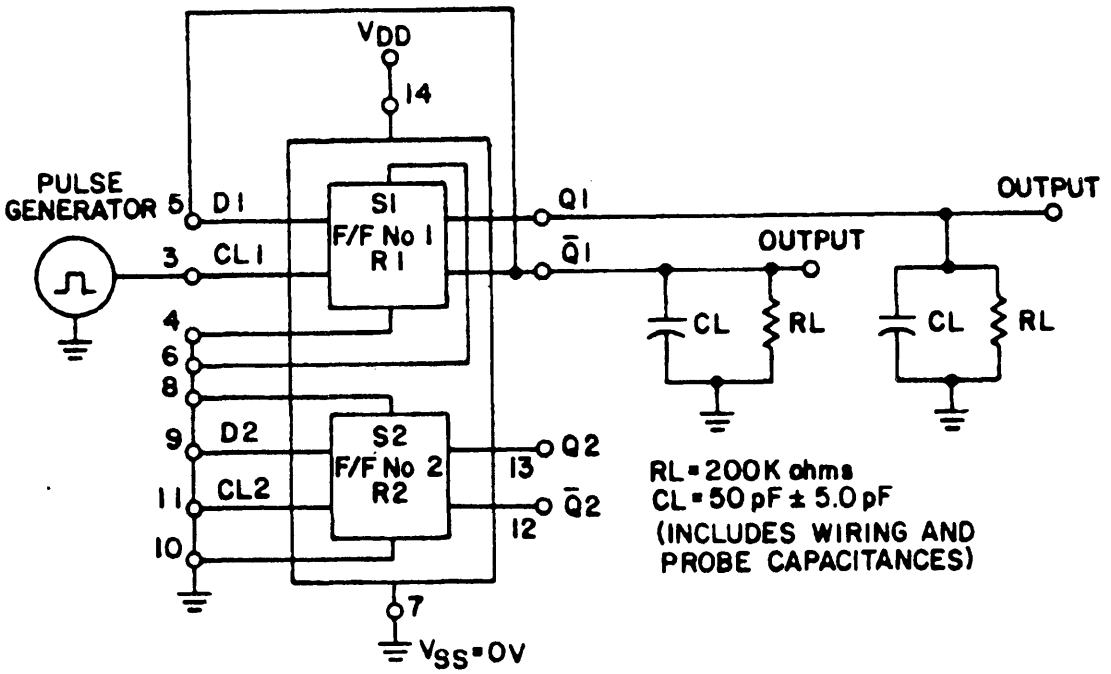
		SW1 POS	SW2 POS	SW3 POS	Q OUT	$\overline{Q}$ OUT	
STEP	1	2*	1	-	"1"	"0"	*Denotes momentary contact.
	2	1	1	1*	"0"	"1"	
	3	1	2	1*	"1"	"0"	
	4	1	1	1*	"0"	"1"	
	5	1	2	2*	"0"	"1"	
	6	1	2	2*	"0"	"1"	
	7	1	2	1*	"1"	"0"	
STEP	8	1	1	1*	"0"	"1"	

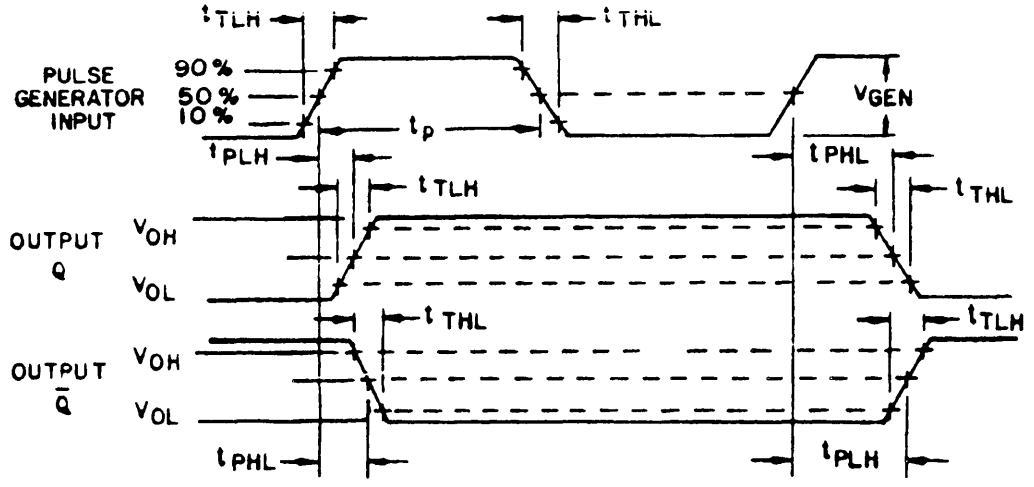
Monitor either Q or $\overline{Q}$ of the flip-flop under test. Compliance with table III limits is established by a change of logic levels at the Q or $\overline{Q}$ output in going from step 1 to step 2, step 2 to step 3, step 3 to step 4, step 6 to step 7, and step 7 to step 8, while no change shall occur in going from step 4 to step 5 or step 5 to step 6.

2. Identical measurements are obtained from F/F No. 1 and F/F No. 2.
3. V_{OH} = 1 and V_{OL} = 0.

FIGURE 7. Clock input high and low test circuit for device types 01 and 51.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:03Z
Check the source to verify that this is the current version before use.

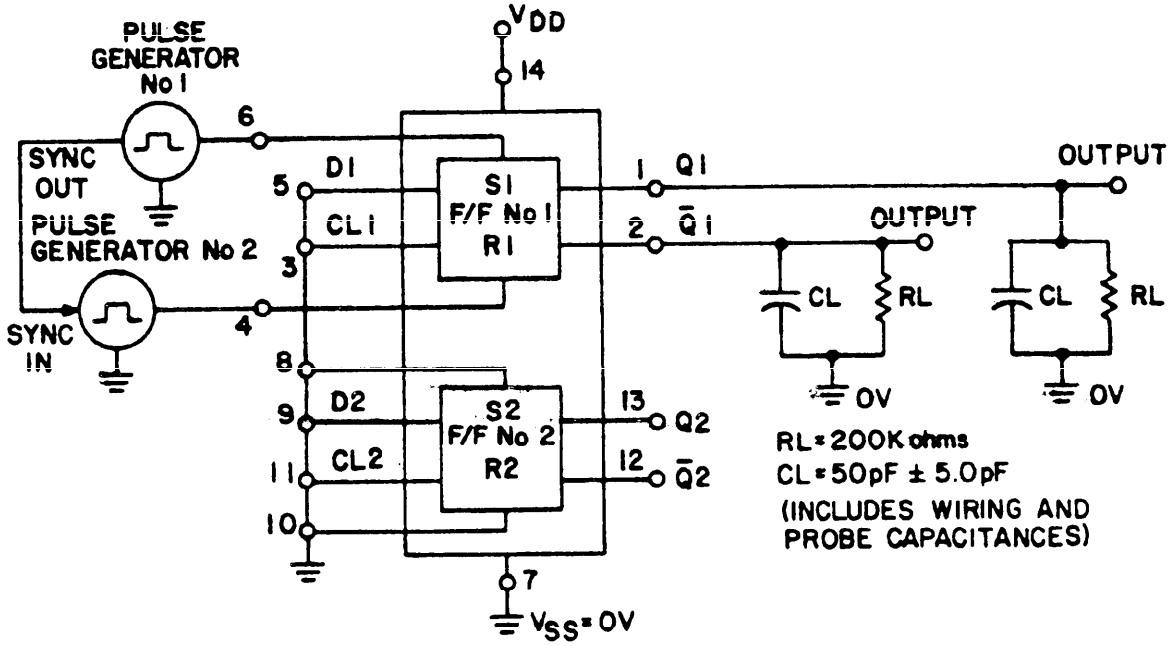


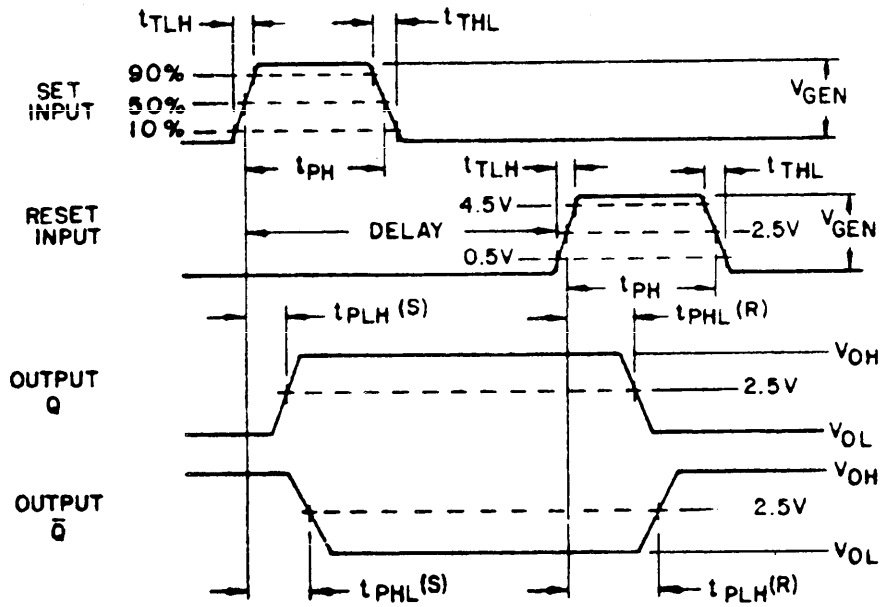


NOTES:
1. The pulse generator has the following characteristics: V_{gen} = V_{DD} ±1%, duty cycle = 50%, t_{TLH} and t_{THL} = 20 ± 2.0 ns and pulse repetition period = 5.0 ± 0.5 µs.
2. Identical switching measurements are obtained from F/F No. 1 and F/F No. 2.
3. For f_{CL} and tp the pulse repetition period is variable.
4. Requirements for max clock frequency (f_{CL}), max clock rise time and minimum clock pulse width are established by setting the parameter to the limits given in table III and observing proper output state changes.

FIGURE 8. Switching time test circuit and waveforms for device types 01 and 51.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:03Z
Check the source to verify that this is the current version before use.

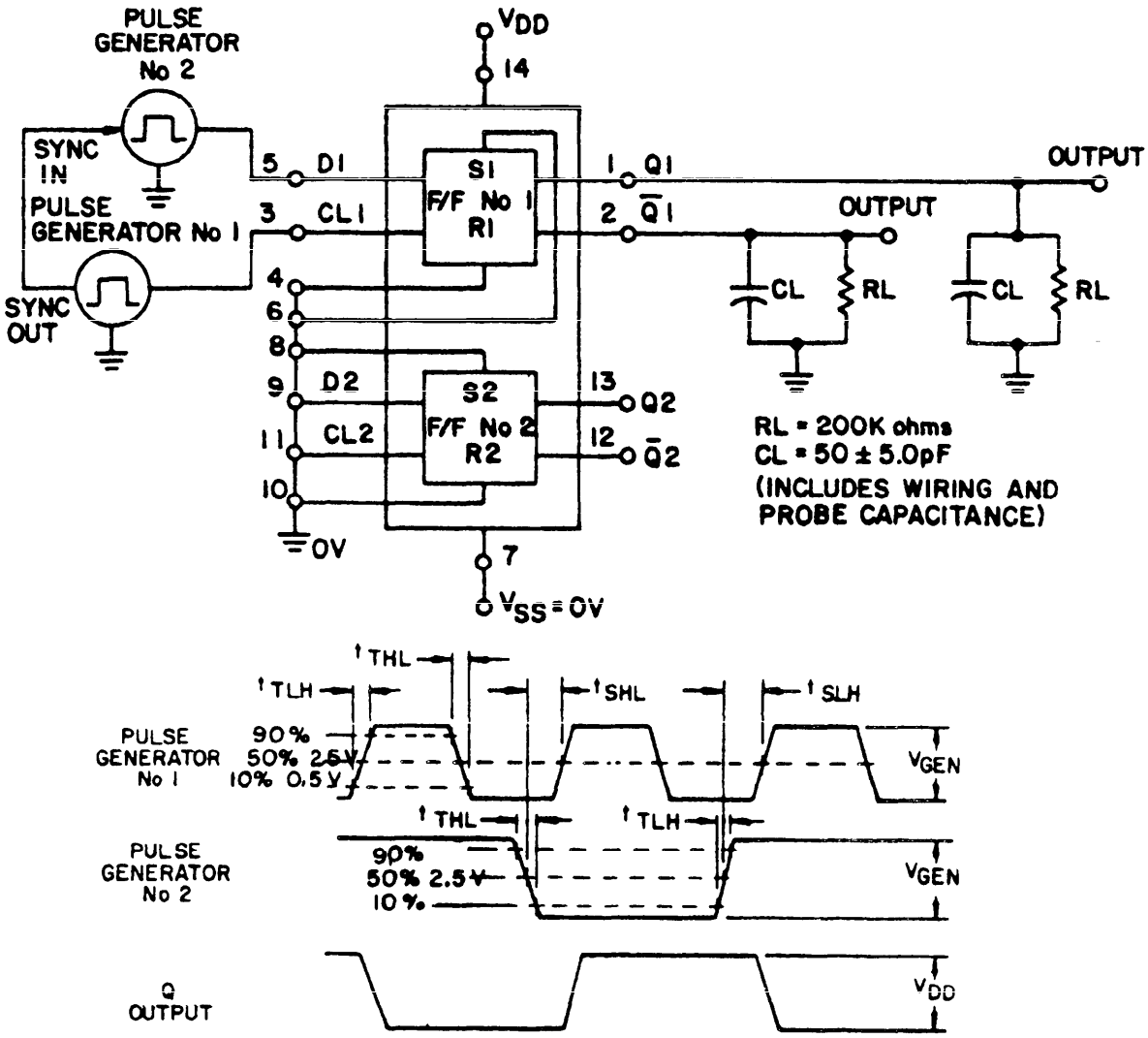




NOTES:
1. The pulse generators have the following characteristics: V_{gen} = VDD ±1%, t_{PH} = 1.0 ±0.1 µs, t_{THL} and t_{TLH} = 20 ±2.0 ns and pulse repetition period = 5.0 ±0.5 µs.
2. The reset pulse delay is 2.5 ±0.25 µs.
3. Identical switching measurements are obtained from F/F No. 1 and F/F No. 2.

FIGURE 9. Set-reset switching test circuit and waveforms for device types 01 and 51.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:03Z
Check the source to verify that this is the current version before use.

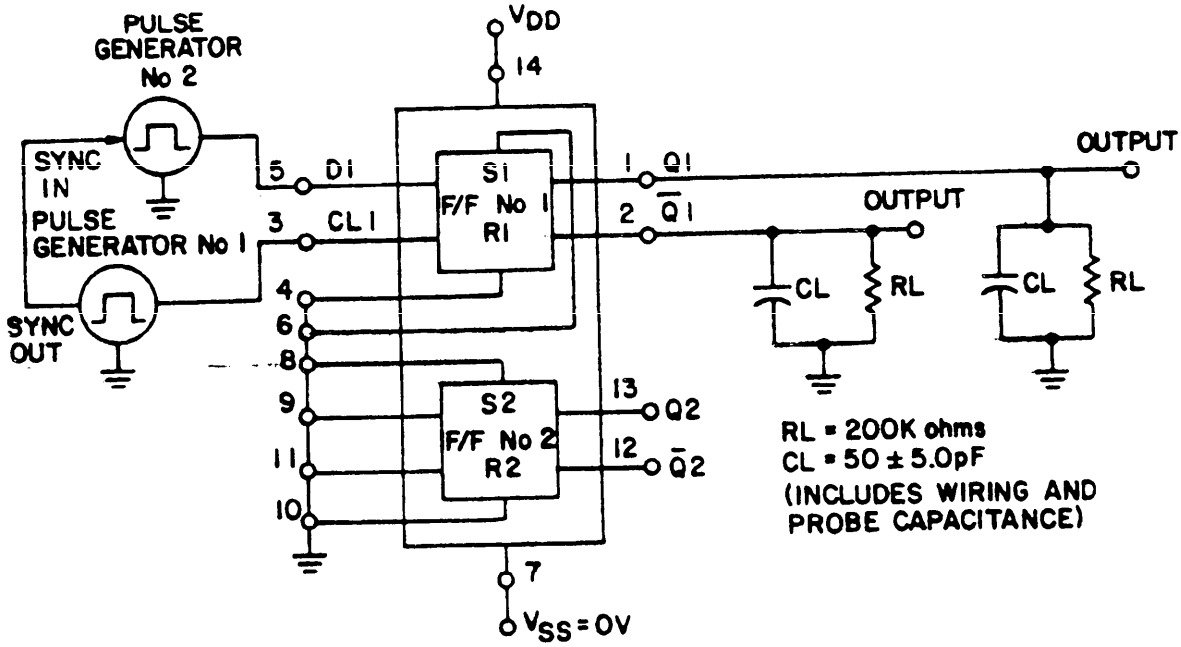


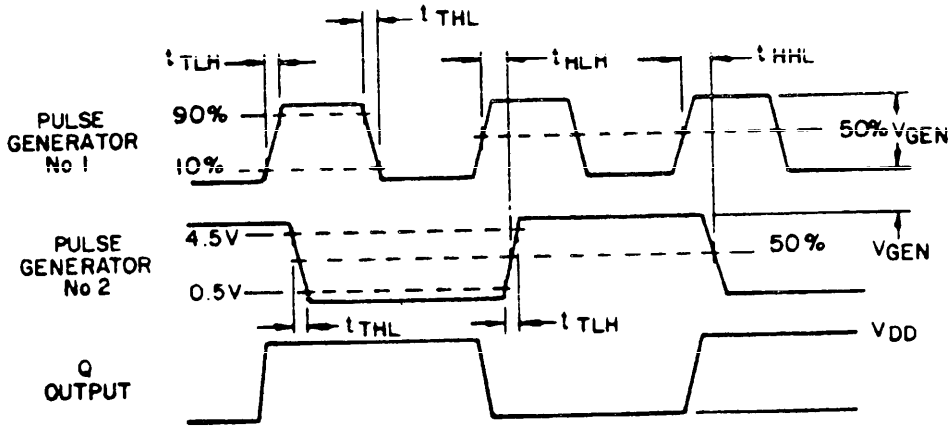
NOTES:

1. Pulse generator No. 1 has the following characteristics: V_{gen} = V_{DD} ±1%, duty cycle = 50%, t_{TLH} and t_{THL} = 20 ± 2 ns and pulse repetition period = 5.0 ± 0.5 µs.
2. Pulse generator No. 2 has the following characteristics: V_{gen} = V_{DD} ±1.0%, duty cycle = variable, t_{TLH} and t_{THL} = 20 ± 2.0 ns and pulse repetition period = twice that of pulse generator No. 1.
3. Identical measurements are obtained from F/F No. 1 and F/F No. 2.
4. Requirements for setup times are considered met if proper output state changes occur with t_{SETUP} set to that given in the limits column of table III.

FIGURE 10. Setup time test circuit and waveforms for device types 01 and 51.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:03Z
Check the source to verify that this is the current version before use.

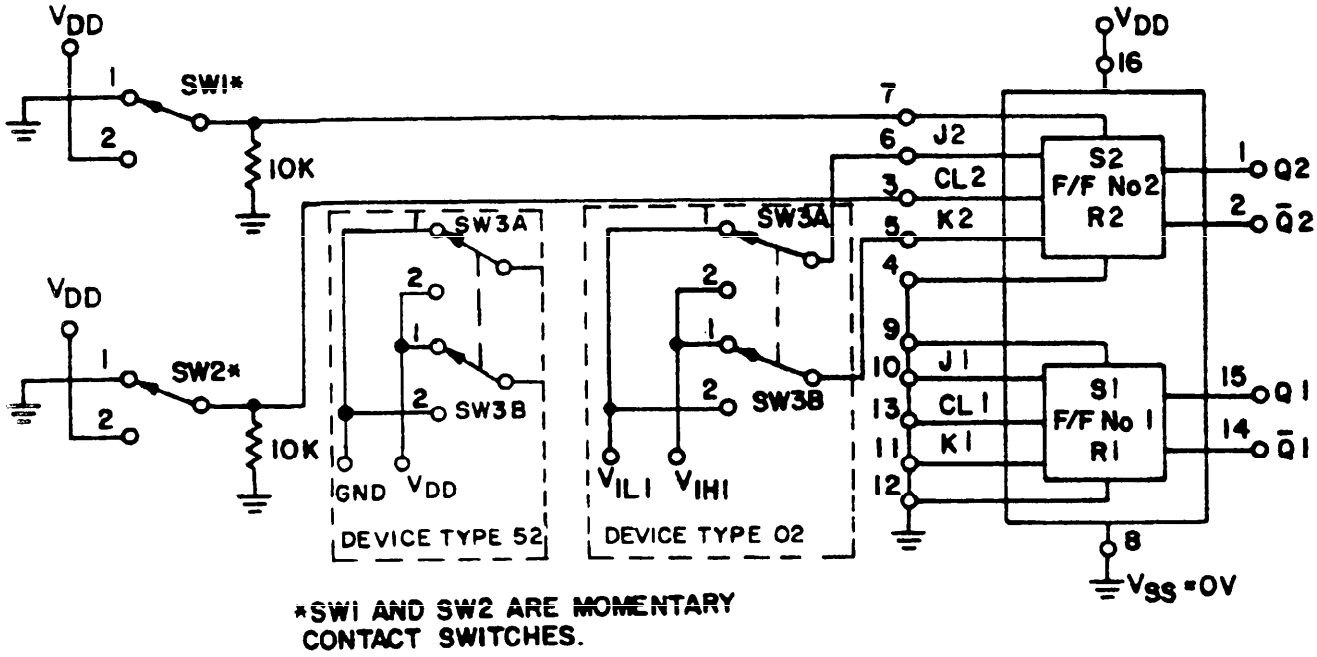




NOTES:
1. Pulse generator No. 1 has the following characteristics: $V_{gen} = V_{DD} \pm 1\%$, duty cycle = 50%, t_{TLH} and $t_{THL} = 20 \pm 2.0$ ns and pulse repetition period = 5.0 ± 0.5 µs.
2. Pulse generator No. 2 has the following characteristics: $V_{gen} = V_{DD} \pm 1\%$, duty cycle = variable, t_{TLH} and $t_{THL} = 20 \pm 2.0$ ns and pulse repetition period = twice that of pulse generator No. 1.
3. Identical measurements are obtained from either F/F No. 1 and F/F No. 2.
4. Requirements for hold times are considered met if proper output state changes occur with t_{HOLD} set to that given in the limits column of table III.

FIGURE 11. Hold time test circuit and waveforms for device types 01 and 51.

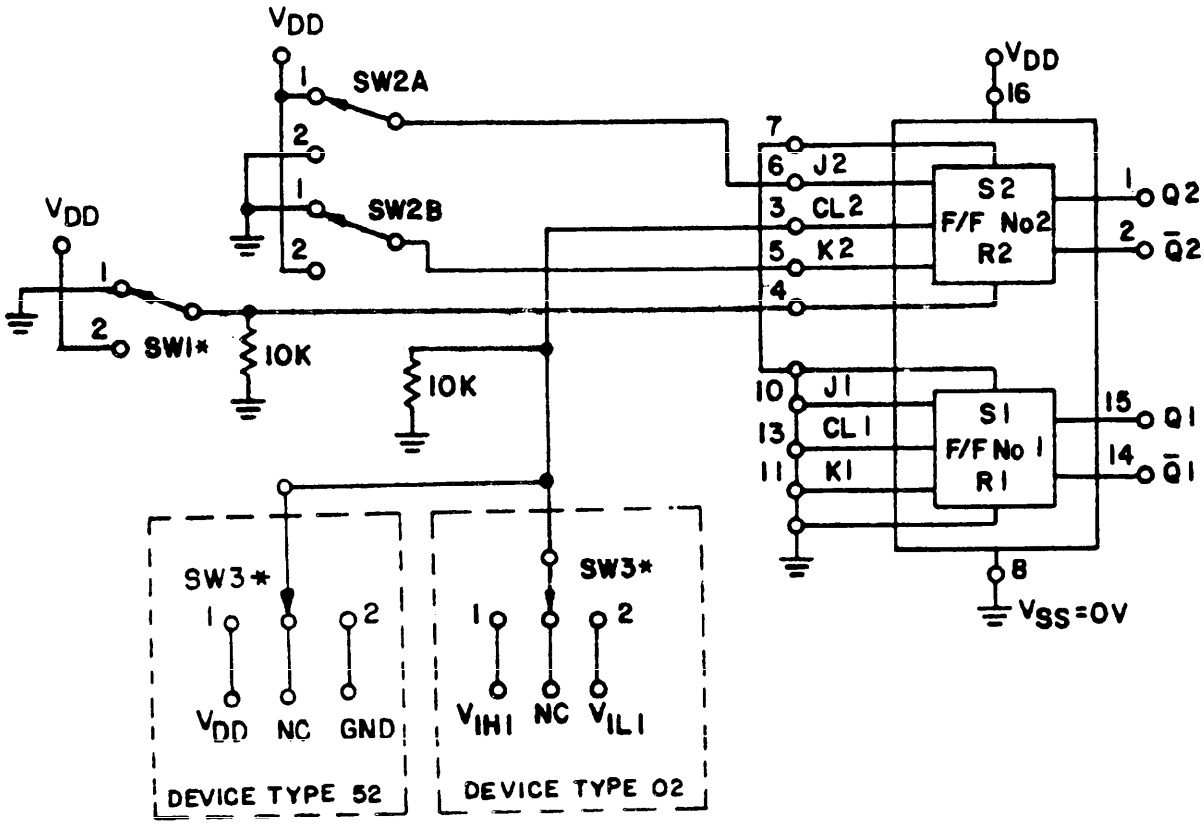
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:03Z
Check the source to verify that this is the current version before use.



NOTES:
1. To implement test numbers 59 thru 62 (device type 02) and 43 thru 46 (device type 52) place SW3 in position 1. Set the flip-flop by momentarily placing SW1 in position 2. Following the return of SW1 to position 1 momentarily place SW2 in position 2. Measure the output levels at Q and $\overline{Q}$ to insure compliance with table III limits.
2. To implement test numbers 63 thru 66 (device type 02) and 47 thru 50 (device type 52) set the flip-flop as described in note 1. Place SW3 in position 2. Momentarily place SW2 in position 2. Following the return of SW2 to position 1 measure the output levels at Q and $\overline{Q}$ to insure compliance with table III limits.
3. Identical measurements are obtained from either F/F No. 1 or F/F No. 2.

FIGURE 12. J and K input voltage high and low test circuit for device types 02 and 52.

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*SW1 AND SW3 ARE MOMENTARY CONTACT SWITCHES.

NOTES:

1. Test numbers 67 thru 70 (device type 02) and 51 thru 54 (device type 52) are implemented by the following step by step sequence:

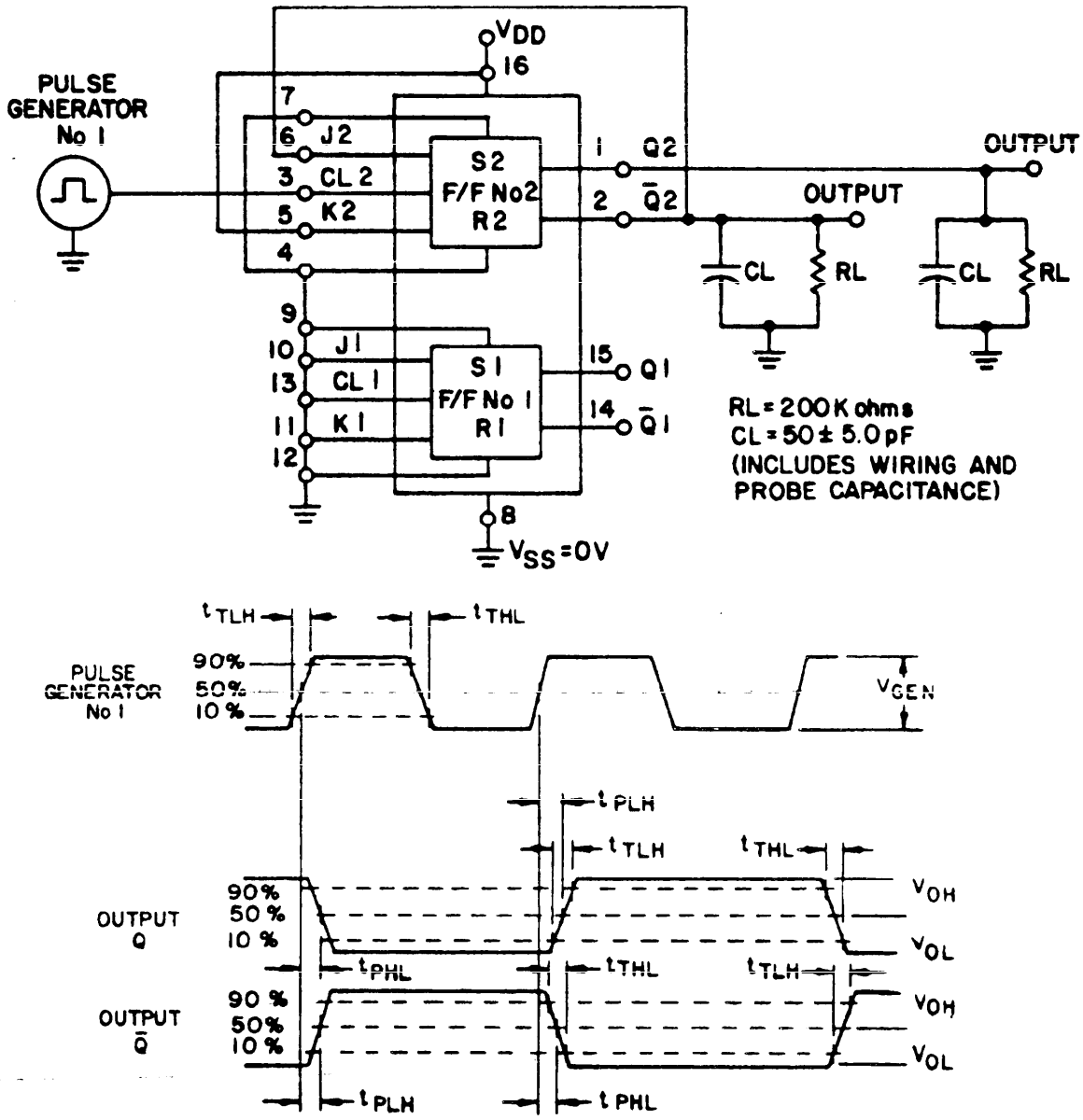
		SW1 POS	SW2 POS	SW3 POS	Q OUT	$\overline{Q}$ OUT
STEP	1	2	1	-	"0"	"1"
	2	1	1	1	"1"	"0"
	3	1	2	1	"0"	"1"
	4	1	1	2	"0"	"1"
	5	1	2	2	"0"	"1"
	6	1	1	1	"1"	"0"
STEP	7	1	2	1	"0"	"1"

Monitor either Q or $\overline{Q}$ of the flip-flop under test. Compliance with table III limits is established by a change of logic levels at the Q or $\overline{Q}$ output in going from step 1 to step 2, step 2 to step 3, step 5 to step 6 and step 6 to step 7, while no change shall occur in going from step 3 to step 4 to step 4 to step 5.

2. Identical measurements are obtained from either F/F No. 1 or F/F No. 2.
3. V_{OH} = "1" and V_{OL} = "0".

FIGURE 13. Clock input high and low test circuit for device types 02 and 52.

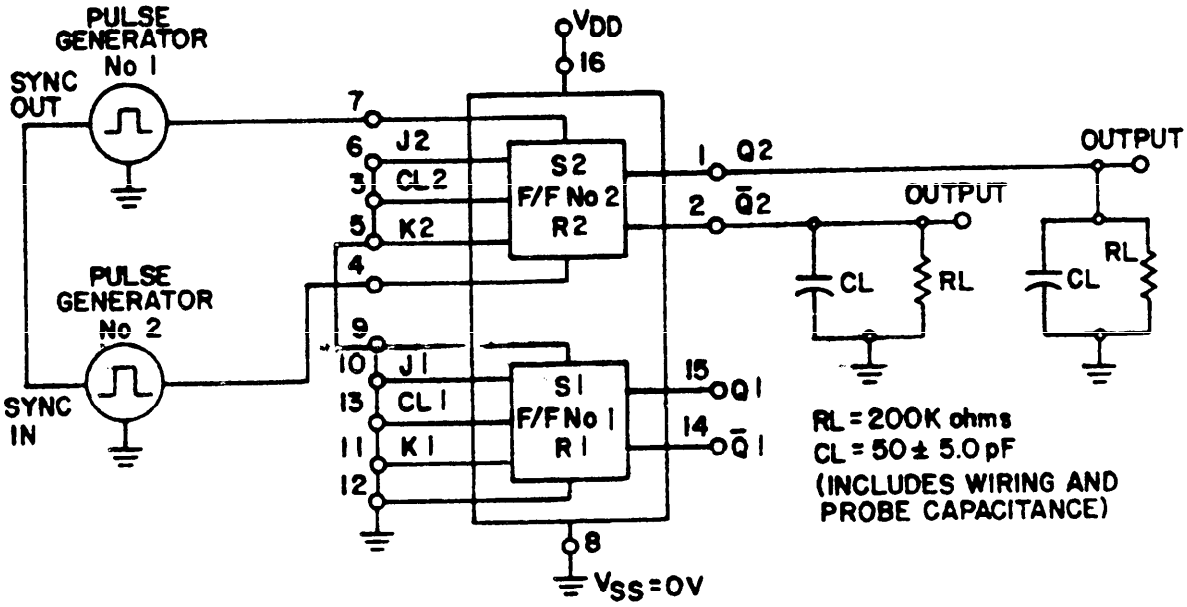
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:03Z
Check the source to verify that this is the current version before use.

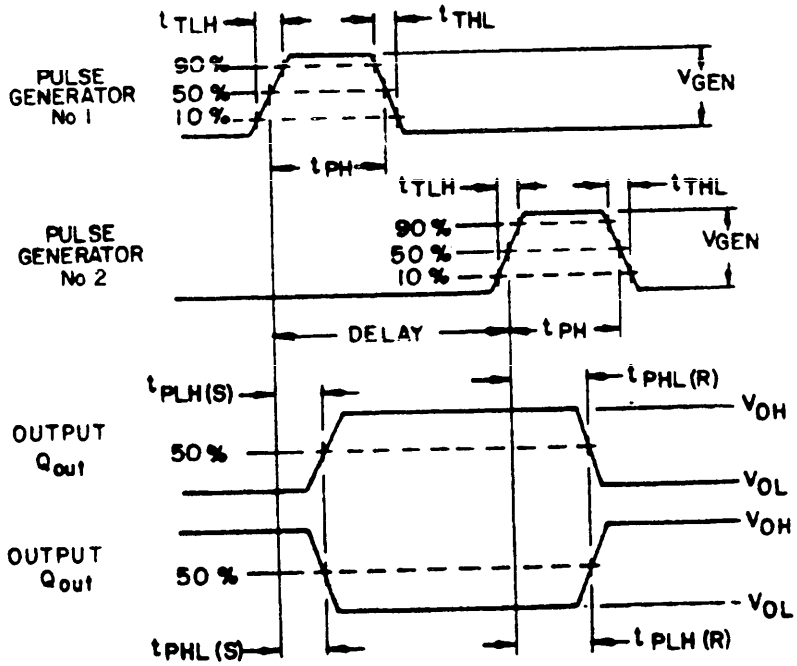


NOTES:
1. The pulse generator has the following characteristics: $V_{gen} = V_{DD} \pm 1.0\%$, duty cycle = 50%, t_{TLH} and t_{THL} = 20 ±2.0 ns and pulse repetition period = 5.0 ±0.5 µs maximum.
2. Identical switching measurements are obtained from F/F No. 1 and F/F No. 2.
3. For K input test connect terminal 6 to terminal 16 and terminal 5 to terminal 1. Similar connections are required for measurements on F/F No. 2.
4. For f_{CL} and tp the pulse repetition period is variable.
5. Requirements for max clock frequency (f_{CL}), max clock rise time and minimum clock pulse width are established by setting the parameter to the limit given in table III and observing proper output state changes.

FIGURE 14. Switching time test circuit and waveforms for device types 02 and 52.

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Check the source to verify that this is the current version before use.

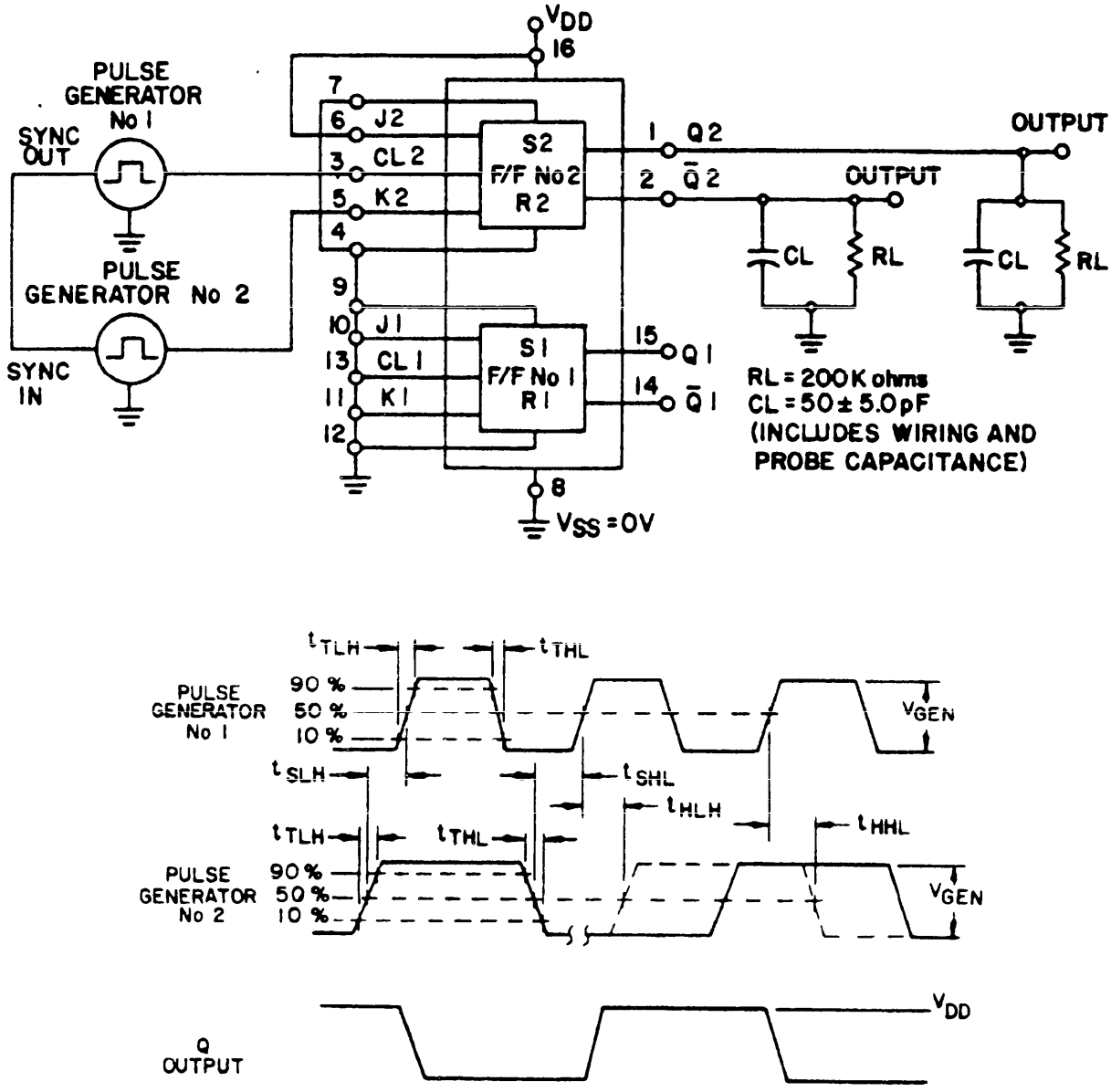




NOTES:
1. The pulse generators have the following characteristics: $V_{gen} = V_{DD} \pm 1.0\%$, $t_P = 1.0 \pm 0.1$ µs, t_{TLH} and $t_{THL} = 20 \pm 2.0$ ns and pulse repetition period = 5.0 ± 0.5 µs.
2. The reset pulse delay is 2.5 ± 0.25 µs.
3. Identical switching measurements are obtained from F/F No. 1 and F/F No. 2.

FIGURE 15. Set-reset switching test circuit and waveforms for device types 02 and 52.

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Check the source to verify that this is the current version before use.

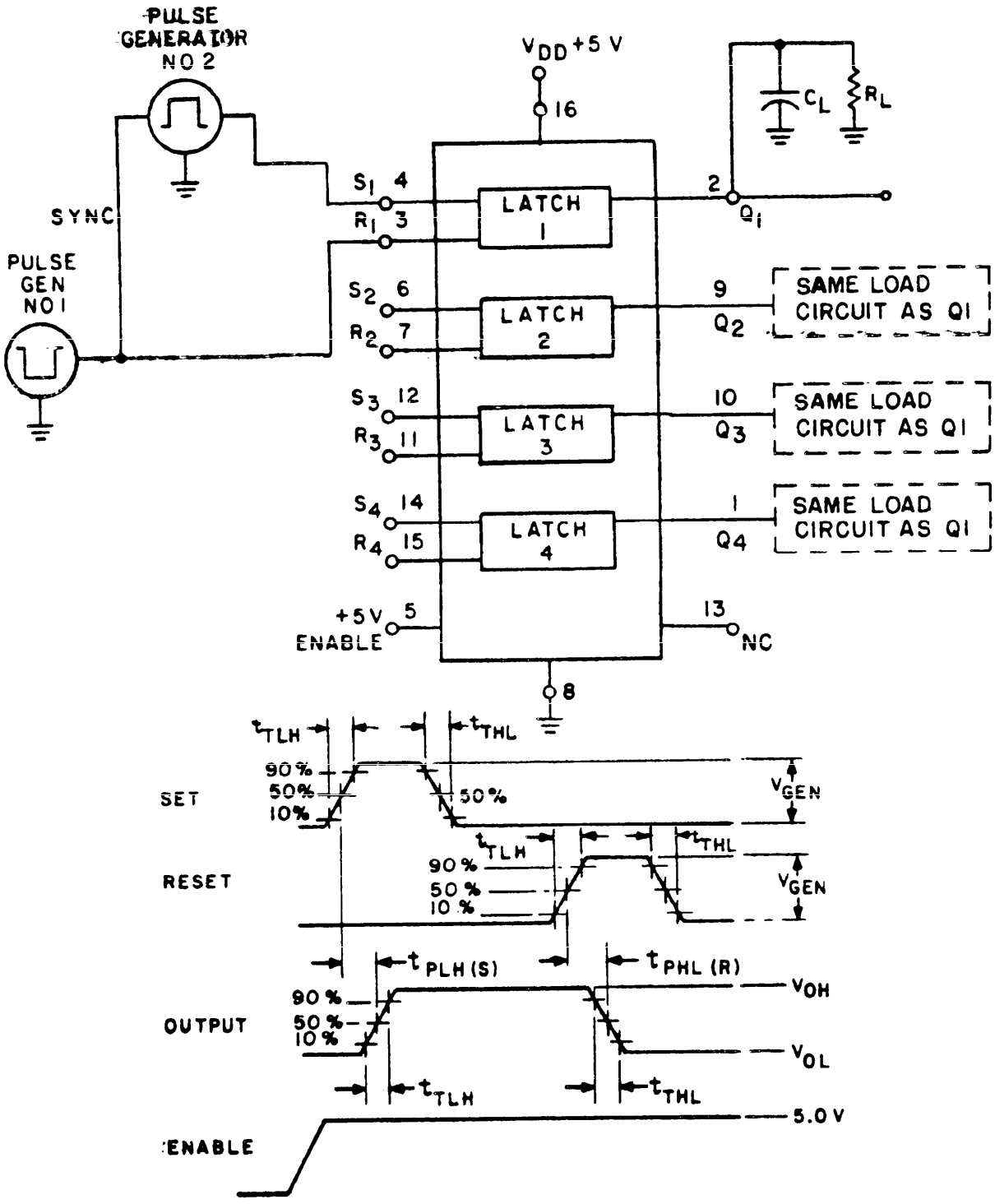


NOTES:

1. Pulse generator No. 1 has the following characteristics: $V_{gen} = V_{DD} \pm 1.0\%$, duty cycle = 50%, t_{THL} and t_{TLH} = 20 ±2.0 ns and pulse repetition period = 5.0 ±0.5 µs.
2. Pulse generator No. 2 has the following characteristics: $V_{gen} = V_{DD} \pm 1.0\%$, duty cycle = variable, t_{THL} and t_{TLH} = 20 ±2.0 ns and pulse repetition period = 5.0 ±0.5 µs.
3. Identical switching measurements are obtained from F/F No. 1 and F/F No. 2.
4. For J input test, connect terminal 5 to terminal 16 and terminal 6 to pulse generator No. 2. Similar terminal connections are required for measurement on F/F No. 2.
5. Requirements for setup times and hold times are established by setting the parameter to the limit given in table III and observing output state changes.

FIGURE 16. Setup and hold time test circuit and waveforms for device types 02 and 52.

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Check the source to verify that this is the current version before use.

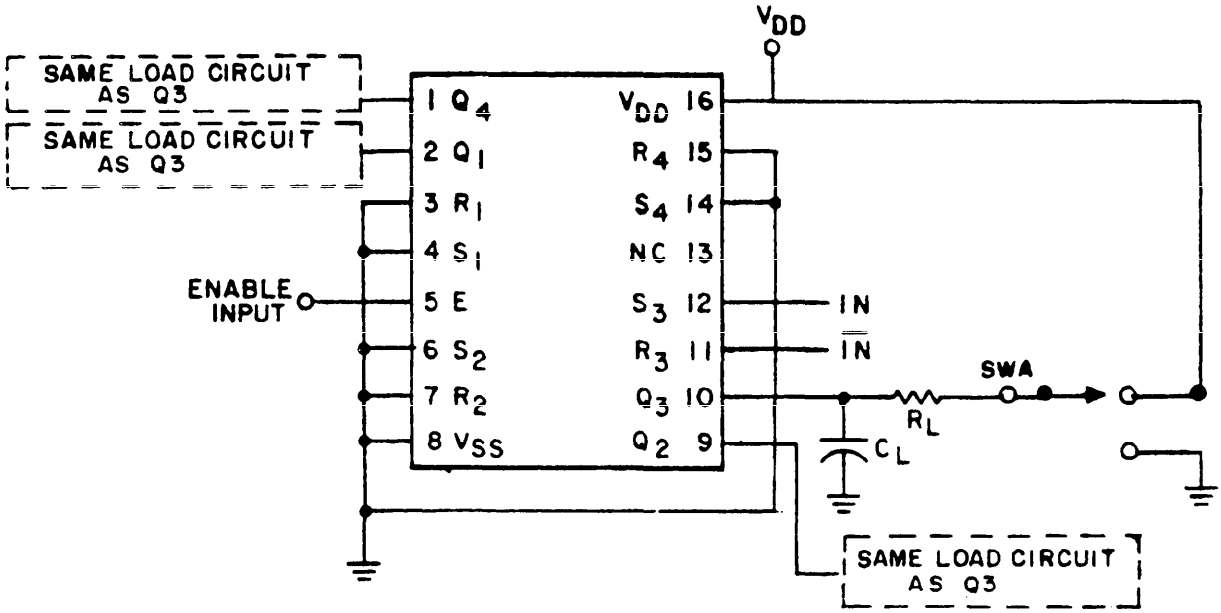


NOTES:
1. Pulse generator No. 1 characteristics: t_{TLH} and $t_{THL} \leq 20$ ns, $t_p = 1.0$ µs, $V_{GEN} = 0$ to 5 V, PRR = 200 kHz.
2. Pulse generator No. 2 characteristics: t_{TLH} and $t_{THL} \leq 20$ ns, $t_p = 1.0$ µs, delayed 2.0 µs after pulse No. 1, $V_{GEN} = 0$ to 5 V, PRR = 200 kHz.
3. Load conditions: $C_L = 50$ pF, $R_L = 200$ k ohms (includes probe and jig impedances).
4. Identical switching measurements are obtained from latch 2, latch 3, and latch 4.

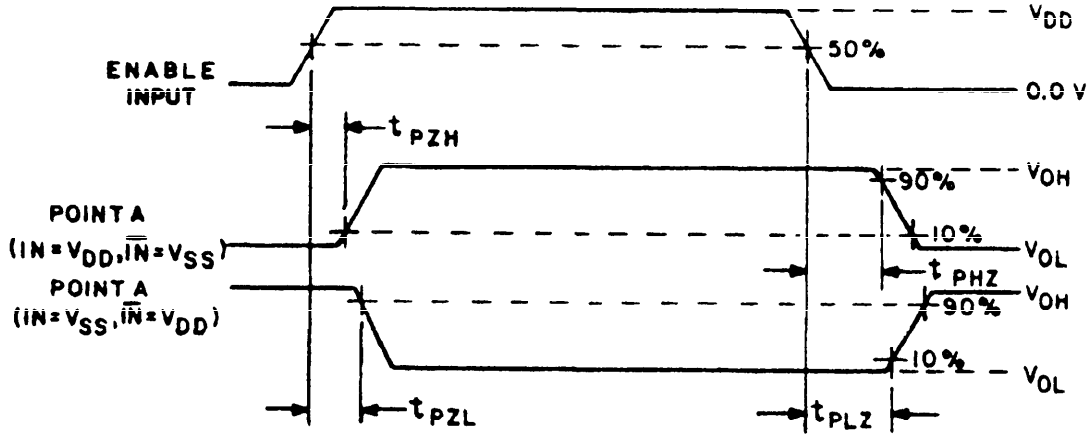
FIGURE 17. Switching test circuit and waveforms for device types 03 and 53.

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Device types 03 and 53



TEST	IN	$\overline{IN}$	SWA
t_{PHZ}	V_{DD}	V_{SS}	V_{SS}
t_{PLZ}	V_{SS}	V_{DD}	V_{DD}
t_{PZH}	V_{DD}	V_{SS}	V_{SS}
t_{PZL}	V_{SS}	V_{DD}	V_{DD}



NOTES:
1. Identical switching measurements are obtained from latch 1, latch 2, latch 3, and latch 4.
2. Load conditions: C_L = 50 pF and R_L = 1 kΩ (includes probe and jig impedances).

FIGURE 18. Enable propagation delay time test circuit and waveforms for device types 03 and 53.

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8. NOTES

TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1				1 mA		1 mA								GND	SET1		1.5					V
		2														"	RS1		"					"
		3					1 mA									"	D1		"					"
		4			1 mA											"	CLK1		"					"
		5								1 mA						"	SET2		"					"
		6										1 mA				"	RS2		"					"
		7									1 mA					"	D2		"					"
		8											1 mA			"	CLK2		"					"
$V_{IC(neg)}$		9						-1 mA	GND								SET1		-6					"
		10				-1 mA			"								RS1		"					"
		11					-1 mA		"								D1		"					"
		12			-1 mA				"								CLK1		"					"
		13							"	-1 mA							SET2		"					"
		14							"		-1 mA						RS2		"					"
		15							"			-1 mA					D2		"					"
		16							"				-1 mA				CLK2		"					"
I_{SS} 2/	3005	17			15 V	GND	GND	GND	"	GND	GND	GND	GND			15 V	None		-.25		-2.5			μA
	"	18			GND	"	"	"	"	"	"	"	"			"	None		"		"			"
	"	19			15 V	"	"	"	"	"	"	"	"			"	None		"		"			"
	"	20			15 V	"	15 V	"	"	"	"	"	"			"	V_{SS}		"		"			"
	"	21			GND	"	"	"	"	"	"	"	"			"	V_{SS}		"		"			"
	"	22			15 V	"	"	"	"	"	"	"	"			"	None		"		"			"
	"	23			15 V	"	GND	"	"	"	"	"	"			"	None		"		"			"
	"	24			GND	"	"	"	"	"	"	"	"			"	V_{SS}		"		"			"
	"	25			15 V	"	"	"	"	"	"	"	"			"	None		"		"			"
	"	26			15 V	15 V	15 V	15 V	"	"	"	"	"			"	V_{SS}		"		"			"
	"	27			GND	15 V	15 V	15 V	"	"	"	"	"			"	V_{SS}		"		"			"
	"	28			"	GND	GND	GND	"	"	"	"	15 V			"	None		"		"			"
	"	29			"	"	"	"	"	"	"	"	GND			"	None		"		"			"
	"	30			"	"	"	"	"	"	"	"	15 V			"	None		"		"			"
	"	31			"	"	"	"	"	"	15 V	"	15 V			"	V_{SS}		"		"			"
	"	32			"	"	"	"	"	"	"	"	GND			"	V_{SS}		"		"			"
	"	33			"	"	"	"	"	"	GND	"	15 V			"	None		"		"			"
	"	34			"	"	"	"	"	"	"	"	15 V			"	None		"		"			"
	"	35			"	"	"	"	"	"	"	"	GND			"	V_{SS}		"		"			"
	"	36			"	"	"	"	"	"	"	"	15 V			"	None		"		"			"
	"	37			"	"	"	"	"	15 V	15 V	15 V	15 V			"	V_{SS}		"		"			"
	"	38			"	"	"	"	"	15 V	15 V	15 V	GND			"	V_{SS}		"		"			"
V_{OH1}	3006	39	I_{OH}	I_{OH}	"	V_{IL1}	"	V_{IH1}	"	GND	GND	GND	"			5.0 V	Q1	4.5		4.5		4.5		V
	"	40			"	V_{IH1}	"	V_{IL1}	"	GND	"	GND	"			"	$\overline{Q1}$	"		"		"		"
	"	41			"	GND	"	GND	"	V_{IH1}	"	V_{IL1}	"			"	Q2	"		"		"		"
	"	42			"	GND	"	GND	"	V_{IL1}	"	V_{IH1}	"	I_{OH}	I_{OH}	"	$\overline{Q2}$	"		"		"		"
V_{OH2}	"	43			"	V_{IL1}	"	V_{IH1}	"	GND	"	GND	"			"	Q1	4.95		4.95		4.95		"
	"	44			"	V_{IH1}	"	V_{IL1}	"	GND	"	GND	"			"	$\overline{Q1}$	"		"		"		"
	"	45			"	GND	"	GND	"	V_{IH1}	"	V_{IL1}	"			"	Q2	"		"		"		"
	"	46			"	GND	"	GND	"	V_{IL1}	"	V_{IH1}	"			"	$\overline{Q2}$	"		"		"		"
V_{OH3}	"	47			"	V_{IL2}	"	V_{IH2}	"	GND	"	GND	"			12.5 V	Q1	11.25		11.25		11.25		"
	"	48			"	V_{IH2}	"	V_{IL2}	"	GND	"	GND	"			"	$\overline{Q1}$	"		"		"		"
	"	49			"	GND	"	GND	"	V_{IH2}	"	V_{IL2}	"			"	Q2	"		"		"		"
	"	50			"	GND	"	GND	"	V_{IL2}	"	V_{IH2}	"			"	$\overline{Q2}$	"		"		"		"

1/ See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test Limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
V_{OL1}	3007	51	I_{OL}	I_{OL}	GND	V_{IH1}	GND	V_{IL1}	GND	GND	GND	GND	GND	I_{OL}	I_{OL}	5.0 V	Q1		0.5		0.5		0.5	V
	"	52			"	V_{IL1}	"	V_{IH1}	"	GND	"	GND	"			"	$\overline{Q1}$		"		"		"	"
	"	53			"	GND	"	GND	"	V_{IL1}	"	V_{IH1}	"			"	Q2		"		"		"	"
	"	54			"	GND	"	GND	"	V_{IH1}	"	V_{IL1}	"			"	$\overline{Q2}$		"		"		"	"
V_{OL2}	"	55			"	V_{IH1}	"	V_{IL1}	"	GND	"	GND	"			"	Q1		.05		.05		.05	"
	"	56			"	V_{IL1}	"	V_{IH1}	"	GND	"	GND	"			"	$\overline{Q1}$		"		"		"	"
	"	57			"	GND	"	GND	"	V_{IL1}	"	V_{IH1}	"			"	Q2		"		"		"	"
	"	58			"	GND	"	GND	"	V_{IH1}	"	V_{IL1}	"			"	$\overline{Q2}$		"		"		"	"
V_{OL3}	"	59			"	V_{IH2}	"	V_{IL2}	"	GND	"	GND	"			12.5 V	Q1		1.25		1.25		1.25	"
	"	60			"	V_{IL2}	"	V_{IH2}	"	GND	"	GND	"			"	$\overline{Q1}$		"		"		"	"
	"	61			"	GND	"	GND	"	V_{IL2}	"	V_{IH2}	"			"	Q2		"		"		"	"
	"	62			"	GND	"	GND	"	V_{IH2}	"	V_{IL2}	"			"	$\overline{Q2}$		"		"		"	"
V_{OH4}	3006	63			9/	"	V_{IL1}	9/	"	GND	"	GND	"			5.0 V	$\overline{Q1}$	4.95		4.95		4.95		"
V_{OL4}	3007	64			9/	"	V_{IL1}	9/	"	GND	"	"	"			"	Q1		0.05		0.05		0.05	"
V_{OH4}	3006	65			GND	"	GND	GND	"	9/	V_{IL1}	"	9/			"	$\overline{Q2}$	4.95		4.95		4.95		"
V_{OL4}	3007	66			GND	"	GND	GND	"	9/	V_{IL1}	"	9/			"	Q2		0.05		0.05		0.05	"
V_{OL4}	3007	67			9/	"	V_{IH1}	9/	"	GND	GND	"	GND			"	$\overline{Q1}$		0.05		0.05		0.05	"
V_{OH4}	3006	68			9/	"	V_{IH1}	9/	"	GND	GND	"	GND			"	Q1	4.95		4.95		4.95		"
V_{OL4}	3007	69			GND	"	GND	GND	"	9/	V_{IH1}	"	9/			"	$\overline{Q2}$		0.05		0.05		0.05	"
V_{OH4}	3006	70			GND	"	GND	GND	"	9/	V_{IH1}	"	9/			"	Q2	4.95		4.95		4.95		"
V_{ICL1}		71			10/	"	10/	10/	"	GND	GND	"	GND			"	CLK1, 10/		V_{IH1}		V_{IH1}		V_{IH1}	"
V_{ICL1}		72			$\overline{10}$/	"	$\overline{10}$/	$\overline{10}$/	"	GND	GND	"	GND			"	CLK1, $\overline{10}$/	V_{IL1}		V_{IL1}		V_{IL1}		"
V_{ICL2}		73			GND	"	GND	GND	"	10/	10/	"	10/			"	CLK2, 10/		V_{IH1}		V_{IH1}		V_{IH1}	"
V_{ICL2}		74			GND	"	GND	GND	"	$\overline{10}$/	$\overline{10}$/	"	$\overline{10}$/			"	CLK2, $\overline{10}$/	V_{IL1}		V_{IL1}		V_{IL1}		"
I_{IN1} 11/	3010	75			15.0 V	15.0 V	15.0 V	15.0 V	"	15.0 V	15.0 V	15.0 V	15.0 V			15.0 V	All inputs together		8					nA
I_{IH2}	"	76			"	"	"	"	"	"	"	"	"			"	CLK1		1.0		45			"
	"	77			"	"	"	"	"	"	"	"	"			"	RS1		"		"			"
	"	78			"	"	"	"	"	"	"	"	"			"	D1		"		"			"
	"	79			"	"	"	"	"	"	"	"	"			"	SET1		"		"			"
	"	80			"	"	"	"	"	"	"	"	"			"	SET2		"		"			"
	"	81			"	"	"	"	"	"	"	"	"			"	D2		"		"			"
	"	82			"	"	"	"	"	"	"	"	"			"	RS2		"		"			"
	"	83			"	"	"	"	"	"	"	"	"			"	CLK2		"		"			"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test Limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
I_{IL1} 11/	3009	84			GND	GND	GND	GND	GND	GND	GND	GND	GND			15.0 V	All inputs together		-8.0					nA
I_{IL2}	"	85			"	"	"	"	"	"	"	"	"			"	CLK1		-1.0		-45			"
	"	86			"	"	"	"	"	"	"	"	"			"	RS1		"		"			"
	"	87			"	"	"	"	"	"	"	"	"			"	D1		"		"			"
	"	88			"	"	"	"	"	"	"	"	"			"	SET1		"		"			"
	"	89			"	"	"	"	"	"	"	"	"			"	SET2		"		"			"
	"	90			"	"	"	"	"	"	"	"	"			"	D2		"		"			"
	"	91			"	"	"	"	"	"	"	"	"			"	RS2		"		"			"
	"	92			"	"	"	"	"	"	"	"	"			"	CLK2		"		"			"
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C_I	3012	93			12/	12/	12/	12/	GND	12/	12/	12/	12/			GND	CLK1		12					pF
	"	94							"							"	RS1		"					"
	"	95							"							"	D1		"					"
	"	96							"							"	SET1		"					"
	"	97							"							"	SET2		"					"
	"	98							"							"	D2		"					"
	"	99							"							"	RS2		"					"
	"	100							"							"	CLK2		"					"
																		Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
Truth table test	3014	101			5.0 V	GND	GND	GND	GND	GND	GND	GND	5.0 V			5.0 V	None							
	"	102			GND	"	"	"	"	"	"	"	GND			"	None							
	"	103	L	H	5.0 V	"	"	"	"	"	"	"	5.0 V	H	L	"	All outputs							
	"	104	"	"	5.0 V	"	5.0 V	"	"	"	5.0 V	"	5.0 V	"	"	"	"							
	"	105	"	"	GND	"	"	"	"	"	"	"	GND	L	H	"	"							
	"	106	H	L	5.0 V	"	"	"	"	"	"	"	5.0 V	H	L	"	"							
	"	107	L	H	5.0 V	5.0 V	"	"	"	"	"	5.0 V	5.0 V	"	"	"	"							
	"	108	"	"	GND	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	109	"	"	5.0 V	"	"	"	"	"	"	"	5.0 V	"	"	"	"							
	"	110	H	L	5.0 V	GND	GND	5.0 V	"	5.0 V	GND	GND	5.0 V	L	H	"	"							
	"	111	"	"	GND	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	112	"	"	5.0 V	"	"	"	"	"	"	"	5.0 V	"	"	"	"							
	"	113	"	"	5.0 V	"	"	GND	"	GND	"	"	5.0 V	"	"	"	"							
	"	114	"	"	GND	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	115	L	H	5.0 V	"	"	"	"	"	"	"	5.0 V	H	L	"	"							
	"	116	H	"	5.0 V	5.0 V	"	5.0 V	"	5.0 V	"	5.0 V	5.0 V	"	H	"	"							
	"	117	"	"	GND	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	118	"	"	5.0 V	"	"	"	"	"	"	"	5.0 V	"	"	"	"							

See notes 13/ and 14/

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test Limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 (Fig. 8)	119	OUT		IN				GND							5.0 V	CLK1 to Q1	50	500	50	750	35	500	ns
		120		OUT	IN				"							"	CLK1 to $\overline{Q1}$	"	"	"	"	"	"	"
		121							"				IN		OUT	"	CLK2 to Q2	"	"	"	"	"	"	"
		122							"				IN	OUT		"	CLK2 to $\overline{Q2}$	"	"	"	"	"	"	"
t_{PHL} R or S	3003 (Fig. 9)	123		OUT		IN		IN	"							"	SET1 to $\overline{Q1}$	"	550	"	825	"	550	"
		124	OUT						"	IN				OUT		"	RS1 to Q1	"	"	"	"	"	"	"
		125							"			IN			OUT	"	SET2 to $\overline{Q2}$	"	"	"	"	"	"	"
		126							"							"	RS2 to Q2	"	"	"	"	"	"	"
t_{PLH}	3003 (Fig. 8)	127	OUT		IN				"							"	CLK1 to Q1	"	"	"	"	"	"	"
		128		OUT	IN				"							"	CLK1 to $\overline{Q1}$	"	"	"	"	"	"	"
		129							"				IN		OUT	"	CLK2 to Q2	"	"	"	"	"	"	"
		130							"				IN	OUT		"	CLK2 to $\overline{Q2}$	"	"	"	"	"	"	"
t_{PLH} R or S	3003 (Fig. 9)	131	OUT						"							"	SET1 to Q1	"	420	"	630	"	420	"
		132		OUT		IN		IN	"							"	RS1 to $\overline{Q1}$	"	"	"	"	"	"	"
		133							"	IN				OUT		"	SET2 to Q2	"	"	"	"	"	"	"
		134							"			IN			OUT	"	RS2 to $\overline{Q2}$	"	"	"	"	"	"	"
t_{THL}	3004 (Fig. 8)	135	OUT		IN				"							"	Q1	15	300	20	450	10	300	"
		136		OUT	IN				"							"	$\overline{Q1}$	"	"	"	"	"	"	"
		137							"				IN		OUT	"	Q2	"	"	"	"	"	"	"
		138							"				IN	OUT		"	$\overline{Q2}$	"	"	"	"	"	"	"
t_{TLH}	3004 (Fig. 8)	139		OUT	IN				"							"	$\overline{Q1}$	"	350	"	525	"	350	"
		140	OUT		IN				"							"	Q1	"	"	"	"	"	"	"
		141							"				IN		OUT	"	$\overline{Q2}$	"	"	"	"	"	"	"
		142							"				IN	OUT		"	Q2	"	"	"	"	"	"	"
$f_{CL(max)}$ 15/	(Fig. 8)	143	OUT		IN				"							"	CLK1		0.67		1.0		0.67	µs
		144							"				IN		OUT	"	CLK2		0.67		1.0		0.67	"
$T_{TLHCL(Max)}$ 16/	(Fig. 8)	145	OUT		IN				"							"	CLK1	15		15		10		"
		146							"				IN		OUT	"	CLK2	15		15		10		"
t_{p} 17/	(Fig. 8)	147	OUT		IN				"							"	CLK1		300		450		300	ns
		148							"				IN		OUT	"	CLK2		300		450		300	"
t_{SHL} t_{SHL}	(Fig. 10)	149			IN		IN		"							"	D1 to CLK1		165		225		165	"
		150							"		IN		IN			"	D2 to CLK2		"		"		"	"
t_{SLH} t_{SLH}	"	151			IN		IN		"							"	D1 to CLK1		"		"		"	"
	"	152							"		IN		IN			"	D2 to CLK2		"		"		"	"

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
t_{1HL} t_{2HL}	(Fig. 11)	153 154			IN		IN		GND "		IN		IN			5.0 V "	D1 to CLK1 D2 to CLK2		150 "		225 "		150 "	ns "
t_{1LH} t_{2LH}	" "	155 156			IN		IN		" "		IN		IN			" "	D1 to CLK1 D2 to CLK2		" "		" "		" "	" "

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ Test numbers 17 thru 38 shall be run in sequence.

3/ I_{OH} = -0.25 mA at 25°C, -0.175 mA at 125°C, -0.31 mA at -55°C.

4/ V_{IH1} = 3.8 V at 25°C, 3.6 V at 125°C, 3.95 V at -55°C.

5/ V_{IH2} = 9.5 V at 25°C, 9.25 V at 125°C, 9.75 V at -55°C.

6/ I_{OL} = 0.5 mA at 25°C, 0.35 mA at 125°C, 0.65 mA at -55°C.

7/ V_{IL1} = 1.1 V at 25°C, 0.85 V at 125°C, 1.35 V at -55°C.

8/ V_{IL2} = 2.8 V at 25°C, 2.55 V at 125°C, 3.0 V at -55°C.

9/ For input conditions, see figure 6.

10/ For input voltage conditions, see figure 7.

11/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

12/ See 4.4.1c.

13/ Test numbers 101 thru 118 shall be run in sequence and the functional tests shall be performed with V_{IH} and $V_{DD} \leq 5.0$ V and ≥ 15.0 V.

14/ L = V_{SS} + 0.5 V maximum and H = V_{DD} - 0.5 V minimum.

15/ The maximum clock frequency (f_{CL}) requirement is considered met if proper output state changes occur with the pulse repetition period set to that given in the limits column.

16/ Pulse repetition period = 100 µs, 50 percent duty cycle. The maximum clock transition time (t_{TLHCL}) requirement is considered met if proper output state changes occur with the rise time set to that given in the limits column.

17/ The minimum clock pulse width (t_D) requirement is considered met if proper output state changes occur with the pulse width set to that given in the limits column.

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1				1 mA			1 mA									GND	SET2		1.5					V
		2																"	RS2		"					"
		3						1 mA										"	J2		"					"
		4					1 mA											"	K2		"					"
		5			1 mA													"	CLK2		"					"
		6									1 mA							"	SET1		"					"
		7												1 mA				"	RS1		"					"
		8										1 mA						"	J1		"					"
		9											1 mA					"	K1		"					"
		10													1 mA			"	CLK1		"					"
$V_{IC(neg)}$		11				-1 mA			-1 mA	GND									SET2		-6					"
		12								"									RS2		"					"
		13						-1 mA		"									J2		"					"
		14					-1 mA			"									K2		"					"
		15			-1 mA					"									CLK2		"					"
		16								"	-1 mA								SET1		"					"
		17								"				-1 mA					RS1		"					"
		18								"		-1 mA							J1		"					"
		19								"			-1 mA						K1		"					"
		20								"					-1 mA				CLK1		"					"
I_{SS} 2/		21			GND	15.0 V	15.0 V	15.0 V	GND	"	GND	GND	GND	GND	GND			15.0 V	V_{SS}		-0.25		-2.5			uA
		22			15.0 V	15.0 V	"	"	"	"	"	"	"	"	"			"	None		"		"			"
		23			15.0 V	GND	"	"	"	"	"	"	"	"	"			"	V_{SS}		"		"			"
		24			GND	"	"	"	"	"	"	"	"	"	"			"	None		"		"			"
		25			15.0 V	"	"	"	"	"	"	"	"	"	"			"	V_{SS}		"		"			"
		26			"	"	GND	GND	"	"	"	"	"	"	"			"	V_{SS}		"		"			"
		27			"	15.0 V	"	"	15.0 V	"	"	"	"	"	"			"	V_{SS}		"		"			"
		28			GND	GND	"	"	GND	"	"	15.0 V	15.0 V	15.0 V	15.0 V			"	V_{SS}		"		"			"
		29			"	"	"	"	"	"	"	"	"	15.0 V	15.0 V			"	None		"		"			"
		30			"	"	"	"	"	"	"	"	"	"	GND			"	V_{SS}		"		"			"
		31			"	"	"	"	"	"	"	"	"	"	"			"	None		"		"			"
		32			"	"	"	"	"	"	"	"	"	"	"			"	V_{SS}		"		"			"
		33			"	"	"	"	"	"	"	"	"	"	"			"	V_{SS}		"		"			"
		34			"	"	"	"	"	"	15.0 V	"	"	15.0 V	"			"	V_{SS}		"		"			"
V_{OH1}	3006	35	I_{OH}		"	V_{IL1}	"	"	V_{IH1}	"	GND	"	"	GND	GND			5.0 V	Q2	4.5		4.5		4.5			V
	"	36		I_{OH}	"	V_{IH1}	"	"	V_{IL1}	"	GND	"	"	GND	"			"	$\overline{Q2}$	"		"		"			"
	"	37			"	GND	"	"	GND	"	V_{IH1}	"	"	V_{IL1}	"		I_{OH}	"	Q1	"		"		"			"
	"	38			"	GND	"	"	GND	"	V_{IL1}	"	"	V_{IH1}	"	I_{OH}		"	$\overline{Q1}$	"		"		"			"
V_{OH2}	"	39			"	V_{IL1}	"	"	V_{IH1}	"	GND	"	"	GND	"			"	Q2	4.95		4.95		4.95			"
	"	40			"	V_{IH1}	"	"	V_{IL1}	"	GND	"	"	GND	"			"	$\overline{Q2}$	"		"		"			"
	"	41			"	GND	"	"	GND	"	V_{IH1}	"	"	V_{IL1}	"			"	Q1	"		"		"			"
	"	42			"	GND	"	"	GND	"	V_{IL1}	"	"	V_{IH1}	"			"	$\overline{Q1}$	"		"		"			"
V_{OH3}	"	43			"	V_{IL2}	"	"	V_{IH2}	"	GND	"	"	GND	"			12.5 V	Q2	11.25		11.25		11.25			"
	"	44			"	V_{IH2}	"	"	V_{IL2}	"	GND	"	"	GND	"			"	$\overline{Q2}$	"		"		"			"
	"	45			"	GND	"	"	GND	"	V_{IH2}	"	"	V_{IL2}	"			"	Q1	"		"		"			"
	"	46			"	GND	"	"	GND	"	V_{IL2}	"	"	V_{IH2}	"			"	$\overline{Q1}$	"		"		"			"

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL1}	3007	47	I_{OL}	I_{OL}	GND	V_{IH1}	GND	GND	V_{IL1}	GND	GND	GND	GND	GND	GND	I_{OL}	I_{OL}	5.0 V	Q2		0.5		0.5		0.5	V
		48			"	V_{IL1}	"	"	V_{IH1}	"	GND	"	"	GND	"			"	$\overline{Q2}$		"		"		"	"
		49			"	GND	"	"	GND	"	V_{IL1}	"	"	V_{IH1}	"			"	Q1		"		"		"	"
		50			"	GND	"	"	GND	"	V_{IH1}	"	"	V_{IL1}	"			"	$\overline{Q1}$		"		"		"	"
V_{OL2}	"	51			"	V_{IH1}	"	"	V_{IL1}	"	GND	"	"	GND	"			"	Q2		0.05		0.05		0.05	"
		52			"	V_{IL1}	"	"	V_{IH1}	"	GND	"	"	GND	"			"	$\overline{Q2}$		"		"		"	"
		53			"	GND	"	"	GND	"	V_{IL1}	"	"	V_{IH1}	"			"	Q1		"		"		"	"
		54			"	GND	"	"	GND	"	V_{IH1}	"	"	V_{IL1}	"			"	$\overline{Q1}$		"		"		"	"
V_{OL3}	"	55			"	V_{IH2}	"	"	V_{IL2}	"	GND	"	"	GND	"			12.5 V	Q2		1.25		1.25		1.25	"
		56			"	V_{IL2}	"	"	V_{IH2}	"	GND	"	"	GND	"			"	$\overline{Q2}$		"		"		"	"
		57			"	GND	"	"	GND	"	V_{IL2}	"	"	V_{IH2}	"			"	Q1		"		"		"	"
		58			"	"	"	"	"	"	V_{IH2}	"	"	V_{IL2}	"			"	$\overline{Q1}$		"		"		"	"
V_{OH4}	3006	59			9/	"	V_{IH1}	V_{IL1}	9/	"	GND	"	"	GND	"			5.0 V	$\overline{Q2}$	4.95		4.95		4.95		"
V_{OL4}	3007	60			9/	"	V_{IH1}	V_{IL1}	9/	"	GND	"	"	"	"			"	Q2		.05		.05		.05	"
V_{OH4}	3006	61			GND	"	GND	GND	GND	"	9/	V_{IL1}	V_{IH1}	"	9/			"	$\overline{Q1}$	4.95		4.95		4.95		"
V_{OL4}	3007	62			GND	"	GND	GND	GND	"	9/	V_{IL1}	V_{IH1}	"	9/			"	Q1		.05		.05		.05	"
V_{OL4}	3007	63			9/	"	V_{IL1}	V_{IH1}	9/	"	GND	GND	GND	"	GND			"	$\overline{Q2}$		.05		.05		.05	"
V_{OH4}	3006	64			9/	"	V_{IL1}	V_{IH1}	9/	"	GND	GND	GND	"	GND			"	Q2	4.95		4.95		4.95		"
V_{OL4}	3007	65			GND	"	GND	GND	GND	"	9/	V_{IH1}	V_{IL1}	"	9/			"	$\overline{Q1}$		.05		.05		.05	"
V_{OH4}	3006	66			GND	"	GND	GND	"	"	9/	V_{IH1}	V_{IL1}	"	9/			"	Q1	4.95		4.95		4.95		"
V_{ICL1}		67			10/	10/	10/	10/	"	"	GND	GND	GND	"	GND			"	CLK2	V_{IL1}	V_{IH1}	V_{IL1}	V_{IH1}	V_{IL1}	V_{IH1}	V
V_{ICL1}		68			10/	10/	10/	10/	"	"	GND	GND	GND	"	GND			"	CLK2	V_{IL1}	V_{IH1}	V_{IL1}	V_{IH1}	V_{IL1}	V_{IH1}	V
V_{ICL2}		69			GND	GND	GND	GND	"	"	"	10/	10/	10/	10/			"	CLK1	V_{IL1}	V_{IH1}	V_{IL1}	V_{IH1}	V_{IL1}	V_{IH1}	V
V_{ICL2}		70			GND	GND	GND	GND	"	"	"	10/	10/	10/	10/			"	CLK1	V_{IL1}	V_{IH1}	V_{IL1}	V_{IH1}	V_{IL1}	V_{IH1}	V
I_{IH1} 11/	3009	71			15.0 V	15.0 V	15.0 V	15.0 V	15.0 V	"	15.0 V	15.0 V	15.0 V	15.0 V	15.0 V			15.0 V	All inputs together		10					nA
I_{IH2}	"	72			"	"	"	"	"	"	"	"	"	"	"			"	CLK2		1.0		45			"
		73			"	"	"	"	"	"	"	"	"	"	"			"	RS2		"		"			"
		74			"	"	"	"	"	"	"	"	"	"	"			"	K2		"		"			"
		75			"	"	"	"	"	"	"	"	"	"	"			"	J2		"		"			"
		76			"	"	"	"	"	"	"	"	"	"	"			"	SET2		"		"			"
		77			"	"	"	"	"	"	"	"	"	"	"			"	SET1		"		"			"
		78			"	"	"	"	"	"	"	"	"	"	"			"	J1		"		"			"
		79			"	"	"	"	"	"	"	"	"	"	"			"	K1		"		"			"
		80			"	"	"	"	"	"	"	"	"	"	"			"	RS1		"		"			"
		81			"	"	"	"	"	"	"	"	"	"	"			"	CLK1		"		"			"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/															Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{IL1} 11/	3009	82			GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			15.0 V	All inputs together		-10					nA
I_{IL2}	"	83			"	"	"	"	"	"	"	"	"	"	"			"	CLK2		-1.0		-45			"
	"	84			"	"	"	"	"	"	"	"	"	"	"			"	RS2		"		"			"
	"	85			"	"	"	"	"	"	"	"	"	"	"			"	K2		"		"			"
	"	86			"	"	"	"	"	"	"	"	"	"	"			"	J2		"		"			"
	"	87			"	"	"	"	"	"	"	"	"	"	"			"	SET2		"		"			"
	"	88			"	"	"	"	"	"	"	"	"	"	"			"	SET1		"		"			"
	"	89			"	"	"	"	"	"	"	"	"	"	"			"	J1		"		"			"
	"	90			"	"	"	"	"	"	"	"	"	"	"			"	K1		"		"			"
	"	91			"	"	"	"	"	"	"	"	"	"	"			"	RS1		"		"			"
	"	92			"	"	"	"	"	"	"	"	"	"	"			"	CLK1		"		"			"
																				Subgroup 4 T_A = 25°C						
																				Min	Max					
Cl	3012	93			12/					GND								GND	CLK2		12					pF
	"	94				12/				"								"	RS2		"					"
	"	95					12/			"								"	K2		"					"
	"	96						12/		"								"	J2		"					"
	"	97							12/	"								"	SET2		"					"
	"	98								"	12/							"	SET1		"					"
	"	99								"		12/						"	J1		"					"
	"	100								"			12/					"	K1		"					"
	"	101								"				12/				"	RS1		"					"
	"	102								"					12/			"	CLK1		"					"
																				Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table test	3014	103			5.0 V	GND	5.0 V	GND	GND	GND	GND	GND	5.0 V	GND	5.0 V			5.0 V	None							
	"	104			GND	"	"	"	"	"	"	"	"	"	GND			"	None							
	"	105	L	H	5.0 V	"	"	"	"	"	"	"	"	"	5.0 V	H	L	"	All outputs							
	"	106	"	"	5.0 V	"	GND	"	"	"	"	"	GND	"	5.0 V	"	"	"	"							
	"	107	"	"	GND	"	"	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	108	"	"	5.0 V	"	"	"	"	"	"	"	"	"	5.0 V	"	"	"	"							
	"	109	"	"	5.0 V	"	"	5.0 V	"	"	"	5.0 V	"	"	5.0 V	"	"	"	"							
	"	110	"	"	GND	"	"	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	111	H	L	5.0 V	"	"	"	"	"	"	"	"	"	5.0 V	L	H	"	"							
	"	112	"	"	5.0 V	"	"	GND	"	"	"	GND	"	"	5.0 V	"	"	"	"			See 13/ and 14/				
	"	113	"	"	GND	"	"	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	114	"	"	5.0 V	"	"	"	"	"	"	"	"	"	5.0 V	"	"	"	"							
	"	115	"	"	5.0 V	"	5.0 V	5.0 V	"	"	"	5.0 V	5.0 V	"	5.0 V	"	"	"	"							
	"	116	"	"	GND	"	"	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	117	L	H	5.0 V	"	"	"	"	"	"	"	"	"	5.0 V	H	L	"	"							
	"	118	H	L	5.0 V	"	"	GND	5.0 V	"	5.0 V	GND	"	"	5.0 V	L	H	"	"							
	"	119	H	L	GND	"	"	"	5.0 V	"	5.0 V	"	"	"	GND	L	H	"	"							
	"	120	L	H	GND	5.0 V	GND	"	GND	"	GND	"	GND	5.0 V	GND	H	L	"	"							
	"	121	L	H	5.0 V	5.0 V	GND	"	GND	"	GND	"	GND	5.0 V	5.0 V	H	L	"	"							

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}	Min	Max	Min	Max	Min	Max	
Truth table test	3014	122	H	L	5.0 V	GND	GND	GND	5.0 V	GND	5.0 V	GND	GND	GND	5.0 V	L	H	5.0 V	All outputs			See 13/ and 19/			
		123	"	"	5.0 V	"	5.0 V	"	GND	"	GND	"	5.0 V	"	5.0 V	"	"	"	"						
		124	"	"	GND	"	"	"	"	"	"	"	"	"	GND	H	L	"	"						
		125	L	H	5.0 V	"	"	5.0 V	"	"	"	5.0 V	"	"	5.0 V	"	"	"	"						
		126	"	"	5.0 V	"	"	"	"	"	"	"	"	"	GND	"	H	"	"						
		127	"	"	GND	"	"	"	"	"	"	"	"	"	5.0 V	L	"	"	"						
		128	H	L	5.0 V	5.0 V	"	"	"	"	"	"	"	5.0 V	5.0 V	H	L	"	"						
		129	L	H	5.0 V	"	"	"	"	"	"	"	"	"	GND	"	"	"	"						
		130	"	"	GND	"	"	"	"	"	"	"	"	"	5.0 V	"	"	"	"						
		131	"	"	5.0 V	"	GND	GND	5.0 V	"	5.0 V	GND	GND	"	5.0 V	"	H	"	"						
		132	H	"	5.0 V	"	"	"	"	"	"	"	"	"	GND	"	"	"	"						
		133	"	"	GND	"	"	"	"	"	"	"	"	"	5.0 V	"	"	"	"						
		134	"	"	5.0 V	"	"	"	"	"	"	"	"	"		"	"	"	"						

																				Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{PHL}	3003 Fig. 14	135	OUT	OUT	IN					GND					IN	OUT	OUT	5.0 V	CLK2 to Q2	50	575	50	865	35	575	ns
		136			IN					"					"			"	CLK2 to $\overline{Q2}$	"	"	"	"	"	"	"
		137								"								"	CLK1 to Q1	"	"	"	"	"	"	"
		138								"								"	CLK1 to $\overline{Q1}$	"	"	"	"	"	"	"
t_{PHL} R or S	3003 Fig. 15	139	OUT	OUT		IN			IN	"	IN			IN		OUT	OUT	"	RS2 to Q2	"	600	"	900	"	600	"
		140								"								"	SET2 to $\overline{Q2}$	"	"	"	"	"	"	"
		141								"								"	RS1 to Q1	"	"	"	"	"	"	"
		142								"								"	SET1 to $\overline{Q1}$	"	"	"	"	"	"	"
t_{PLH}	3003 Fig. 14	143	OUT	OUT	IN					"					IN	OUT	OUT	"	CLK2 to Q2	"	625	"	940	"	625	"
		144			IN					"					"			"	CLK2 to $\overline{Q2}$	"	"	"	"	"	"	"
		145								"								"	CLK1 to Q1	"	"	"	"	"	"	"
		146								"								"	CLK1 to $\overline{Q1}$	"	"	"	"	"	"	"
t_{PLH} R or S	3003 Fig. 15	147	OUT	OUT		IN			IN	"	IN			IN		OUT	OUT	"	SET2 to Q2	"	400	"	600	"	400	"
		148								"								"	RS2 to $\overline{Q2}$	"	"	"	"	"	"	"
		149								"								"	SET1 to Q1	"	"	"	"	"	"	"
		150								"								"	RS1 to $\overline{Q1}$	"	"	"	"	"	"	"
t_{THL}	3004 Fig. 14	151	OUT	OUT	IN					"					IN	OUT	OUT	"	Q2	15	325	20	490	10	325	"
		152			IN					"					"			"	$\overline{Q2}$	"	"	"	"	"	"	"
		153								"								"	Q1	"	"	"	"	"	"	"
		154								"								"	$\overline{Q1}$	"	"	"	"	"	"	"
t_{TLH}	"	155	OUT	OUT	IN					"					IN	OUT	OUT	"	Q2	"	"	"	"	"	"	"
	"	156			IN					"					"			"	$\overline{Q2}$	"	"	"	"	"	"	"
	"	157								"								"	Q1	"	"	"	"	"	"	"
	"	158								"								"	$\overline{Q1}$	"	"	"	"	"	"	"
$f_{CL(max)}$ 15/	"	159	OUT		IN					"					IN		OUT	"	CLK2		1.0		1.4		1.0	μs
	"	160								"								"	CLK1		1.0		1.4		1.0	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{TLHCL} 16/	Fig. 14	161	OUT		IN					GND								5.0 V	CLK2	15		15		10		µs
	"	162								"					IN		OUT	"	CLK1	15		15		10		µs
t_p 17/	"	163	OUT		IN					"								"	CLK2	300		450		300		ns
	"	164								"					IN		OUT	"	CLK1	300		450		300		"
t_{SHL}	Fig. 16	165	OUT		IN		IN			"								"	K2 to CLK2		165		225		165	"
	"	166	OUT		IN			IN		"								"	J2 to CLK2		"		"		"	"
	"	167								"			IN		IN		OUT	"	K1 to CLK1		"		"		"	"
	"	168								"		IN			IN		OUT	"	J1 to CLK1		"		"		"	"
t_{SLH}	"	169	OUT		IN		IN			"								"	K2 to CLK2		"		"		"	"
	"	170	OUT		IN			IN		"								"	J2 to CLK2		"		"		"	"
	"	171								"			IN		IN		OUT	"	K1 to CLK1		"		"		"	"
	"	172								"		IN			IN		OUT	"	J1 to CLK1		"		"		"	"
t_{HLH}	"	173	OUT		IN		IN			"								"	K2 to CLK2		150		"		150	"
	"	174	OUT		IN			IN		"								"	J2 to CLK2		"		"		"	"
	"	175								"			IN		IN		OUT	"	K1 to CLK1		"		"		"	"
	"	176								"		IN			IN		OUT	"	J1 to CLK1		"		"		"	"
t_{HHL}	"	177	OUT		IN		IN			"								"	K2 to CLK2		"		"		"	"
	"	178	OUT		IN			IN		"								"	J2 to CLK2		"		"		"	"
	"	179								"			IN		IN		OUT	"	K1 to CLK1		"		"		"	"
	"	180								"		IN			IN		OUT	"	J1 to CLK1		"		"		"	"

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ Test numbers 21 thru 34 shall be run in sequence.

3/ I_{OH} = -0.25 mA at 25°C, -0.175 mA at 125°C, -0.31 mA at -55°C.

4/ V_{IH1} = 3.8 V at 25°C, 3.6 V at 125°C, 3.95 V at -55°C.

5/ V_{IH2} = 9.5 V at 25°C, 9.25 V at 125°C, 9.75 V at -55°C.

6/ I_{OL} = 0.5 mA at 25°C, 0.35 mA at 125°C, 0.65 mA at -55°C.

7/ V_{IL1} = 1.1 V at 25°C, 0.85 V at 125°C, 1.35 V at -55°C.

8/ V_{IL2} = 2.8 V at 25°C, 2.55 V at 125°C, 3.05 V at -55°C.

9/ For input voltage conditions, see figure 12.

10/ For input voltage conditions, see figure 13.

11/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

12/ See 4.4.1c.

13/ Test numbers 103 thru 134 shall be run in sequence and the functional tests shall be performed with V_{IH} and $V_{DD} \leq 5.0$ V and ≥ 15.0 V.

14/ L = V_{SS} + 0.5 V maximum and H = V_{DD} - 0.5 V minimum.

15/ The maximum clock frequency (f_{CL}) requirement is considered met if proper output state changes occur with the pulse repetition period set to that given in the limits column.

16/ Pulse repetition period = 100 µs, 50 percent duty cycle. The maximum clock transition time (t_{TLHCL}) requirement is considered met if proper output state changes occur with the rise time set to that given in the limits column.

17/ The minimum clock pulse width (t_p) requirement is considered met if proper output state changes occur with the pulse width set to that given in the limits column.

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 test method	Case E,F,2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q4	Q1	R1	S1	E	S2	R2	V_{SS}	Q2	Q3	R3	S3	MC	S4	R4	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1 2 3 4 5 6 7 8 9			1 mA	1 mA	1 mA	1 mA	1 mA				1 mA	1 mA		1 mA	1 mA	GND " " " " " " "	R1 S1 E S2 R2 R3 S3 S4 R4		1.5 " " " " " " " "					V " " " " " " " "
$V_{IC(neg)}$		10 11 12 13 14 15 16 17 18			-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	GND " " " " " " "			-1 mA	-1 mA		-1 mA	-1 mA		R1 S1 E S2 R2 R3 S3 S4 R4		-6 " " " " " " " "					" " " " " " " " "
I_{SS} 2/	3005 " " " "	19 20 21 22 23 24			GND GND 15.0 V GND 15.0 V GND	15.0 V GND " " 15.0 V GND	15.0 V " " " GND	15.0 V GND " " 15.0 V GND	GND GND 15.0 V GND 15.0 V GND	" " " " " "			GND GND 15.0 V GND 15.0 V GND	15.0 V GND " " 15.0 V GND		15.0 V GND " " 15.0 V GND	GND GND 15.0 V GND 15.0 V GND	15.0 V " " " " "	V_{SS} " " " " "		-1.0 " " " " "		-2.5 " " " "			mA " " " " "
V_{OH1}	3006 " "	25 26 27 28	I_{OH}	I_{OH}	V_{IL1} " "	V_{IH1} V_{IL1} V_{IL1} V_{IL1}	V_{IH1} " "	V_{IL1} V_{IH1} V_{IL1} V_{IL1}	V_{IL1} " "	" " " "	I_{OH}	I_{OH}	V_{IL1} " "	V_{IL1} V_{IL1} V_{IH1} V_{IL1}		V_{IL1} V_{IL1} V_{IL1} V_{IH1}	V_{IL1} " "	5.0 V " "	Q1 Q2 Q3 Q4	4.5 " "		4.5 " "		4.5 " "		V " "
V_{OH2}	" " "	29 30 31 32			" " "	V_{IH1} V_{IL1} V_{IL1} V_{IL1}	" " "	V_{IL1} V_{IH1} V_{IL1} V_{IL1}	" " "	" " "			" " "	V_{IL1} V_{IL1} V_{IH1} V_{IL1}		V_{IL1} V_{IL1} V_{IL1} V_{IH1}	" " "	" " "	Q1 Q2 Q3 Q4	4.95 " "		4.95 " "		4.95 " "		" " "
V_{OH3}	" " "	33 34 35 36			V_{IL2} " "	V_{IH2} V_{IL2} V_{IL2} V_{IL2}	V_{IH2} " "	V_{IL2} V_{IH2} V_{IL2} V_{IL2}	V_{IL2} " "	" " "			V_{IL2} " "	V_{IL2} V_{IL2} V_{IH2} V_{IL2}		V_{IL2} V_{IL2} V_{IL2} V_{IH2}	V_{IL2} " "	12.5 V " "	Q1 Q2 Q3 Q4	11.25 " "		11.25 " "		11.25 " "		" " "
V_{OL1}	3007 " "	37 38 39 40	I_{OL}	I_{OL}	V_{IH1} " "	V_{IL1} V_{IH1} V_{IH1} V_{IH1}	V_{IH1} " "	V_{IH1} V_{IL1} V_{IH1} V_{IH1}	V_{IH1} " "	" " "	I_{OL}	I_{OL}	V_{IH1} " "	V_{IH1} V_{IH1} V_{IL1} V_{IH1}		V_{IH1} V_{IH1} V_{IH1} V_{IL1}	V_{IH1} " "	5.0 V " "	Q1 Q2 Q3 Q4		.5 " "		.5 " "		.5 " "	" " "
V_{OL2}	" " "	41 42 43 44			" " "	V_{IL1} V_{IH1} V_{IH1} V_{IH1}	" " "	V_{IH1} V_{IL1} V_{IH1} V_{IH1}	" " "	" " "			" " "	V_{IH1} V_{IH1} V_{IL1} V_{IH1}		V_{IH1} V_{IH1} V_{IH1} V_{IL1}	" " "	" " "	Q1 Q2 Q3 Q4		.05 " "		.05 " "		.05 " "	" " "

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q4	Q1	R1	S1	E	S2	R2	Y_{SS}	Q2	Q3	R3	S3	NC	S4	R4	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL3}	3007	45			V_{IH2}	V_{IL2}	V_{IH2}	V_{IH2}	V_{IH2}	GND			V_{IH2}	V_{IH2}		V_{IH2}	V_{IH2}	12.5 V	Q1		1.25		1.25		1.25	V
		46				V_{IH2}		V_{IL2}						V_{IH2}		V_{IH2}			Q2							
		47				V_{IH2}		V_{IH2}						V_{IL2}		V_{IH2}			Q3							
		48				V_{IH2}		V_{IH2}						V_{IH2}		V_{IL2}			Q4							
I_{IH1} 9/	3010	49			15.0 V	15.0 V	15.0 V	15.0 V	15.0 V	"			15.0 V	15.0 V		15.0 V	15.0 V	15.0 V	All inputs together		9					nA
I_{IH2}	"	50			"	"	"	"	"	"			"	"		"	"	"	R1		1.0		45			"
		51																	S1							
		52																	E							
		53																	S2							
		54																	R2							
		55																	R3							
		56																	S3							
		57																	S4							
		58																	R4							
I_{IL1} 9/	3009	59			GND	GND	GND	GND	GND	"			GND	GND		GND	GND	"	All inputs together		-9					"
I_{IL2}	"	60			"	"	"	"	"	"			"	"		"	"	"	R1		-1.0		-45			"
		61																	S1							
		62																	E							
		63																	S2							
		64																	R2							
		65																	R3							
		66																	S3							
		67																	S4							
		68																	R4							
																				Subgroup 4 T_A = 25°C						
																				Min	Max					
C1	3012	69			10/	10/	10/	10/	10/	GND			10/	10/		10/	10/	GND	R1		12					pF
		70																	S1							
		71																	E							
		72																	S2							
		73																	R2							
		74																	R3							
		75																	S3							
		76																	S4							
		77																	R4							

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits Subgroup 7 T_A = 25°C Min	Max	Subgroup 8 T_A = 125°C Min	Max	T_A = -55°C Min	Max	Unit
		Test no.	Q4	Q1	R1	S1	E	S2	R2	V_{SS}	Q2	Q3	R3	S3	NC	S4	R4	V_{DD}								
Truth table test	3014	78	L	H	L	H	5.0 V	L	H	GND	L	H	L	H		L	H	5.0 V	All outputs	See 11/ and 12/						
"	"	79	L	H	L	L	"	L	L	"	L	H	L	L		L	L	"	"							
"	"	80	H	L	H	L	"	H	L	"	H	L	H	L		H	L	"	"							
"	"	81	H	L	L	L	"	L	L	"	H	L	L	L		L	L	"	"							
"	"	82	L	H	H	H	"	L	H	"	L	H	H	H		L	H	"	"							
"	"	83	H	L	H	L	"	H	H	"	H	L	H	L		H	H	"	"							
																				Subgroup 9 T_A = 25°C Min	Max	Subgroup 10 T_A = 125°C Min	Max	Subgroup 11 T_A = -55°C Min	Max	
t_{PHL} R	3003 Fig. 17	84		OUT	IN		5.0 V		IN	GND	OUT		IN					5.0 V	R1 to Q1	16	320	19	370	14	270	ns
"	"	85		"	"		"		"	"	"		"					"	R2 to Q2	"	"	"	"	"	"	"
"	"	86		"	"		"		"	"	"		"					"	R3 to Q3	"	"	"	"	"	"	"
"	"	87	OUT	"	"		"		"	"	"	OUT	"				IN	"	R4 to Q4	"	"	"	"	"	"	"
t_{PLH}	"	88		OUT		IN	"			"	OUT							"	S1 to Q1	10	200	12	245	9	185	"
"	"	89		"		"	"	IN		"	"	OUT						"	S2 to Q2	"	"	"	"	"	"	"
"	"	90		"		"	"	"		"	"	"		IN				"	S3 to Q3	"	"	"	"	"	"	"
"	"	91	OUT	"		"	"	"		"	"	"		"		IN		"	S4 to Q4	"	"	"	"	"	"	"
t_{PZH}	Fig. 18	92		OUT	GND	5.0 V	IN	GND	GND	"			GND	GND		GND	GND	"	E to Q1		230		340		230	"
t_{PHZ}	"	93			"	GND	"	5.0 V	"	"	OUT		"	GND		"	"	"	E to Q2		"		"		"	"
"	"	94			"	"	"	GND	"	"	"	OUT	"	5.0 V		"	"	"	E to Q3		"		"		"	"
"	"	95	OUT		"	"	"	"	"	"	"	"	"	GND		5.0 V	"	"	E to Q4		"		"		"	"
t_{PZL}	"	96		OUT	5.0 V	"	"	"	"	"			"	"		GND	"	"	E to Q1		180		240		180	"
t_{PLZ}	"	97			GND	"	"	"	5.0 V	"	OUT		"	"		"	"	"	E to Q2		"		"		"	"
"	"	98			"	"	"	"	GND	"	"	OUT	5.0 V	"		"	"	"	E to Q3		"		"		"	"
"	"	99	OUT		"	"	"	"	GND	"	"	"	GND	"		"	5.0 V	"	E to Q4		"		"		"	"
t_{THL}	3004 Fig. 17	100		OUT		IN	5.0 V	IN		"	OUT							"	Q1	10	200	12	245	9	185	"
"	"	101		"		"	"	"		"	"	OUT						"	Q2	"	"	"	"	"	"	"
"	"	102		"		"	"	"		"	"	"		IN				"	Q3	"	"	"	"	"	"	"
"	"	103	OUT	"		"	"	"		"	"	"		"		IN		"	Q4	"	"	"	"	"	"	"
t_{TLH}	"	104		OUT		IN	"	IN		"	OUT							"	Q1	15	300	18	360	13	250	"
"	"	105		"		"	"	"		"	"	OUT						"	Q2	"	"	"	"	"	"	"
"	"	106		"		"	"	"		"	"	"		IN				"	Q3	"	"	"	"	"	"	"
"	"	107	OUT	"		"	"	"		"	"	"		"		IN		"	Q4	"	"	"	"	"	"	"

See footnotes at end of device type 03.

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1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ Test numbers 19 thru 24 shall be run in sequence.

3/ I_{OH} = -0.175 mA at 25°C, -0.12 mA at 125°C, -0.22 mA at -55°C.

4/ V_{IH1} = 3.8 V at 25°C, 3.6 V at 125°C, 3.95 V at -55°C.

5/ V_{IH2} = 9.5 V at 25°C, 9.25 V at 125°C, 9.75 V at -55°C.

6/ I_{OL} = 0.20 mA at 25°C, 0.14 mA at 125°C, 0.25 mA at -55°C.

7/ V_{IL1} = 1.10 V at 25°C, 0.8 V at 125°C, 1.35 V at -55°C.

8/ V_{IL2} = 2.8 V at 25°C, 2.55 V at 125°C, 3.0 V at -55°C.

9/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

10/ See 4.4.1c.

11/ Test numbers 78 thru 83 shall be run in sequence and the functional tests shall be performed with V_{IH} and $V_{DD} \leq 5.0$ V and ≥ 15.0 V.

12/ L = V_{SS} + 0.5 V maximum and H = V_{DD} - 0.5 V minimum.

MIL-M-38510/51E

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TABLE III. Group A inspection for device type 51.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
$V_{IC(pos)}$		1				1 mA		1 mA								GND	SET1		1.5					V
		2														"	RS1		"					"
		3			1 mA		1 mA			1 mA						"	D1		"					"
		4									1 mA					"	CLK1		"					"
		5										1 mA				"	SET2		"					"
		6														"	RS2		"					"
		7														"	D2		"					"
		8											1 mA			"	CLK2		"					"
$V_{IC(neg)}$		9				-1 mA		-1 mA	GND								SET1		-6					"
		10							"								RS1		"					"
		11			-1 mA		-1 mA		"	-1 mA							D1		"					"
		12							"		-1 mA						CLK1		"					"
		13							"			-1 mA					SET2		"					"
		14							"								RS2		"					"
		15							"								D2		"					"
		16							"				-1 mA				CLK2		"					"
I_{SS}	3005	17			18.0 V	GND	GND	GND		GND	GND	GND	GND			18.0 V	None		-.25		-2.5			uA
	"	18			GND	"	"	"		"	"	"	"			"	None		"		"			"
	"	19			18.0 V	"	"	"		"	"	"	"			"	None		"		"			"
	"	20			18.0 V	"	18.0 V	"		"	"	"	"			"	V_{SS}		"		"			"
	"	21			GND	"	"	"		"	"	"	"			"	V_{SS}		"		"			"
	"	22			18.0 V	"	"	"		"	"	"	"			"	None		"		"			"
	"	23			18.0 V	"	GND	"		"	"	"	"			"	None		"		"			"
	"	24			GND	"	"	"		"	"	"	"			"	V_{SS}		"		"			"
	"	25			18.0 V	"	"	"		"	"	"	"			"	None		"		"			"
	"	26			18.0 V	18.0 V	"	18.0 V		"	"	"	18.0 V			"	V_{SS}		"		"			"
	"	27			GND	"	"	"		"	"	"	GND			"	V_{SS}		"		"			"
	"	28			"	"	"	"		"	"	"	"			"	None		"		"			"
	"	29			"	"	"	"		"	"	"	"			"	None		"		"			"
	"	30			"	"	"	"		"	"	"	"			"	None		"		"			"
	"	31			"	"	"	"		"	"	"	"			"	V_{SS}		"		"			"
	"	32			"	"	"	"		"	"	"	"			"	V_{SS}		"		"			"
	"	33			"	"	"	"		"	"	"	"			"	None		"		"			"
	"	34			"	"	"	"		"	"	"	"			"	None		"		"			"
	"	35			"	"	"	"		"	"	"	"			"	None		"		"			"
	"	36			"	"	"	"		"	"	"	"			"	V_{SS}		"		"			"
	"	37			"	"	"	"		18.0 V	18.0 V	18.0 V	18.0 V			"	V_{SS}		"		"			"
	"	38			"	"	"	"		18.0 V	18.0 V	18.0 V	GND			"	V_{SS}		"		"			"
V_{OH5}	3006	39			"	"	"	15.0 V	"	GND	GND	GND	"			15.0 V	Q1	14.95		14.95		14.95		V
	"	40			"	"	"	GND	"	GND	"	"	"			"	$\overline{Q1}$	"		"		"		"
	"	41			"	"	"	GND	"	15.0 V	"	"	"			"	Q2	"		"		"		"
	"	42			"	"	"	"	"	GND	"	15.0 V	"			"	$\overline{Q2}$	"		"		"		"
V_{OL5}	3007	43			"	15.0 V	"	"	"	"	"	GND	"			"	Q1		0.05		0.05		0.05	"
	"	44			"	GND	"	15.0 V	"	"	"	GND	"			"	$\overline{Q1}$		"		"		"	"
	"	45			"	"	"	GND	"	"	"	15.0 V	"			"	Q2		"		"		"	"
	"	46			"	"	"	GND	"	15.0 V	"	GND	"			"	$\overline{Q2}$		"		"		"	"
V_{OH6}	3006	47			3/	"	"	3/	"	GND	"	"	"			"	$\overline{Q1}$	14.95		14.95		14.95		"
V_{OL6}	3007	48			3/	"	"	3/	"	GND	"	"	"			"	Q1		0.05		0.05		0.05	"

See footnotes at end of device type 51.

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/													Measured terminal	Test limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
V_{OH6}	3006	49			GND	GND	GND	GND	GND	3/	GND	GND	3/			15.0 V	$\overline{Q2}$	14.95		14.95		14.95		V
V_{OL6}	3007	50			GND	"	GND	GND	"	3/	"	"	3/			"	Q2		0.05		0.05		0.05	"
V_{OL6}	3007	51			3/	"	15.0 V	3/	"	GND	"	"	GND			"	$\overline{Q1}$		0.05		0.05		0.05	"
V_{OH6}	3006	52			3/	"	15.0 V	3/	"	GND	"	"	GND			"	Q1	14.95		14.95		14.95		"
V_{OL6}	3007	53			GND	"	GND	GND	"	3/	15.0 V	"	3/			"	$\overline{Q2}$		0.05		0.05		0.05	"
V_{OH6}	3006	54			GND	"	GND	GND	"	3/	15.0 V	"	3/			"	Q2	14.95		14.95		14.95		"
V_{ICL1}		55			4/	"	4/	4/	"	GND	GND	"	GND			"	CLK1	4/	4/	4/	4/	4/	4/	"
		56			4/	"	4/	4/	"	GND	GND	"	GND			"	CLK1							"
V_{ICL2}		57			GND	"	GND	GND	"	4/	4/	"	4/			"	CLK2	4/	4/	4/	4/	4/	4/	"
		58			"	"	GND	GND	"	4/	4/	"	4/			"	CLK2							"
V_{IH1}		59			"	1.5 V	1.5 V	3.5 V	"	GND	GND	"	GND			5.0 V	Q1	4.5		4.5		4.5		"
		60			"	3.5 V	1.5 V	1.5 V	"	"	"	"	"			"	$\overline{Q1}$							"
		61			5/	1.5 V	3.5 V	"	"	"	"	"	"			"	Q1							"
		62			5/	1.5 V	1.5 V	"	"	"	"	"	"			"	$\overline{Q1}$							"
		63			GND	GND	GND	GND	"	3.5 V	1.5 V	1.5 V	"			"	Q2							"
		64			"	"	"	"	"	1.5 V	1.5 V	3.5 V	"			"	$\overline{Q2}$							"
		65			"	"	"	"	"	"	3.5 V	1.5 V	5/			"	Q2							"
		66			"	"	"	"	"	"	1.5 V	1.5 V	5/			"	$\overline{Q2}$							"
V_{IH2}		67			"	3.0 V	3.0 V	7.0 V	"	GND	GND	GND	GND			10.0 V	Q1	9.0		9.0		9.0		"
		68			"	7.0 V	3.0 V	3.0 V	"	"	"	"	"			"	$\overline{Q1}$							"
		69			5/	3.0 V	7.0 V	"	"	"	"	"	"			"	Q1							"
		70			5/	3.0 V	3.0 V	"	"	"	"	"	"			"	$\overline{Q1}$							"
		71			GND	GND	GND	GND	"	7.0 V	3.0 V	3.0 V	"			"	Q2							"
		72			"	"	"	"	"	3.0 V	3.0 V	7.0 V	"			"	$\overline{Q2}$							"
		73			"	"	"	"	"	"	7.0 V	3.0 V	5/			"	Q2							"
		74			"	"	"	"	"	"	3.0 V	3.0 V	5/			"	$\overline{Q2}$							"
V_{IH3}		75			"	4.0 V	4.0 V	11.0 V	"	GND	GND	GND	GND			15.0 V	Q1	13.5		13.5		13.5		"
		76			"	11.0 V	4.0 V	4.0 V	"	"	"	"	"			"	$\overline{Q1}$							"
		77			5/	4.0 V	11.0 V	"	"	"	"	"	"			"	Q1							"
		78			5/	4.0 V	4.0 V	"	"	"	"	"	"			"	$\overline{Q1}$							"
		79			GND	GND	GND	GND	"	11.0 V	4.0 V	4.0 V	"			"	Q2							"
		80			"	"	"	"	"	4.0 V	4.0 V	11.0 V	"			"	$\overline{Q2}$							"
		81			"	"	"	"	"	"	11.0 V	4.0 V	5/			"	Q2							"
		82			"	"	"	"	"	"	4.0 V	4.0 V	5/			"	$\overline{Q2}$							"

See footnotes at end of device type 51.

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test Limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
V_{IL1}		83			GND	3.5 V	3.5 V	1.5 V	GND	GND	GND	GND	GND			5.0 V	Q1		0.5		0.5		0.5	V
		84			GND	1.5 V	3.5 V	1.5 V	"	"	"	"	"			"	$\overline{Q1}$		"		"		"	
		85			5/	"	1.5 V	1.5 V	"	"	"	"	"			"	Q1		"		"		"	
		86			5/	GND	3.5 V	1.5 V	"	"	"	"	"			"	$\overline{Q1}$		"		"		"	
		87			GND	"	GND	GND	"	1.5 V	3.5 V	3.5 V	"			"	Q2		"		"		"	
		88			"	"	"	"	"	3.5 V	3.5 V	1.5 V	5/			"	$\overline{Q2}$		"		"		"	
		89			"	"	"	"	"	1.5 V	1.5 V	"	5/			"	Q2		"		"		"	
		90			"	"	"	"	"	3.5 V	3.5 V	"	"			"	$\overline{Q2}$		"		"		"	
V_{IL2}		91			"	7.0 V	7.0 V	3.0 V	"	GND	GND	GND	GND			10.0 V	Q1		1.0		1.0		1.0	V
		92			"	3.0 V	7.0 V	3.0 V	"	"	"	"	"			"	$\overline{Q1}$		"		"		"	
		93			5/	"	3.0 V	3.0 V	"	"	"	"	"			"	Q1		"		"		"	
		94			5/	GND	7.0 V	3.0 V	"	"	"	"	"			"	$\overline{Q1}$		"		"		"	
		95			GND	"	GND	GND	"	3.0 V	7.0 V	7.0 V	"			"	Q2		"		"		"	
		96			"	"	"	"	"	7.0 V	7.0 V	3.0 V	5/			"	$\overline{Q2}$		"		"		"	
		97			"	"	"	"	"	3.0 V	3.0 V	"	5/			"	Q2		"		"		"	
		98			"	"	"	"	"	7.0 V	7.0 V	"	"			"	$\overline{Q2}$		"		"		"	
V_{IL3}		99			"	11.0 V	11.0 V	4.0 V	"	GND	GND	GND	GND			15.0 V	Q1		1.5		1.5		1.5	V
		100			"	4.0 V	11.0 V	4.0 V	"	"	"	"	"			"	$\overline{Q1}$		"		"		"	
		101			5/	"	4.0 V	4.0 V	"	"	"	"	"			"	Q1		"		"		"	
		102			5/	GND	11.0 V	4.0 V	"	"	"	"	"			"	$\overline{Q1}$		"		"		"	
		103			GND	"	GND	GND	"	4.0 V	11.0 V	11.0 V	"			"	Q2		"		"		"	
		104			"	"	"	"	"	11.0 V	11.0 V	4.0 V	5/			"	$\overline{Q2}$		"		"		"	
		105			"	"	"	"	"	4.0 V	4.0 V	"	5/			"	Q2		"		"		"	
		106			"	"	"	"	"	11.0 V	11.0 V	"	"			"	$\overline{Q2}$		"		"		"	
I_{OL1}		107	0.4 V	0.4 V	"	5.0 V	"	"	"	GND	GND	GND	GND	0.4 V	0.4 V	5.0 V	Q1	0.51		0.36		0.64		mA
		108	"	"	"	GND	"	5.0 V	"	"	"	GND	"	"	"	"	$\overline{Q1}$	"		"		"		
		109	"	"	"	"	"	GND	"	5.0 V	"	5.0 V	"	"	"	"	Q2	"		"		"		
		110	"	"	"	"	"	"	"	"	"	GND	"	"	"	"	$\overline{Q2}$	"		"		"		
I_{OL2}		111	1.5 V	1.5 V	"	15.0 V	"	"	"	GND	"	"	"	1.5 V	1.5 V	15.0 V	Q1	3.4		2.4		4.2		"
		112	"	"	"	GND	"	15.0 V	"	"	"	"	"	"	"	"	$\overline{Q1}$	"		"		"		
		113	"	"	"	"	"	GND	"	15.0 V	"	15.0 V	"	"	"	"	Q2	"		"		"		
		114	"	"	"	"	"	GND	"	"	"	GND	"	"	"	"	$\overline{Q2}$	"		"		"		
I_{OH1}		115	4.6 V	4.6 V	"	"	"	"	"	GND	"	"	"	4.6 V	4.6 V	5.0 V	Q1	-0.51		-0.36		-0.64		"
		116	"	"	"	5.0 V	"	5.0 V	"	GND	"	"	"	"	"	"	$\overline{Q1}$	"		"		"		
		117	"	"	"	GND	"	GND	"	5.0 V	"	"	"	"	"	"	Q2	"		"		"		
		118	"	"	"	"	"	"	"	GND	"	5.0 V	"	"	"	"	$\overline{Q2}$	"		"		"		
I_{OH2}		119	13.5 V	13.5 V	"	"	"	15.0 V	"	"	"	GND	"	13.5 V	13.5 V	15.0 V	Q1	-3.4		-2.4		-4.2		"
		120	"	"	"	15.0 V	"	GND	"	"	"	"	"	"	"	"	$\overline{Q1}$	"		"		"		
		121	"	"	"	GND	"	"	"	15.0 V	"	15.0 V	"	"	"	"	Q2	"		"		"		
		122	"	"	"	"	"	"	"	GND	"	"	"	"	"	"	$\overline{Q2}$	"		"		"		
I_{IH1} 6/	3010	123			18.0 V	18.0 V	18.0 V	18.0 V	"	18.0 V	18.0 V	18.0 V	18.0 V			18.0 V	All inputs together		8					nA

See footnotes at end of device type 51.

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test Limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14		Min	Max	Min	Max	Min	Max	
I_{IH2}	3010	124			18.0 V	18.0 V	18.0 V	18.0 V	GND	18.0 V	18.0 V	18.0 V	18.0 V			18.0 V	CLK1		1.0		45			nA
"	"	125			"	"	"	"	"	"	"	"	"			"	RS1		"		"			"
"	"	126			"	"	"	"	"	"	"	"	"			"	D1		"		"			"
"	"	127			"	"	"	"	"	"	"	"	"			"	SET1		"		"			"
"	"	128			"	"	"	"	"	"	"	"	"			"	SET2		"		"			"
"	"	129			"	"	"	"	"	"	"	"	"			"	D2		"		"			"
"	"	130			"	"	"	"	"	"	"	"	"			"	RS2		"		"			"
"	"	131			"	"	"	"	"	"	"	"	"			"	CLK2		"		"			"
I_{IL1} 6/	3009	132			GND	GND	GND	GND	"	GND	GND	GND	GND			"	All inputs together		-8.0					"
I_{IL2}	"	133			"	"	"	"	"	"	"	"	"			"	CLK1		-1.0		-45			"
"	"	134			"	"	"	"	"	"	"	"	"			"	RS1		"		"			"
"	"	135			"	"	"	"	"	"	"	"	"			"	D1		"		"			"
"	"	136			"	"	"	"	"	"	"	"	"			"	SET1		"		"			"
"	"	137			"	"	"	"	"	"	"	"	"			"	SET2		"		"			"
"	"	138			"	"	"	"	"	"	"	"	"			"	D2		"		"			"
"	"	139			"	"	"	"	"	"	"	"	"			"	RS2		"		"			"
"	"	140			"	"	"	"	"	"	"	"	"			"	CLK2		"		"			"
																		Subgroup 4 T_A = 25°C						
																		Min	Max					
C1	3012	141			7/				GND							GND	CLK1		12					pF
"	"	142				7/			"							"	RS1		"					"
"	"	143					7/		"							"	D1		"					"
"	"	144						7/	"							"	SET1		"					"
"	"	145							"	7/						"	SET2		"					"
"	"	146							"		7/					"	D2		"					"
"	"	147							"			7/				"	RS2		"					"
"	"	148							"				7/			"	CLK2		"					"
																		Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
Truth table test	3014	149			5.0 V	GND	GND	GND	GND	GND	GND	GND	5.0 V			5.0 V	None	See notes 8/ and 9/					
"	"	150			GND	"	"	"	"	"	"	"	GND			"	None							
"	"	151	L	H	5.0 V	"	"	"	"	"	"	"	5.0 V	H	L	"	All							
"	"	152	"	"	5.0 V	"	5.0 V	"	"	"	5.0 V	"	5.0 V	"	"	"	outputs							
"	"	153	"	"	GND	"	"	"	"	"	"	"	GND	"	"	"	"							
"	"	154	H	L	5.0 V	"	"	"	"	"	"	5.0 V	5.0 V	L	H	"	"							
"	"	155	L	H	5.0 V	5.0 V	"	"	"	"	"	"	5.0 V	H	L	"	"							
"	"	156	"	"	GND	"	"	"	"	"	"	"	GND	"	"	"	"							
"	"	157	"	"	5.0 V	"	"	"	"	"	"	"	5.0 V	"	"	"	"							
"	"	158	H	L	5.0 V	GND	GND	5.0 V	"	5.0 V	GND	GND	5.0 V	L	H	"	"							
"	"	159	"	"	GND	"	"	"	"	"	"	"	GND	"	"	"	"							
"	"	160	"	"	5.0 V	"	"	"	"	"	"	"	5.0 V	"	"	"	"							
"	"	161	"	"	5.0 V	"	"	GND	"	GND	"	"	5.0 V	"	"	"	"							
"	"	162	"	"	GND	"	"	"	"	"	"	"	GND	"	"	"	"							
"	"	163	L	H	5.0 V	"	"	"	"	"	"	"	5.0 V	H	L	"	"							
"	"	164	H	"	5.0 V	5.0 V	"	5.0 V	"	5.0 V	"	5.0 V	5.0 V	"	H	"	"							
"	"	165	"	"	GND	"	"	"	"	"	"	"	GND	"	"	"	"							
"	"	166	"	"	5.0 V	"	"	"	"	"	"	"	5.0 V	"	"	"	"							

See footnotes at end of device type 51.

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test Limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
t_{PHL}	3003 (Fig. 8)	167 168 169 170	OUT	OUT	IN IN				GND " "				IN IN	OUT	OUT	5.0 V " "	CLK1 to Q1 CLK1 to $\overline{Q1}$ CLK2 to Q2 CLK2 to $\overline{Q2}$	25 " "	500 " "	38 " "	750 " "	25 " "	500 " "	ns " "
t_{PHL} R or S	3003 (Fig. 9)	171 172 173 174	OUT	OUT		IN		IN	" " " "	IN		IN		OUT	OUT	" " " "	SET1 to $\overline{Q1}$ RS1 to Q1 SET2 to $\overline{Q2}$ RS2 to Q2	28 " "	550 " "	41 " "	825 " "	28 " "	550 " "	" " "
t_{PLH}	3003 (Fig. 8)	175 176 177 178	OUT	OUT	IN IN				" " " "				IN IN	OUT	OUT	" " " "	CLK1 to Q1 CLK1 to $\overline{Q1}$ CLK2 to Q2 CLK2 to $\overline{Q2}$	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "	" " " "
t_{PLH} R or S	3003 (Fig. 9)	179 180 181 182	OUT	OUT		IN		IN	" " " "	IN		IN		OUT	OUT	" " " "	SET1 to Q1 RS1 to $\overline{Q1}$ SET2 to Q2 RS2 to $\overline{Q2}$	21 " "	420 " "	32 " "	630 " "	21 " "	420 " "	" " "
t_{THL}	3004 (Fig. 8)	183 184 185 186	OUT	OUT	IN IN				" " " "				IN IN	OUT	OUT	" " " "	Q1 $\overline{Q1}$ Q2 $\overline{Q2}$	15 " "	300 " "	23 " "	450 " "	15 " "	300 " "	" " "
t_{TLH}	" " "	187 188 189 190	OUT	OUT	IN IN				" " " "				IN IN	OUT	OUT	" " " "	$\overline{Q1}$ Q1 $\overline{Q2}$ Q2	18 " "	350 " "	26 " "	525 " "	18 " "	350 " "	" " "
$f_{CL(MAX)}$ 10/	(Fig. 8)	191 192	OUT		IN				" "				IN		OUT	" "	CLK1 CLK2		0.67 0.67		1.0 1.0		0.67 0.67	µs "
$T_{TLHCL(max)}$ 11/	" "	193 194	OUT		IN				" "				IN		OUT	" "	CLK1 CLK2	15 15		15 15		10 10		" "
t_p 12/	" "	195 196	OUT		IN				" "				IN		OUT	" "	CLK1 CLK2		300 300		450 450		300 300	ns "
t_{SHL} t_{SHL}	(Fig. 10)	197 198			IN		IN		" "		IN		IN			" "	D1 to CLK1 D2 to CLK2		165 "		225 "		165 "	" "
t_{SLH} t_{SLH}	" "	199 200			IN		IN		" "		IN		IN			" "	D1 to CLK1 D2 to CLK2		" "		" "		" "	" "

See footnotes at end of device type 51.

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TABLE III. Group A inspection for device type 51 - Continued.

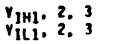
Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	Terminal conditions 1/														Measured terminal	Test limits						Unit
			Q1	$\overline{Q1}$	CLK1	RS1	D1	SET1	V_{SS}	SET2	D2	RS2	CLK2	$\overline{Q2}$	Q2	V_{DD}		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
																		Min	Max	Min	Max	Min	Max	
		Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14								
t_{HHL} t_{HHL}	(Fig. 11) "	201 202			IN		IN		GND "		IN		IN			5.0 V "	D1 to CLK1 D2 to CLK2		150 "		225 "		150 "	ns "
t_{HLH} t_{HLH}	" "	203 204			IN		IN		" "		IN		IN			" "	D1 to CLK1 D2 to CLK2		" "		" "		" "	" "

1/ Pins not designated may be "high" level logic "low" logic, or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ Test numbers 17 thru 38 shall be run in sequence.

3/ For input conditions, see figure 8.

4/ For input conditions, see figure 9.

5/ Apply a clock pulse  - - - V_{IH1}, 2, 3
- - - V_{IL1}, 2, 3

6/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

7/ See 4.4.1c.

8/ Test numbers 144 thru 166 shall be run in sequence and the functional tests shall be performed with V_{IH} and $V_{DD} \leq 5.0$ V and ≥ 18.0 V.

9/ L = V_{SS} + 0.5 V maximum and H = V_{DD} - 0.5 V minimum.

10/ The maximum clock frequency (f_{CL}) requirement is considered met if proper output state changes occur with the pulse repetition period set to that given is the limits column.

11/ Pulse repetition period = 100 µs, 50 percent duty cycle. The maximum clock transition time (t_{TLHCL}) requirement is considered met if proper output state changes occur with the rise time set to that given in the limits column.

12/ The minimum clock pulse width (t_p) requirement is considered met if proper output state changes occur with the pulse width set to that given is the limits column.

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TABLE III. Group A inspection for device type 52.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/															Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1 2 3 4 5 6 7 8 9 10			1 mA	1 mA	1 mA	1 mA	1 mA		1 mA	1 mA	1 mA	1 mA	1 mA			GND " " " " " " " "	SET2 RS2 J2 K2 CLK2 SET1 RS1 J1 K1 CLK1		1.5 " " " " " " " "					V " " " " " " " "
$V_{IC(neg)}$		11 12 13 14 15 16 17 18 19 20			-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	GND " " " " " " " "	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA				SET2 RS2 J2 K2 CLK2 SET1 RS1 J1 K1 CLK1		-6 " " " " " " " "					" " " " " " " " "
I_{SS} 2/		21 22 23 24 25 26 27 28 29 30 31 32 33 34			GND 18.0 V 18.0 V GND 18.0 V " " GND " " " " " "	18.0 V 18.0 V GND " " " 18.0 V GND " " " " " "	18.0 V " " " " GND " " " " " " " "	18.0 V " " " " GND " " " " " " " "	GND " " " " " " 18.0 V GND " " " " "	" " " " " " " " " " " " " "	GND " " " " " " " " " " " " 18.0 V	GND " " " " " 18.0 V " " " " " " GND	GND " " " " " " " " " " " " "	GND " " " " " 18.0 V " " " " " " 18.0 V	GND " " " " " " " " " " " " "			18.0 V " " " " " " " " " " " " "	V_{SS} None V_{SS} None V_{SS} V_{SS} V_{SS} V_{SS} None V_{SS} None V_{SS} V_{SS} V_{SS}		-0.25 " " " " " " " " " " " " "		-2.5 " " " " " " " " " " " " "			mA " " " " " " " " " " " " "
V_{OH5}	3006 " "	35 36 37 38			" " " "	" 15.0 V GND GND	" " " "	" " " "	15.0 V GND " "	" " " "	GND GND 15.0 V GND	" " " "	" " " "	GND " " 15.0 V	GND " " "			15.0 V " " "	Q2 $\overline{Q2}$ Q1 $\overline{Q1}$	14.95 " " "		14.95 " " "		14.95 " " "		V " " "
V_{OL5}	3007 " "	39 40 41 42			" " " "	15.0 V GND " "	" " " "	" " " "	15.0 V GND GND "	" " " "	" " " 15.0 V	" " " "	" " " "	GND GND 15.0 V GND	" " " "			" " " "	Q2 $\overline{Q2}$ Q1 $\overline{Q1}$		0.05 " " "		0.05 " " "		0.05 " " "	" " " "
V_{OH6}	3006	43			3/	"	15.0 V	"	3/	"	GND	"	"	"	"			"	$\overline{Q2}$	14.95		14.95		14.95		"
V_{OL6}	3007	44			3/	"	15.0 V	"	3/	"	GND	"	"	"	"			"	Q2		0.05		0.05		0.05	"
V_{OH6}	3006	45			GND	"	GND	"	GND	"	3/	"	15.0 V	"	3/			"	$\overline{Q1}$	14.95		14.95		14.95		"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 test method	Case E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{OL6}	3007	46			GND	GND	GND	GND	GND	GND	3/	GND	15.0 V	GND	3/			15.0 V	Q1		0.05		0.05		0.05	V
V_{OL6}	3007	47			3/	"	"	15.0 V	3/	"	GND	"	GND	"	GND			"	$\overline{Q2}$		0.05		0.05		0.05	"
V_{OH6}	3006	48			3/	"	"	15.0 V	3/	"	GND	"	"	"	GND			"	Q2	14.95		14.95		14.95		"
V_{OL6}	3007	49			GND	"	"	GND	GND	"	3/	15.0 V	"	"	3/			"	$\overline{Q1}$		0.05		0.05		0.05	"
V_{OH6}	3006	50			GND	"	"	GND	GND	"	3/	15.0 V	"	"	3/			"	Q1	14.95		14.95		14.95		"
V_{ICL3}		51			4/	4/	4/	4/	"	"	GND	GND						"	CLK2		4/		4/		4/	"
V_{ICL3}		52			4/	4/	4/	4/	"	"	"	GND						"	CLK2	4/		4/		4/		"
V_{ICL4}		53			GND	GND	GND	GND	"	"	"	4/	4/	4/	4/			"	CLK1		4/		4/		4/	"
V_{ICL4}		54			"	GND	"	"	"	"	"	4/	4/	4/	4/			"	CLK1	4/		4/		4/		"
V_{IH1}		55			"	1.5 V	"	"	3.5 V	"	"	GND	GND	GND	GND			5.0 V	Q2	4.5		4.5		4.5		"
		56			"	3.5 V	"	"	1.5 V	"	"	"	"	"	"			"	$\overline{Q2}$	"		"		"		"
		57			5/	1.5 V	1.5 V	3.5 V	"	"	"	"	"	"	"			"	Q2	"		"		"		"
		58			5/	1.5 V	3.5 V	1.5 V	"	"	"	"	"	"	"			"	$\overline{Q2}$	"		"		"		"
		59			GND	GND	GND	GND	GND	"	3.5 V	"	"	"	"			"	Q1	"		"		"		"
		60			"	"	"	"	"	"	1.5 V	"	"	"	3.5 V			"	$\overline{Q1}$	"		"		"		"
		61			"	"	"	"	"	"	"	3.5 V	1.5 V	1.5 V	5/			"	Q1	"		"		"		"
		62			"	"	"	"	"	"	"	1.5 V	3.5 V	1.5 V	5/			"	$\overline{Q1}$	"		"		"		"
V_{IH2}		63			"	3.0 V	"	"	7.0 V	"	GND	GND	GND	GND	GND			10.0 V	Q2	9.0		9.0		9.0		"
		64			"	7.0 V	"	"	3.0 V	"	"	"	"	"	"			"	$\overline{Q2}$	"		"		"		"
		65			5/	3.0 V	3.0 V	7.0 V	"	"	"	"	"	"	"			"	Q2	"		"		"		"
		66			5/	3.0 V	7.0 V	3.0 V	"	"	"	"	"	"	"			"	$\overline{Q2}$	"		"		"		"
		67			GND	GND	GND	GND	GND	"	7.0 V	"	"	3.0 V	"			"	Q1	"		"		"		"
		68			"	"	"	"	"	"	3.0 V	"	"	7.0 V	"			"	$\overline{Q1}$	"		"		"		"
		69			"	"	"	"	"	"	"	7.0 V	3.0 V	3.0 V	5/			"	Q1	"		"		"		"
		70			"	"	"	"	"	"	"	3.0 V	7.0 V	3.0 V	5/			"	$\overline{Q1}$	"		"		"		"
V_{IH3}		71			"	4.0 V	"	"	11.0 V	"	GND	GND	GND	GND	GND			15.0 V	Q2	13.5		13.5		13.5		"
		72			"	11.0 V	"	"	4.0 V	"	"	"	"	"	"			"	$\overline{Q2}$	"		"		"		"
		73			5/	4.0 V	4.0 V	11.0 V	"	"	"	"	"	"	"			"	Q2	"		"		"		"
		74			5/	4.0 V	11.0 V	4.0 V	"	"	"	"	"	"	"			"	$\overline{Q2}$	"		"		"		"
		75			GND	GND	GND	GND	GND	"	11.0 V	"	"	4.0 V	"			"	Q1	"		"		"		"
		76			"	"	"	"	"	"	4.0 V	"	"	11.0 V	"			"	$\overline{Q1}$	"		"		"		"
		77			"	"	"	"	"	"	"	11.0 V	4.0 V	4.0 V	5/			"	Q1	"		"		"		"
		78			"	"	"	"	"	"	"	4.0 V	11.0 V	4.0 V	5/			"	$\overline{Q1}$	"		"		"		"
V_{IL1}		79			"	3.5 V	"	"	1.5 V	"	GND	GND	GND	GND	GND			5.0 V	Q2		0.5		0.5		0.5	"
		80			"	1.5 V	"	"	3.5 V	"	"	"	"	"	"			"	$\overline{Q2}$		"		"		"	"
		81			5/	"	3.5 V	1.5 V	1.5 V	"	"	"	"	"	"			"	Q2		"		"		"	"
		82			5/	"	1.5 V	3.5 V	1.5 V	"	"	"	"	"	"			"	$\overline{Q2}$		"		"		"	"
		83			GND	GND	GND	GND	GND	"	1.5 V	"	"	3.5 V	"			"	Q1		"		"		"	"
		84			"	"	"	"	"	"	3.5 V	"	"	1.5 V	"			"	$\overline{Q1}$		"		"		"	"
		85			"	"	"	"	"	"	1.5 V	1.5 V	3.5 V	"	5/			"	Q1		"		"		"	"
		86			"	"	"	"	"	"	1.5 V	3.5 V	1.5 V	"	5/			"	$\overline{Q1}$		"		"		"	"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IL2}		87			GND	7.0 V	GND	GND	3.0 V	GND	GND	GND	GND	GND	GND			10.0 V	Q2		1.0		1.0		1.0	V
		88			GND	3.0 V	GND	GND	7.0 V	"	"	"	"	"	"			"	$\overline{Q2}$		"		"		"	
		89			5/	"	7.0 V	3.0 V	3.0 V	"	"	"	"	"	"			"	Q2		"		"		"	
		90			5/	"	3.0 V	7.0 V	3.0 V	"	"	"	"	"	"			"	$\overline{Q2}$		"		"		"	
		91			GND	GND	GND	GND	GND	"	3.0 V	"	"	7.0 V	"			"	Q1		"		"		"	
		92			"	"	"	"	"	"	7.0 V	3.0 V	7.0 V	3.0 V	"			"	$\overline{Q1}$		"		"		"	
		93			"	"	"	"	"	"	3.0 V	7.0 V	3.0 V	"	5/			"	Q1		"		"		"	
		94			"	"	"	"	"	"	3.0 V	"	"	"	5/			"	$\overline{Q1}$		"		"		"	
V_{IL3}		95			"	11.0 V	"	"	4.0 V	"	GND	GND	GND	GND	GND			15.0 V	Q2		1.5		1.5		1.5	"
		96			"	4.0 V	"	"	11.0 V	"	"	"	"	"	"			"	$\overline{Q2}$		"		"		"	
		97			5/	"	11.0 V	4.0 V	4.0 V	"	"	"	"	11.0 V	"			"	Q2		"		"		"	
		98			5/	"	4.0 V	11.0 V	4.0 V	"	"	"	"	4.0 V	"			"	$\overline{Q2}$		"		"		"	
		99			GND	GND	GND	GND	GND	"	4.0 V	"	"	"	"			"	Q1		"		"		"	
		100			"	"	"	"	"	"	11.0 V	4.0 V	11.0 V	"	"			"	$\overline{Q1}$		"		"		"	
		101			"	"	"	"	"	"	4.0 V	11.0 V	4.0 V	"	5/			"	Q1		"		"		"	
		102			"	"	"	"	"	"	4.0 V	"	"	"	5/			"	$\overline{Q1}$		"		"		"	
I_{OL1}		103	0.4 V		"	5.0 V	"	"	"	"	GND	GND	GND	GND	GND			5.0 V	Q2	0.51		0.36		0.64		mA
		104		0.4 V	"	GND	"	"	5.0 V	"	"	"	"	GND	"			"	$\overline{Q2}$	"		"		"		"
		105			"	"	"	"	GND	"	"	"	"	5.0 V	"	0.4 V	0.4 V	"	Q1	"		"		"		"
		106			"	"	"	"	"	"	5.0 V	"	"	GND	"			"	$\overline{Q1}$	"		"		"		"
I_{OL2}		107	1.5 V		"	15.0 V	"	"	"	"	GND	"	"	"	"			15.0 V	Q2	3.4		2.4		4.2		"
		108		1.5 V	"	GND	"	"	15.0 V	"	"	"	"	15.0 V	"			"	$\overline{Q2}$	"		"		"		"
		109			"	"	"	"	GND	"	"	"	"	GND	"	1.5 V	1.5 V	"	Q1	"		"		"		"
		110			"	"	"	"	GND	"	15.0 V	"	"	"	"			"	$\overline{Q1}$	"		"		"		"
I_{OH1}		111	4.6 V		"	"	"	"	5.0 V	"	GND	"	"	"	"			5.0 V	Q2	-0.51		-0.36		-0.64		"
		112		4.6 V	"	5.0 V	"	"	GND	"	GND	"	"	"	"	4.6 V	4.6 V	"	$\overline{Q2}$	"		"		"		"
		113			"	GND	"	"	"	"	5.0 V	"	"	5.0 V	"			"	Q1	"		"		"		"
		114			"	"	"	"	"	"	GND	"	"	"	"			"	$\overline{Q1}$	"		"		"		"
I_{OH2}		115	13.5 V		"	"	"	"	15.0 V	"	"	"	"	GND	"			15.0 V	Q2	-3.4		-2.4		-4.2		"
		116		13.5 V	"	15.0 V	"	"	GND	"	"	"	"	"	"	13.5 V	13.5 V	"	$\overline{Q2}$	"		"		"		"
		117			"	GND	"	"	"	"	15.0 V	"	"	15.0 V	"			"	Q1	"		"		"		"
		118			"	GND	"	"	"	"	GND	"	"	"	"			"	$\overline{Q1}$	"		"		"		"
I_{IH1} 6/	3009	119			18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	"	18.0 V	18.0 V	18.0 V	18.0 V	18.0 V			18.0 V	All inputs together		10					nA
I_{IH2}	"	120			"	"	"	"	"	"	"	"	"	"	"			"	CLK2		1.0		45			"
		121			"	"	"	"	"	"	"	"	"	"	"			"	RS2		"		"			"
		122			"	"	"	"	"	"	"	"	"	"	"			"	K2		"		"			"
		123			"	"	"	"	"	"	"	"	"	"	"			"	J2		"		"			"
		124			"	"	"	"	"	"	"	"	"	"	"			"	SET2		"		"			"
		125			"	"	"	"	"	"	"	"	"	"	"			"	SET1		"		"			"
		126			"	"	"	"	"	"	"	"	"	"	"			"	J1		"		"			"
		127			"	"	"	"	"	"	"	"	"	"	"			"	K1		"		"			"
		128			"	"	"	"	"	"	"	"	"	"	"			"	RS1		"		"			"
		129			"	"	"	"	"	"	"	"	"	"	"			"	CLK1		"		"			"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 test method	Case E, F, Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{IL1} 6/	3009	130			GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND			18.0 V	All inputs together		-10					nA
I_{IL2}	"	131			"	"	"	"	"	"	"	"	"	"	"			"	CLK2		-1.0		-45			"
	"	132			"	"	"	"	"	"	"	"	"	"	"			"	RS2		"		"			"
	"	133			"	"	"	"	"	"	"	"	"	"	"			"	K2		"		"			"
	"	134			"	"	"	"	"	"	"	"	"	"	"			"	J2		"		"			"
	"	135			"	"	"	"	"	"	"	"	"	"	"			"	SET2		"		"			"
	"	136			"	"	"	"	"	"	"	"	"	"	"			"	SET1		"		"			"
	"	137			"	"	"	"	"	"	"	"	"	"	"			"	J1		"		"			"
	"	138			"	"	"	"	"	"	"	"	"	"	"			"	K1		"		"			"
	"	139			"	"	"	"	"	"	"	"	"	"	"			"	RS1		"		"			"
	"	140			"	"	"	"	"	"	"	"	"	"	"			"	CLK1		"		"			"

																				Subgroup 4 T_A = 25°C				
																				Min	Max			
C1	3012	141			7/	7/	7/	7/	7/	GND	7/	7/	7/	7/	7/			GND	CLK2		12			pF
	"	142								"								"	RS2		"			"
	"	143								"								"	K2		"			"
	"	144								"								"	J2		"			"
	"	145								"								"	SET2		"			"
	"	146								"								"	SET1		"			"
	"	147								"								"	J1		"			"
	"	148								"								"	K1		"			"
	"	149								"								"	RS1		"			"
	"	150								"								"	CLK1		"			"

																				Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		Subgroup 8 T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table test	3014	151			5.0 V	GND	5.0 V	GND	GND	GND	GND	GND	GND	5.0 V	GND	5.0 V			5.0 V	None	See 8/ and 9/					
	"	152			GND	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	None							
	"	153	L	H	5.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	All outputs							
	"	154	"	"	5.0 V	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"							
	"	155	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
	"	156	"	"	5.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
	"	157	"	"	5.0 V	"	"	"	5.0 V	"	"	"	"	"	"	"	"	"	"	"							
	"	158	"	"	GND	"	"	"	"	"	"	"	5.0 V	"	"	"	"	"	"	"							
	"	159	H	L	5.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
	"	160	"	"	5.0 V	"	"	"	GND	"	"	"	GND	"	"	"	"	"	"	"							
	"	161	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
	"	162	"	"	5.0 V	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
	"	163	"	"	5.0 V	"	5.0 V	5.0 V	"	"	"	"	5.0 V	5.0 V	"	"	"	"	"	"							
	"	164	"	"	GND	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"							
	"	165	L	H	5.0 V	"	"	GND	5.0 V	"	5.0 V	GND	"	"	"	"	"	"	"	"							
	"	166	H	L	5.0 V	"	"	"	5.0 V	"	5.0 V	"	"	"	"	"	"	"	"	"							
	"	167	H	L	GND	"	"	"	GND	"	GND	"	"	"	"	"	"	"	"	"							
	"	168	L	H	GND	5.0 V	GND	"	GND	"	GND	"	GND	5.0 V	GND	H	L	H	L	"							
	"	169	L	H	5.0 V	5.0 V	GND	"	GND	"	GND	"	GND	5.0 V	5.0 V	H	L	L	H	"							

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits: Subgroup 7 T_A = 25°C Min	Max	Subgroup 8 T_A = 125°C Min	Max	Subgroup 8 T_A = -55°C Min	Max	Unit
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}								
Truth table test	3014	170	H	L	5.0 V	GND	GND	GND	5.0 V	GND	5.0 V	GND	GND	GND	5.0 V	L	H	5.0 V	All outputs	See 8/ and 9/						
	"	171	"	"	5.0 V	"	5.0 V	"	GND	"	GND	"	5.0 V	"	5.0 V	"	"	"	"							
	"	172	"	"	GND	"	"	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	173	L	H	5.0 V	"	"	"	"	"	"	5.0 V	"	"	5.0 V	H	L	"	"							
	"	174	"	"	5.0 V	"	"	5.0 V	"	"	"	"	"	"	5.0 V	"	"	"	"							
	"	175	"	"	GND	"	"	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	176	H	L	5.0 V	"	"	"	"	"	"	"	"	5.0 V	5.0 V	L	H	"	"							
	"	177	L	H	5.0 V	5.0 V	"	"	"	"	"	"	"	"	5.0 V	H	L	"	"							
	"	178	"	"	GND	"	"	"	"	"	"	"	"	"	GND	"	"	"	"							
	"	179	"	"	5.0 V	"	"	"	"	"	"	"	"	"	5.0 V	"	"	"	"							
	"	180	H	"	5.0 V	"	GND	GND	5.0 V	"	5.0 V	GND	GND	"	GND	"	H	"	"							
	"	181	"	"	GND	"	"	"	"	"	"	"	"	"	5.0 V	"	"	"	"							
	"	182	"	"	5.0 V	"	"	"	"	"	"	"	"	"													

																				Subgroup 9 T_A = 25°C Min	Max	Subgroup 10 T_A = 125°C Min	Max	Subgroup 11 T_A = -55°C Min	Max	
t_{PHL}	3003 Fig. 14	183	OUT	OUT	IN					GND					IN	OUT	OUT	5.0 V	CLK2 to Q2	29	575	43	865	29	575	ns
		184			IN					"								"	CLK2 to $\overline{Q2}$	"	"	"	"	"	"	"
		185								"								"	CLK1 to Q1	"	"	"	"	"	"	"
		186								"								"	CLK1 to $\overline{Q1}$	"	"	"	"	"	"	"
t_{PHL} R or S	3003 Fig. 15	187	OUT	OUT		IN			IN	"	IN			IN		OUT	OUT	"	RS2 to $\overline{Q2}$	30	600	45	900	30	600	"
		188								"								"	SET2 to $\overline{Q2}$	"	"	"	"	"	"	"
		189								"								"	RS1 to Q1	"	"	"	"	"	"	"
		190								"								"	SET1 to $\overline{Q1}$	"	"	"	"	"	"	"
t_{PLH}	3003 Fig. 14	191	OUT	OUT	IN					"					IN	OUT	OUT	"	CLK2 to Q2	31	625	47	940	35	625	"
		192			IN					"								"	CLK2 to $\overline{Q2}$	"	"	"	"	"	"	"
		193								"								"	CLK1 to Q1	"	"	"	"	"	"	"
		194								"								"	CLK1 to $\overline{Q1}$	"	"	"	"	"	"	"
t_{PLH} R or S	3003 Fig. 15	195	OUT	OUT		IN			IN	"	IN			IN		OUT	OUT	"	SET2 to $\overline{Q2}$	20	400	30	600	20	400	"
		196								"								"	RS2 to $\overline{Q2}$	"	"	"	"	"	"	"
		197								"								"	SET1 to Q1	"	"	"	"	"	"	"
		198								"								"	RS1 to $\overline{Q1}$	"	"	"	"	"	"	"
t_{THL}	3004 Fig. 14	199	OUT	OUT	IN					"					IN	OUT	OUT	"	Q2	16	325	25	490	16	325	"
		200			IN					"								"	$\overline{Q2}$	"	"	"	"	"	"	"
		201								"								"	Q1	"	"	"	"	"	"	"
		202								"								"	$\overline{Q1}$	"	"	"	"	"	"	"
t_{TLH}	"	203	OUT	OUT	IN					"					IN	OUT	OUT	"	Q2	"	"	"	"	"	"	"
	"	204			IN					"								"	$\overline{Q2}$	"	"	"	"	"	"	"
	"	205								"								"	Q1	"	"	"	"	"	"	"
	"	206								"								"	$\overline{Q1}$	"	"	"	"	"	"	"
$f_{CL(max)}$ 10/	"	207	OUT		IN					"					IN		OUT	"	CLK2		1.0		1.4		1.0	μs
	"	208								"								"	CLK1		1.0		1.4		1.0	"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	Q2	$\overline{Q2}$	CLK2	RS2	K2	J2	SET2	V_{SS}	SET1	J1	K1	RS1	CLK1	$\overline{Q1}$	Q1	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{TLHCL} 11/	Fig. 14	209	OUT		IN					GND					IN		OUT	5.0 V	CLK2	15		15		10		µs
	"	210								"								"	CLK1	15		15		10		µs
t_{p} 12/	"	211	OUT		IN					"					IN		OUT	"	CLK2	300		450		300		ns
	"	212								"								"	CLK1	300		450		300		"
t_{SHL}	Fig. 16	213	OUT		IN		IN			"								"	K2 to CLK2		165		225		165	"
	"	214	OUT		IN			IN		"								"	J2 to CLK2		"		"		"	"
	"	215								"			IN		IN		OUT	"	K1 to CLK1		"		"		"	"
	"	216								"		IN			IN		OUT	"	J1 to CLK1		"		"		"	"
t_{SLH}	"	217	OUT		IN		IN			"								"	K2 to CLK2		"		"		"	"
	"	218	OUT		IN			IN		"								"	J2 to CLK2		"		"		"	"
	"	219								"			IN		IN		OUT	"	K1 to CLK1		"		"		"	"
	"	220								"		IN			IN		OUT	"	J1 to CLK1		"		"		"	"
t_{HLH}	"	221	OUT		IN		IN			"								"	K2 to CLK2		150		"		150	"
	"	222	OUT		IN			IN		"								"	J2 to CLK2		"		"		"	"
	"	223								"			IN		IN		OUT	"	K1 to CLK1		"		"		"	"
	"	224								"		IN			IN		OUT	"	J1 to CLK1		"		"		"	"
t_{HHL}	"	225	OUT		IN		IN			"								"	K2 to CLK2		"		"		"	"
	"	226	OUT		IN			IN		"								"	J2 to CLK2		"		"		"	"
	"	227								"			IN		IN		OUT	"	K1 to CLK1		"		"		"	"
	"	228								"		IN			IN		OUT	"	J1 to CLK1		"		"		"	"

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ Test numbers 21 thru 34 shall be run in sequence.

3/ For input voltage conditions, see figure 12.

4/ For input voltage conditions, see figure 13.

5/ Apply a clock pulse ⎍⎍ - - - - V_{IH1}, 2, 3; - - - - V_{IL1}, 2, 3

6/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

7/ See 4.4.1c.

8/ Test numbers 151 thru 182 shall be run in sequence and the functional tests shall be performed with V_{IH} and $V_{DD} \leq 5.0$ V and ≥ 18.0 V.

9/ L = V_{SS} + 0.5 V maximum and H = V_{DD} - 0.5 V minimum.

10/ The maximum clock frequency (f_{CL}) requirement is considered met if proper output state changes occur with the pulse repetition period set to that given in the limits column.

11/ Pulse repetition period = 100 µs, 50 percent duty cycle. The maximum clock transition time (t_{TLHCL}) requirement is considered met if proper output state changes occur with the rise time set to that given in the limits column.

12/ The minimum clock pulse width (t_p) requirement is considered met if proper output state changes occur with the pulse width set to that given in the limits column.

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TABLE III. Group A inspection for device type 53.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test Limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q4	Q1	R1	S1	E	S2	R2	V_{SS}	Q2	Q3	R3	S3	NC	S4	R4	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1			1 mA													GND	R1		1.5					V
		2				1 mA												"	S1		"					"
		3					1 mA											"	E		"					"
		4						1 mA										"	S2		"					"
		5							1 mA									"	R2		"					"
		6											1 mA					"	R3		"					"
		7												1 mA				"	S3		"					"
		8														1 mA		"	S4		"					"
		9															1 mA	"	R4		"					"
$V_{IC(neg)}$		10			-1 mA					GND									R1		-6					V
		11				-1 mA				"									S1		"					"
		12					-1 mA			"									E		"					"
		13						-1 mA		"									S2		"					"
		14							-1 mA	"									R2		"					"
		15								"			-1 mA						R3		"					"
		16								"				-1 mA					S3		"					"
		17								"						-1 mA			S4		"					"
		18								"							-1 mA		R4		"					"
I_{SS} 2/		19			GND	18.0 V	18.0 V	18.0 V	GND	"			GND	18.0 V		18.0 V	GND		V_{SS}		-1.0		-2.5			μA
		20			GND	GND	"	GND	GND	"			GND	GND		GND	GND		"		"		"			"
		21			18.0 V	"	"	"	18.0 V	"			18.0 V	"		"	18.0 V		"		"		"			"
		22			GND	18.0 V	"	18.0 V	GND	"			GND	18.0 V		18.0 V	GND		"		"		"			"
		23			18.0 V	GND	GND	GND	18.0 V	"			18.0 V	"		GND	18.0 V		"		"		"			"
		24			GND				GND	"			GND	GND			GND		"		"		"			"
V_{OH5}	3006	25			"	15.0 V	15.0 V	GND	"	"			"	"		"	"	15.0 V	Q1	14.95		14.95		14.95		V
	"	26			"	GND	"	15.0 V	"	"			"	"		"	"	"	Q2	"		"		"		"
	"	27			"	"	"	GND	"	"			"	15.0 V		"	"	"	Q3	"		"		"		"
	"	28			"	"	"	GND	"	"			"	GND		15.0 V	"	"	Q4	"		"		"		"
V_{OL5}	3007	29			15.0 V	"	"	15.0 V	15.0 V	"			15.0 V	15.0 V		"	15.0 V	"	Q1		0.05		0.05		0.05	"
	"	30			"	15.0 V	"	GND	"	"			"	15.0 V		"	"	"	Q2		"		"		"	"
	"	31			"	"	"	15.0 V	"	"			"	GND		"	"	"	Q3		"		"		"	"
	"	32			"	"	"	15.0 V	"	"			"	15.0 V		GND	"	"	Q4		"		"		"	"
V_{IH1}		33			1.5 V	3.5 V	3.5 V	GND	GND	"			GND	GND		"	GND	5.0 V	Q1	4.5		4.5		4.5		"
		34			3.5 V	"	"	GND	GND	"			"	"		"	"	"	Q1	"		"		"		"
		35			GND	"	"	"	1.5 V	"			"	"		"	"	"	Q2	"		"		"		"
		36			"	"	"	"	3.5 V	"			"	"		"	"	"	Q2	"		"		"		"
		37			"	"	"	"	GND	"			"	"		"	"	"	Q3	"		"		"		"
		38			"	"	"	"	"	"			1.5 V	3.5 V		3.5 V	1.5 V	"	Q3	"		"		"		"
		39			"	"	"	"	"	"			3.5 V	GND		3.5 V	3.5 V	"	Q4	"		"		"		"
		40			"	"	"	"	"	"			GND	"		"	"	"	Q4	"		"		"		"
V_{IH2}		41			3.0 V	7.0 V	7.0 V	"	"	"			"	"		GND	GND	10.0 V	Q1	9.0		9.0		9.0		"
		42			7.0 V	7.0 V	"	"	"	"			"	"		"	"	"	Q1	"		"		"		"
		43			GND	GND	"	"	"	"			"	"		"	"	"	Q2	"		"		"		"
		44			"	"	"	"	"	"			"	"		"	"	"	Q2	"		"		"		"
		45			"	"	"	"	"	"			"	"		"	"	"	Q3	"		"		"		"
		46			"	"	"	"	"	"			"	"		"	"	"	Q3	"		"		"		"
		47			"	"	"	"	"	"			3.0 V	7.0 V		7.0 V	3.0 V	"	Q4	"		"		"		"
		48			"	"	"	"	"	"			7.0 V	GND		7.0 V	7.0 V	"	Q4	"		"		"		"
													GND	GND												

See footnotes at end of device type 53.

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q4	Q1	R1	S1	E	S2	R2	V_{SS}	Q2	Q3	R3	S3	NC	S4	R4	V_{DD}		Min	Max	Min	Max	Min	Max	
V_{IH3}		49			4.0 V	11.0 V	11.0 V	GND	GND	GND			GND	GND		GND	GND	15.0 V	Q1	13.5		13.5		13.5		V
		50			11.0 V	11.0 V	"	GND	GND	"			"	"		"	"	"	Q1	"		"		"		"
		51			GND	GND	"	11.0 V	4.0 V	"			"	"		"	"	"	Q2	"		"		"		"
		52			"	"	"	11.0 V	11.0 V	"			"	"		"	"	"	Q2	"		"		"		"
		53			"	"	"	GND	GND	"			4.0 V	11.0 V		"	"	"	Q3	"		"		"		"
		54			"	"	"	"	"	"			11.0 V	11.0 V		"	"	"	Q3	"		"		"		"
		55			"	"	"	"	"	"			GND	GND		11.0 V	4.0 V	"	Q4	"		"		"		"
		56			"	"	"	"	"	"			"	"		11.0 V	11.0 V	"	Q4	"		"		"		"
V_{IL1}		57			3.5 V	1.5 V	3.5 V	"	"	"			"	"		GND	GND	5.0 V	Q1		0.5		0.5		0.5	"
		58			GND	GND	"	1.5 V	3.5 V	"			"	"		"	"	"	Q2		"		"		"	"
		59			"	"	"	GND	GND	"			3.5 V	1.5 V		"	"	"	Q3		"		"		"	"
		60			"	"	"	"	"	"			GND	GND		1.5 V	3.5 V	"	Q4		"		"		"	"
V_{IL2}		61			7.0 V	3.0 V	7.0 V	"	"	"			"	"		GND	GND	10.0 V	Q1		1.0		1.0		1.0	"
		62			GND	GND	"	3.0 V	7.0 V	"			"	"		"	"	"	Q2		"		"		"	"
		63			"	"	"	GND	GND	"			7.0 V	3.0 V		"	"	"	Q3		"		"		"	"
		64			"	"	"	"	"	"			GND	GND		3.0 V	7.0 V	"	Q4		"		"		"	"
V_{IL3}		65			11.0 V	4.0 V	11.0 V	"	"	"			"	"		GND	GND	15.0 V	Q1		1.5		1.5		1.5	"
		66			GND	GND	"	4.0 V	11.0 V	"			"	"		"	"	"	Q2		"		"		"	"
		67			"	"	"	GND	GND	"			11.0 V	4.0 V		"	"	"	Q3		"		"		"	"
		68			"	"	"	"	"	"			GND	GND		4.0 V	11.0 V	"	Q4		"		"		"	"
I_{OL1}		69		0.4 V	5.0 V	"	5.0 V	"	"	"	0.4 V		"	"		GND	GND	5.0 V	Q1	0.51		0.36		0.64		mA
		70			GND	"	"	"	5.0 V	"			"	"		"	"	"	Q2	"		"		"		"
		71			"	"	"	"	GND	"		0.4 V	5.0 V	"		"	"	"	Q3	"		"		"		"
		72	0.4 V		"	"	"	"	"	"			GND	"		"	5.0 V	"	Q4	"		"		"		"
I_{OL2}		73		1.5 V	15.0 V	"	15.0 V	"	"	"	1.5 V		"	"		"	GND	15.0 V	Q1	3.4		2.4		4.2		"
		74			GND	"	"	"	15.0 V	"			"	"		"	"	"	Q2	"		"		"		"
		75			"	"	"	"	GND	"		1.5 V	15.0 V	"		"	"	"	Q3	"		"		"		"
		76	1.5 V		"	"	"	"	"	"			GND	"		"	15.0 V	"	Q4	"		"		"		"
I_{OH1}		77		4.6 V	"	5.0 V	5.0 V	"	"	"	4.6 V		"	"		"	GND	5.0 V	Q1	-0.51		-0.36		-0.64		"
		78			"	GND	"	5.0 V	"	"			"	"		"	"	"	Q2	"		"		"		"
		79			"	"	"	GND	"	"		4.6 V	"	5.0 V		"	"	"	Q3	"		"		"		"
		80	4.6 V		"	"	"	"	"	"			"	GND		5.0 V	"	"	Q4	"		"		"		"
I_{OH2}		81		13.5 V	"	15.0 V	15.0 V	"	"	"	13.5 V		"	"		GND	"	15.0 V	Q1	-3.4		-2.4		-4.2		"
		82			"	GND	"	15.0 V	"	"			"	"		"	"	"	Q2	"		"		"		"
		83			"	"	"	GND	"	"		13.5 V	"	15.0 V		"	"	"	Q3	"		"		"		"
		84	13.5 V		"	"	"	GND	"	"			"	GND		15.0 V	"	"	Q4	"		"		"		"
I_{IH1} 3/	3010	85			18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	"			18.0 V	18.0 V		18.0 V	18.0 V	18.0 V	All inputs together		9					nA

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 1 T_A = 25°C		Subgroup 2 T_A = 125°C		Subgroup 3 T_A = -55°C		
		Test no.	Q4	Q1	R1	S1	E	S2	R2	V_{SS}	Q2	Q3	R3	S3	NC	S4	R4	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{IH2}	3010	86			18.0 V	18.0 V	18.0 V	18.0 V	18.0 V	GND			18.0 V	18.0 V		18.0 V	18.0 V	18.0 V	R1		1.0		45			mA
	"	87			"	"	"	"	"	"			"	"		"	"	"	S1		"		"			
	"	88			"	"	"	"	"	"			"	"		"	"	"	E		"		"			
	"	89			"	"	"	"	"	"			"	"		"	"	"	S2		"		"			
	"	90			"	"	"	"	"	"			"	"		"	"	"	R2		"		"			
	"	91			"	"	"	"	"	"			"	"		"	"	"	R3		"		"			
	"	92			"	"	"	"	"	"			"	"		"	"	"	S3		"		"			
	"	93			"	"	"	"	"	"			"	"		"	"	"	S4		"		"			
	"	94			"	"	"	"	"	"			"	"		"	"	"	R4		"		"			
I_{IL1} 3/	3009	95			GND	GND	GND	GND	GND	"			GND	GND		GND	GND	"	All inputs together		-9					"
I_{IL2}	"	96			"	"	"	"	"	"			"	"		"	"	"	R1		-1.0		-45			"
	"	97			"	"	"	"	"	"			"	"		"	"	"	S1		"		"			
	"	98			"	"	"	"	"	"			"	"		"	"	"	E		"		"			
	"	99			"	"	"	"	"	"			"	"		"	"	"	S2		"		"			
	"	100			"	"	"	"	"	"			"	"		"	"	"	R2		"		"			
	"	101			"	"	"	"	"	"			"	"		"	"	"	R3		"		"			
	"	102			"	"	"	"	"	"			"	"		"	"	"	S3		"		"			
	"	103			"	"	"	"	"	"			"	"		"	"	"	S4		"		"			
	"	104			"	"	"	"	"	"			"	"		"	"	"	R4		"		"			
																				Subgroup 4 T_A = 25°C						
																				Min	Max					
C1	3012	105			4/	4/	4/	4/	4/	GND			4/	4/		4/	4/	GND	R1		12					pF
	"	106								"								"	S1		"					"
	"	107								"								"	E		"					"
	"	108								"								"	S2		"					"
	"	109								"								"	R2		"					"
	"	110								"								"	R3		"					"
	"	111								"								"	S3		"					"
	"	112								"								"	S4		"					"
	"	113								"								"	R4		"					"
																				Subgroup 7 T_A = 25°C		Subgroup 8 T_A = 125°C		T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
Truth table test	3014	114	L	H	L	H	H	5.0 V	L	H	GND	L	H	L	H	L	H	5.0 V	All outputs	} See 5/ and 6/						
	"	115	L	H	L	H	L	"	H	L	"	H	L	H	L	H	L	"	"							
	"	116	H	L	H	L	H	"	L	H	"	L	H	L	H	L	H	"	"							
	"	117	H	L	H	L	L	"	H	L	"	H	L	H	L	H	L	"	"							
	"	118	L	H	L	H	H	"	L	H	"	L	H	L	H	L	H	"	"							
	"	119	L	H	L	H	L	"	H	L	"	H	L	H	L	H	L	"	"							

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TABLE III. Group A inspection for device type 53 - Continued.

Symbol	MIL-STD-883 test method	Case E,F,2	Terminal conditions 1/																Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgroup 9 T_A = 25°C		Subgroup 10 T_A = 125°C		Subgroup 11 T_A = -55°C		
		Test no.	Q4	Q1	R1	S1	E	S2	R2	V_{SS}	Q2	Q3	R3	S3	NC	S4	R4	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PHL} R	3003 Fig. 17 "	120 121 122 123	OUT	OUT	IN		5.0 V " " "		IN	GND " " "	OUT	OUT	IN				IN	5.0 V " " "	R1 to Q1 R2 to Q2 R3 to Q3 R4 to Q4	16 " " "	320 " " "	19 " " "	370 " " "	14 " " "	270 " " "	ns " " "
t_{PLH}	" " " "	124 125 126 127	OUT	OUT		IN	" " " "	IN		" " " "	OUT	OUT		IN		IN		" " " "	S1 to Q1 S2 to Q2 S3 to Q3 S4 to Q4	10 " " "	200 " " "	12 " " "	245 " " "	9 " " "	185 " " "	" " " "
t_{PZH}	Fig. 18	128		OUT	GND	5.0 V	IN	GND	GND	"			GND	GND		GND	GND	"	E to Q1		230		340		230	"
t_{PHZ}	" " "	129 130 131	OUT		" " "	GND " "	" " "	5.0 V GND "	" " "	" " "	OUT	OUT	" " "	GND 5.0 V GND		" " 5.0 V	" " "	" " "	E to Q2 E to Q3 E to Q4		" " "		" " "		" " "	" " "
t_{PZL}	"	132		OUT	5.0 V	"	"	"	"	"			"	"		GND	"	"	E to Q1		180		240		180	"
t_{PLZ}	" " "	133 134 135	OUT		GND " "	" " "	" " "	" " "	5.0 V GND GND	" " "	OUT	OUT	" 5.0 V GND	" " "		" " "	" " 5.0 V	" " "	E to Q2 E to Q3 E to Q4		" " "		" " "		" " "	" " "
t_{THL}	3004 Fig. 17 "	136 137 138 139	OUT	OUT		IN	5.0 V " " "	IN		" " " "	OUT	OUT		IN		IN		" " " "	Q1 Q2 Q3 Q4	10 " " "	200 " " "	12 " " "	245 " " "	9 " " "	185 " " "	" " " "
t_{TLH}	" " " "	140 141 142 143	OUT	OUT		IN	" " " "	IN		" " " "	OUT	OUT		IN		IN		" " " "	Q1 Q2 Q3 Q4	15 " " "	300 " " "	18 " " "	360 " " "	13 " " "	250 " " "	" " " "

1/ Pins not designated may be "high" level logic, "low" level logic, or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ Test numbers 19 thru 24 shall be run in sequence.

3/ The device manufacturer may, at his options, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

4/ See 4.4.1c.

5/ Test numbers 114 thru 119 shall be run in sequence and the functional tests shall be performed with V_{IH} and $V_{DD} \leq 5.0$ V and ≥ 18.0 V.

6/ L = V_{SS} + 0.5 V maximum and H = V_{DD} - 0.5 V minimum.

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_I measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 5, or equivalent, shall be used.

b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and shall be performed on each inspection lot for initially qualified device types 01, 02, 03, and measured only for initial qualification and after process or design changes for initially qualified device types 51, 52, 53. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

c. End point electrical parameters shall be as specified in table II herein and shall consist of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein also. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection, and shall consist of tests specified in table IV herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions.

1. Test condition D or E, as specified in 4.5.2, and as shown on figure 5, or equivalent.

2. T_A = 125°C minimum.

3. Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883; end-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed, it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

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4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25°C ±3°C, and the electrical parameter end-point measurements shall then be performed.

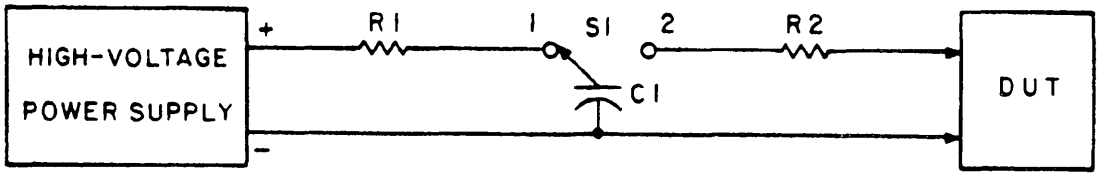
TABLE IV. Delta limits at 25°C.

Parameter	Device types			
	01, 02	03	51, 52	53
I_{SS}	±75 nA	±250 nA	±75 nA	±250 nA
V_{OL1}	±.04 V	±.04 V	---	---
V_{OH1}	±.08 V	±.08 V	---	---
I_{OL1}	---	---	±15%	±15%
I_{OH1}	---	---	±15%	±15%

NOTE: Each of the above parameters shall be recorded before and after the required burn-in or life tests to determine delta's (Δ).

4.5.3 High voltage (V_{ZAP}) test of input protection circuits. All input terminals (up to a maximum of four) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 volts. This destructive test shall be conducted as follows using the test circuit on figure 19.

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD}. Measure I_{SS} on the DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at the specified V_{DD}.



V_{ZAP} = 400 V charge on C1.
1 MΩ ≤R1 ≤50 MΩ.
R2 = 1.5 kΩ.
C1 = 100 pF.
S1 = Hg-wetted "bounceless" relay.

FIGURE 19. High voltage (V_{ZAP}) test circuit.

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b. V_{ZAP} is applied to the DUT in the following modes (see table V) by charging C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high-voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated output

c. Within 24 hours repeat the I_{SS}, I_{IH}, and I_{IL} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS}) test. When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end-point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end-point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table VIII and figure 20. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardness end-point electrical parameters at 25°C.

Parameter	Test limits (all device types)	V_{DD}	
		Device types 01, 02, 03	Device types 51, 52, 53
V_{TN}	0.3 V min	10 V dc	10 V dc
V_{TP}	2.8 V max	10 V dc	10 V dc
ΔV_T	1.4 V max	10 V dc	10 V dc
I_{SS}	100 x max limit	15 V dc	18 V dc
t_{PLH}	1.35 x max limit	5 V dc	5 V dc
t_{PHL}	1.35 x max limit	5 V dc	5 V dc

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TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	V_{DD} = 10 V dc (through a 30 kΩ to 60 kΩ resistor)	V_{SS} = GND	V_{DD} = 10 V dc
01, 51	3, 4, 5, 6, 8, 9, 10, 11	7	14
02, 52	3, 4, 5, 6, 7, 9, 10, 11, 12, 13	8	16
03, 53	3, 4, 5, 6, 7, 11, 12, 14, 15	8	16

Pins not designated are open or connected to 10 V dc through a 30 kΩ to 60 kΩ resistor.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be specified:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady state life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parmeters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths or lead forming, if applicable.

h. Requirements for "JAN" marking.

i. Requirement for total dose radiation testing (see 3.6.1 and 4.5.5), if applicable.

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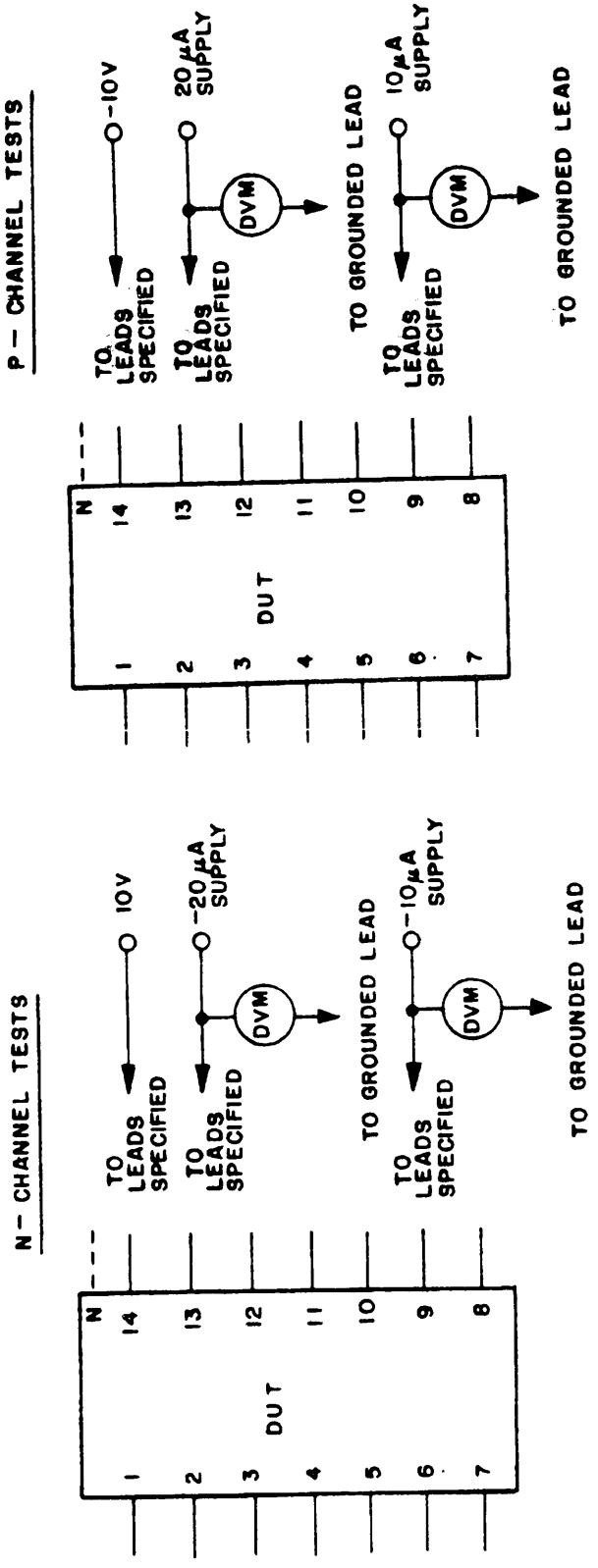


FIGURE 20. Threshold-voltage test circuit.

TABLE VIII. Threshold-voltage test circuit conditions.

Device type	GND	10V	V_{IN} measured at		GND	-10V	V_{TP} measured at	
			-20 μA supply	-10 μA supply			20 μA supply	10 μA supply
01 51	3	14		4-11	3	4-11		14
02 52	13	3-7, 9-12, 16		8	13	3-12		16
03 53	5	16		3, 4, 6-8, 11, 12, 14, 15	5	3, 4, 6-8, 11, 12, 14, 15		16

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{ZAP}	Input test voltage.
C_I	Input terminal-to-V_{SS} capacitance.
GND	Ground. Zero voltage potential.
T_A	Free air temperature.
V_{IC}	Input clamp voltage.
V_{ICL}	Clock input voltage.
V_{DD}	Positive supply voltage.
V_{SS}	Negative supply voltage.
I_{SS}	Quiescent supply voltage.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support shall be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	4013A
02	4027A
03	4043A
51	4013B
52	4027B
53	4043B

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Navy - EC
Air Force - 17
NASA - NA

Review activities:
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Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM, AR
Navy - AS, OS, CG, MC, SH

Agent:
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Preparing activity:
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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/51E	Microcircuits, Digital, CMOS, Flip-Flops and Latches, Monolithic Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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